

Pengantar Pemboran Geotermal

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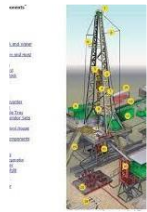
Referensi

- NZ Standard. Code of Practice for Deep Geothermal Wells. 2015
- Sandia Report: Handbook of Best Practices for Geothermal Drilling
- Best Practices Guide for Geothermal Exploration. IFC & IGA
- Pedoman Efisiensi Biaya Pengeboran Sumur Panas Bumi. KESDM 2023.

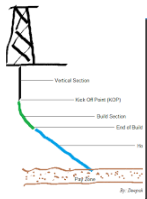
Outline



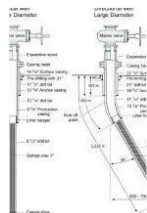
An overview: Indonesian Geothermal Wells



Drilling Equipment & Well Components



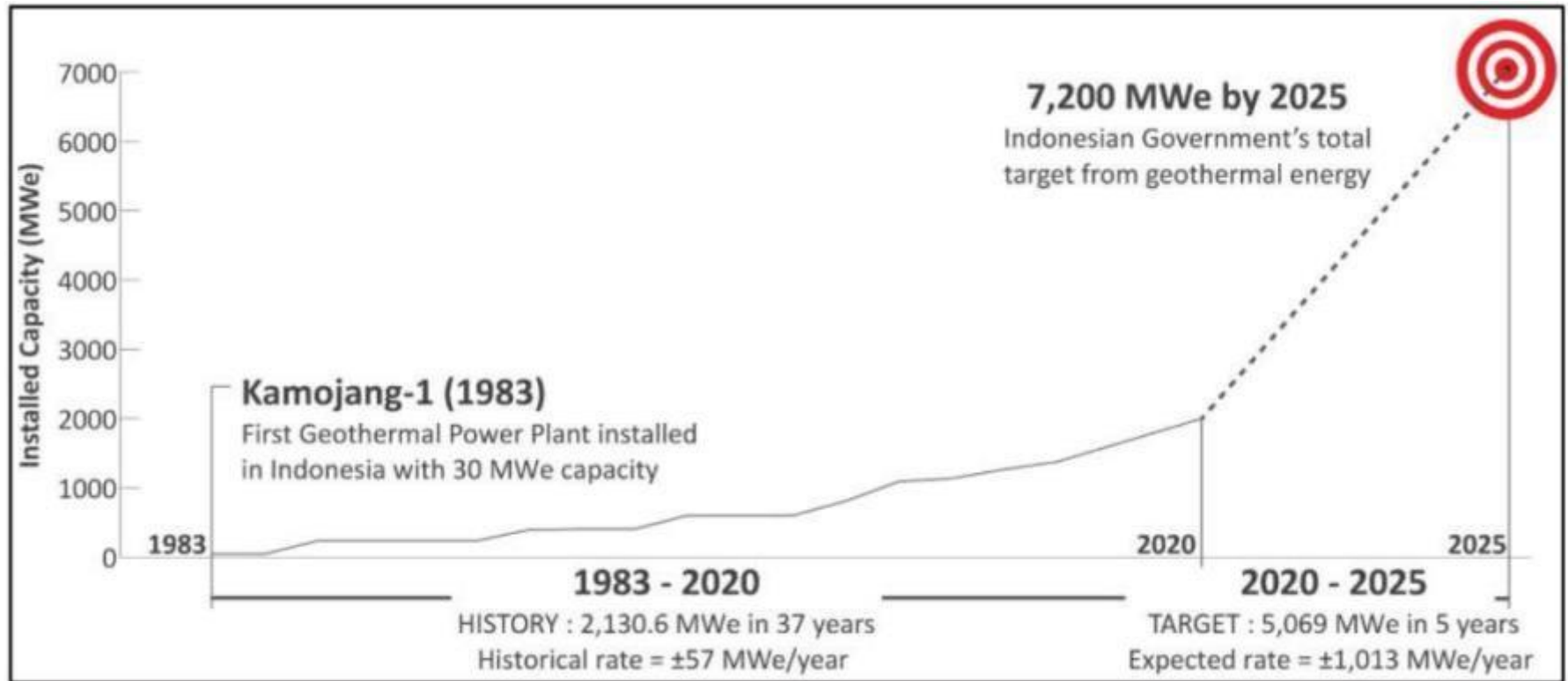
Well Types



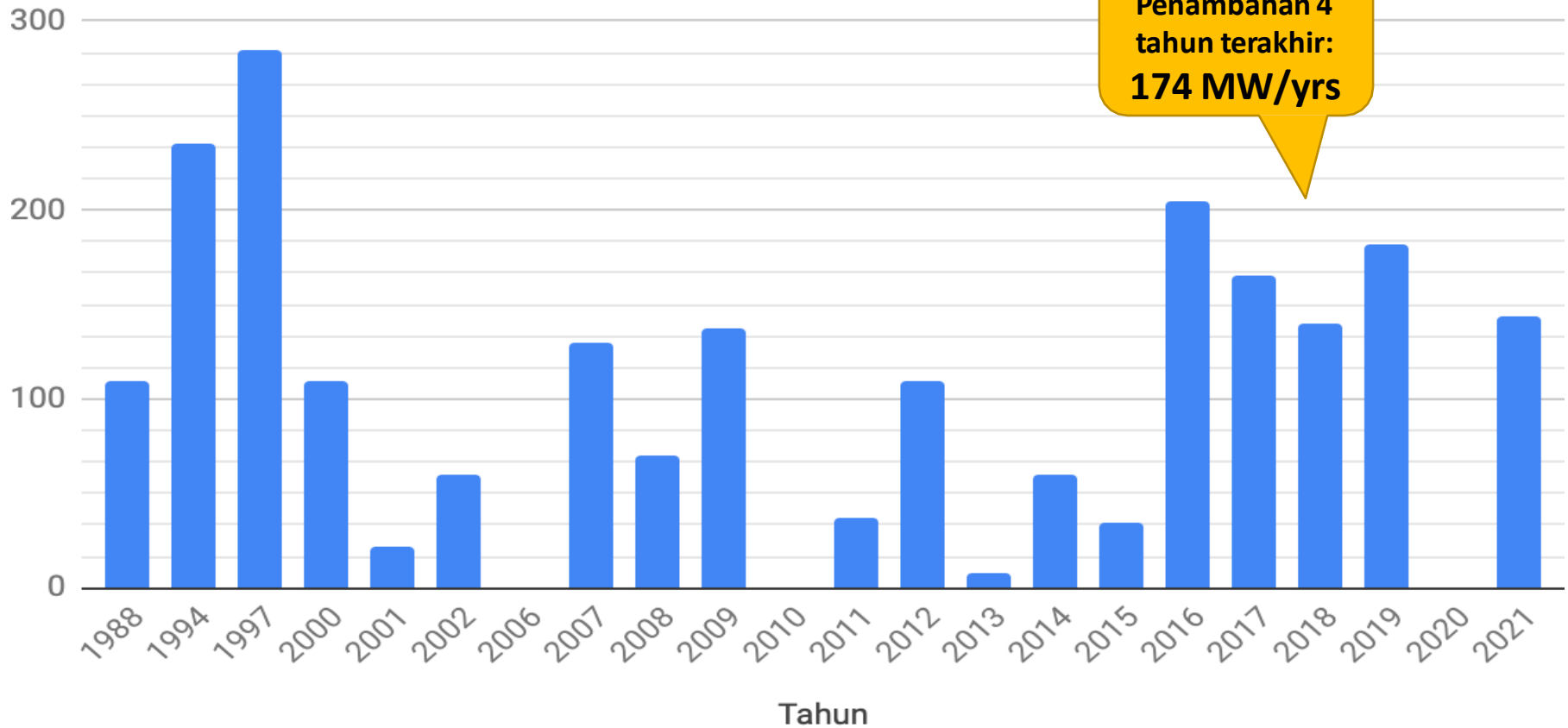
Well Design & Program



Pertumbuhan Pemanfaatan Energi Geotermal di Indonesia



Penambahan MW per Tahun



Penambahan 2021:
Rantau Dedap: 98.4 MW
Sokoria : 5 MW

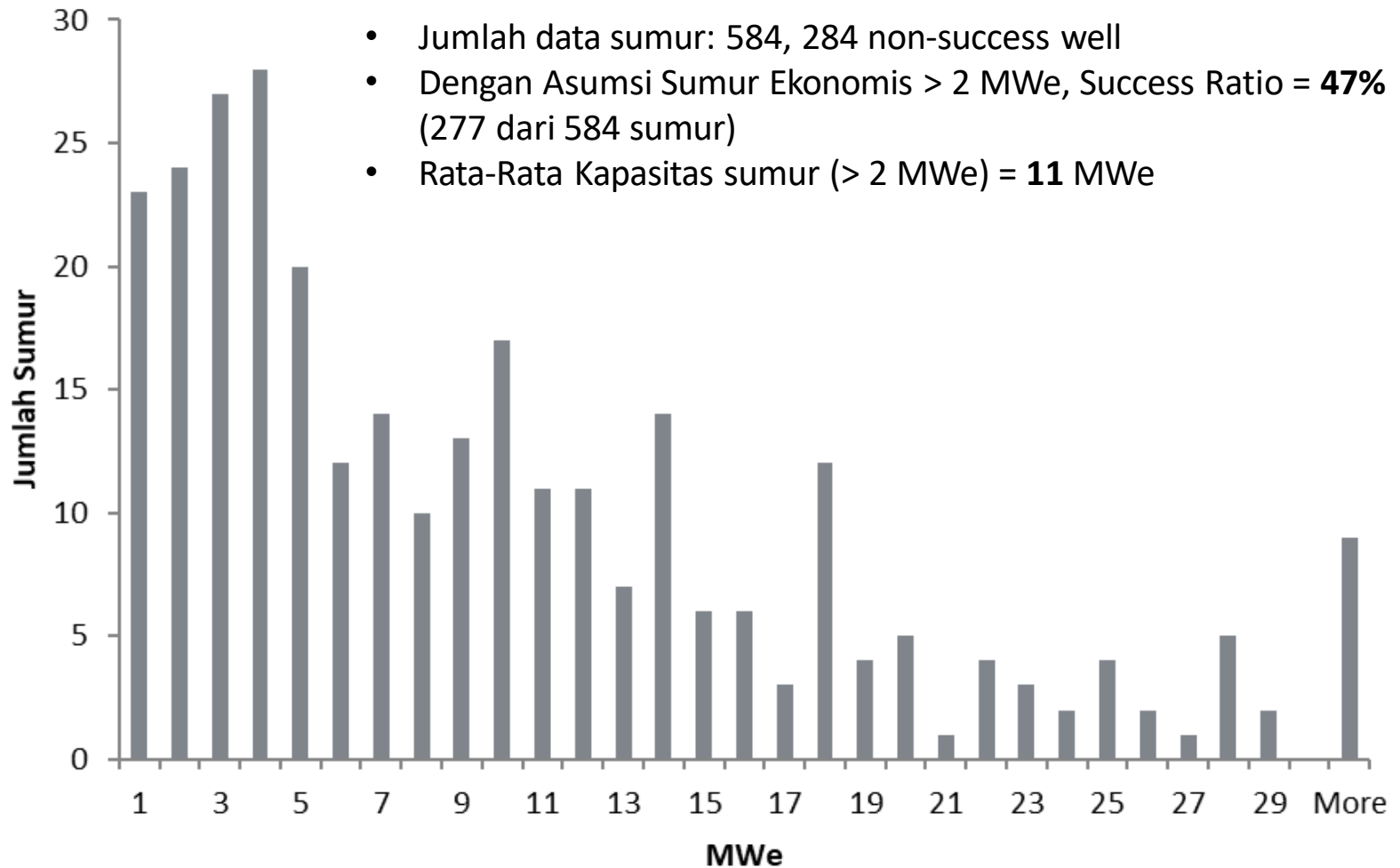
Jumlah Sumur Geotermal di Indonesia

Penambahan		110	235	285	110	22	60	0	130	70	137	0	37	110	7.5	60	35	205	165	140	182.3	0	155.45	6.58	
No.	PLTP	Kapasitas Terpasang (MW)																							
		1983	1988	1994	1997	2000	2001	2002	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
1	Kamojang	30	140	140	140	140	140	140	140	140	200	200	200	200	200	200	235	235	235	235	235	235	235	235	235
2	G. Salak			180	375	375	375	375	375	375	375	375	377	377	377	377	377	377	377	377	377	377	377	377	377
3	Darajat			55	145	145	145	145	255	255	255	255	270	270	270	270	270	270	270	270	270	270	270	270	270
4	W. Windu					110	110	110	110	110	227	227	227	227	227	227	227	227	227	227	227	227	227	227	227
5	Lahendong						20	20	20	40	40	60	60	80	80	80	80	80	120	120	120	120	120	120	120
6	Sibayak						2	2	2	2	12	12	12	12	12	12	12	12	12	12	12	12	12	12	12
7	Dieng							60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60
8	Ulubelu													110	110	110	110	165	220	220	220	220	220	220	220
9	Ulumbu														5	10	10	10	10	10	10	10	10	10	10
10	Mataloko														2.5	2.5	2.5	2.5	2.5	2.5	2.5	2.5	2.5	2.5	2.5
11	Patuha															55	55	55	55	55	55	55	55	55	55
12	Sarulla																		110	220	330	330	330	330	330
13	Karaha																				30	30	30	30	30
14	Muara Labuh																					85	85	85	85
15	Lumut Balai																					55	55	55	55
16	Sorik Marapi																					42.3	42.3	99.35	99.35
17	Rantau Dedap																							98.4	98.4
18	Sokoria																								6.58
	Total	30	140	375	660	770	792	852	852	982	1052	1189	1189	1226	1336	1343.5	1403.5	1438.5	1643.5	1808.5	1948.5	2130.8	2130.8	2286.25	2292.83



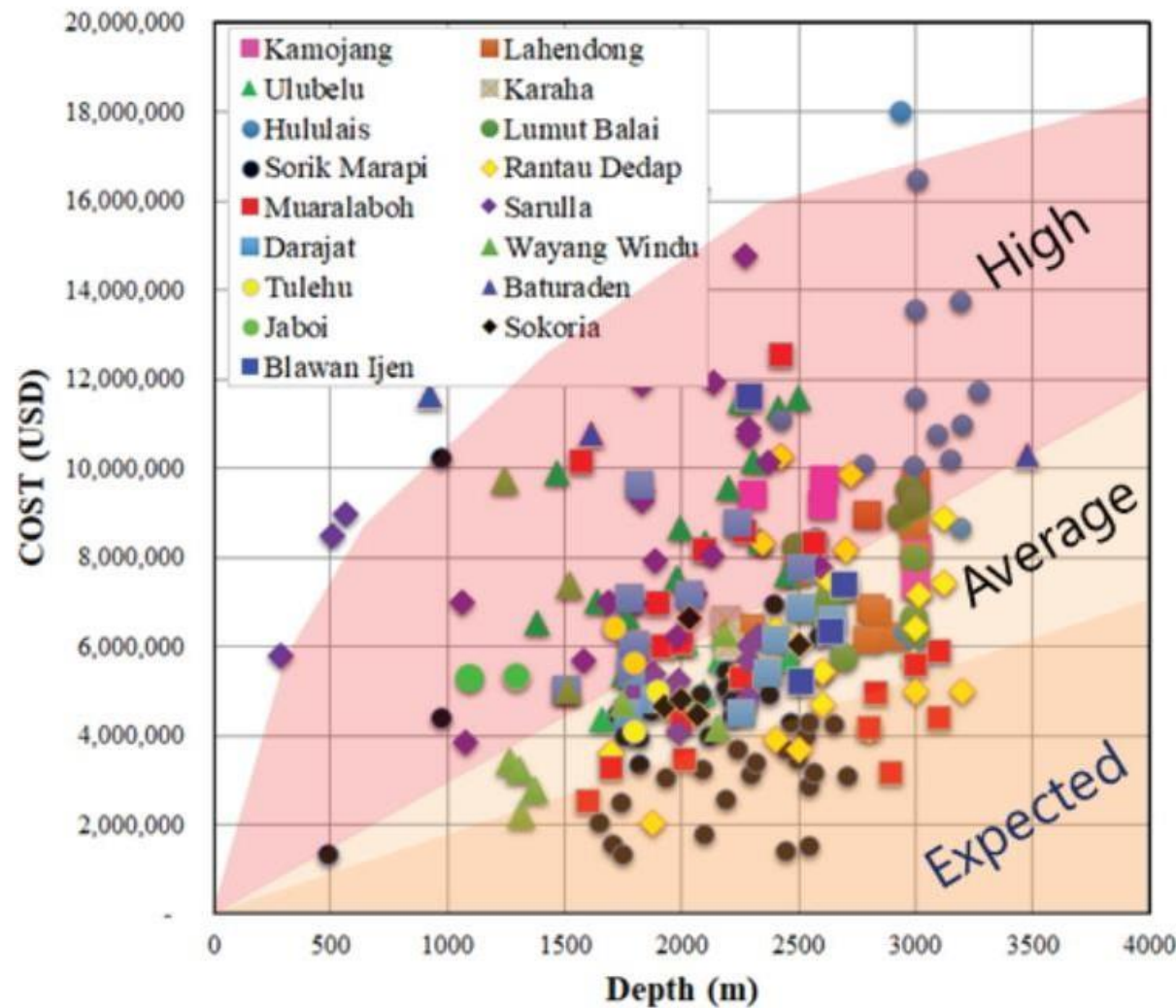
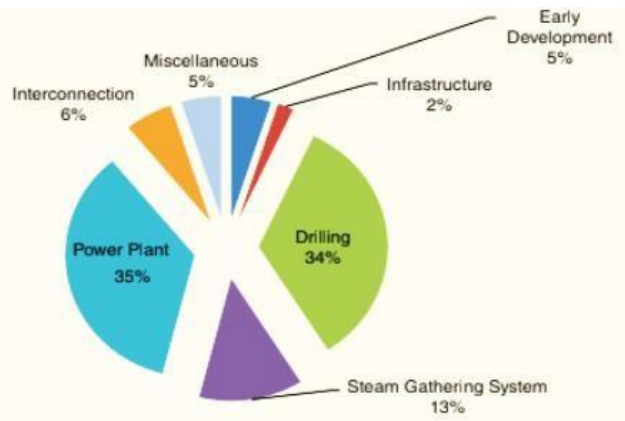
Berapa banyak sumur yang harus di-bor untuk mencapai target RUEN?

Kapasitas MWe Sumur



- Jumlah data sumur: 584, 284 non-success well
- Dengan Asumsi Sumur Ekonomis > 2 MWe, Success Ratio = **47%** (277 dari 584 sumur)
- Rata-Rata Kapasitas sumur (> 2 MWe) = **11 MWe**

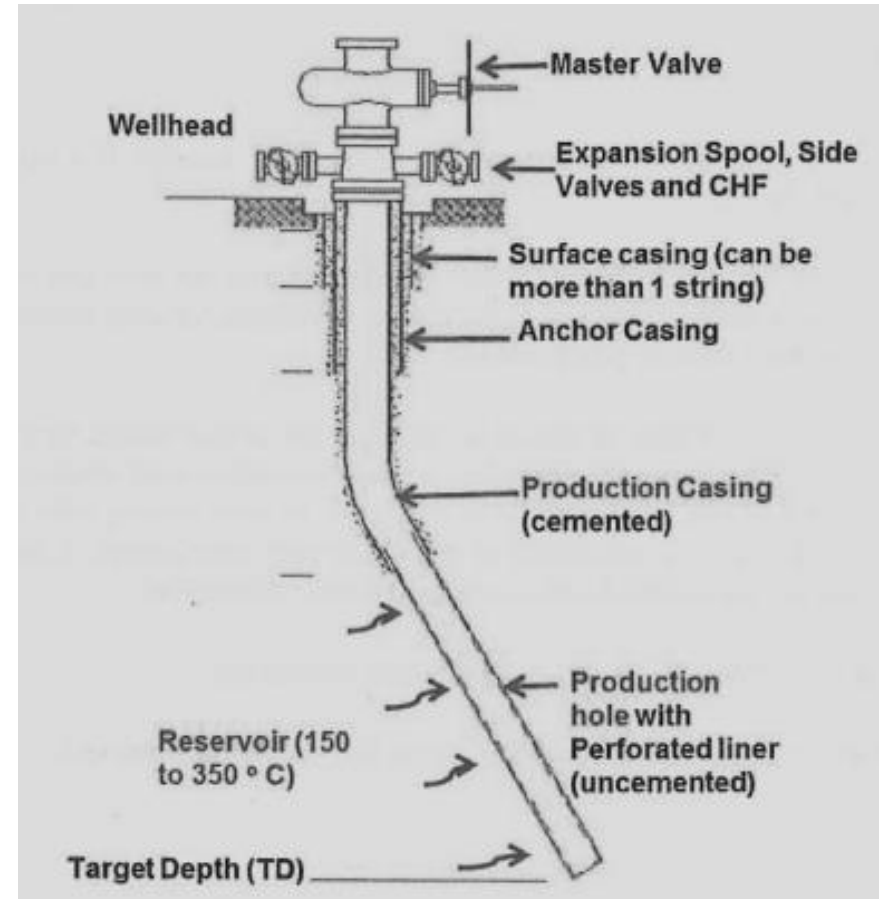
Biaya Pengeboran Sumur Geotermal di Indonesia



Pengeboran Sumur Geotermal

Target Pengeboran Sumur Geotermal

- Menembus zona temperatur tinggi
- Memiliki rekahan-rekahan (permeabilitas baik)
- Kedalaman ekonomis pengeboran (hingga ± 2 km)



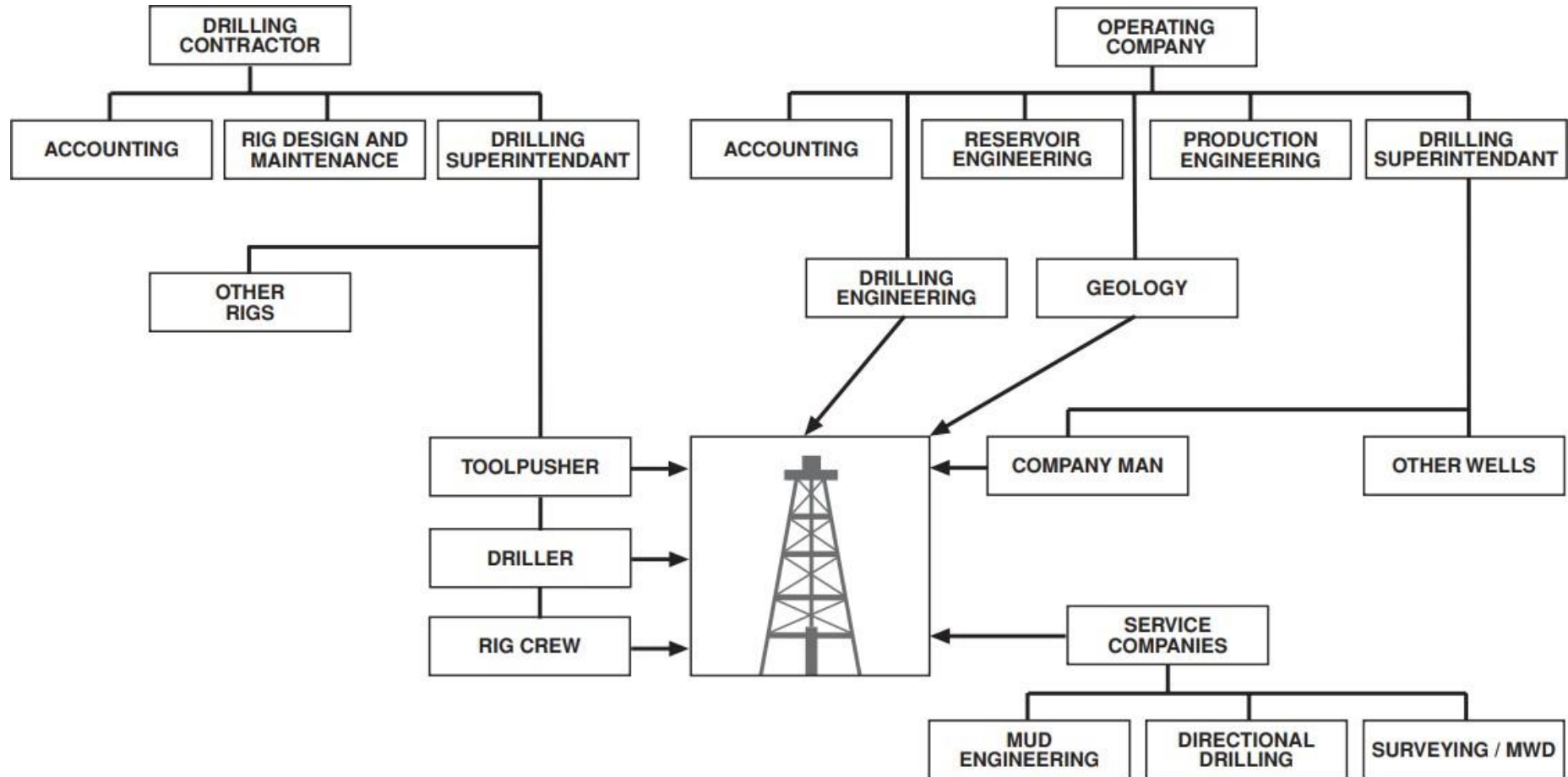
Konfigurasi Sumur Geotermal

Perencanaan dan Desain Pengeboran Sumur Geothermal

Key Question

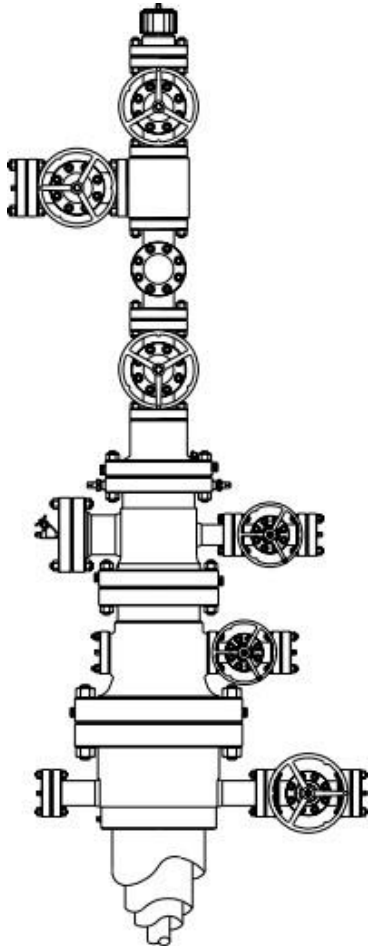
- What is the **Purpose** of the Geothermal Well?
- What **Type of Geothermal Resource** is Targeted?
- What are the **Geological Conditions**?
- What is the **Required Well Depth**?
- What **Drilling Techniques** will be Used?
- What is the **Drilling Fluid Composition**?
- How is **Casing Design** Determined?
- What are the **Environmental Considerations**?
- How is Well **Maintenance and Monitoring** Handled?

Drilling Personnel



Source: Herriot-Watt

Drilling Standard



ANSI/ASME B16.5:2013	Pipe flanges and flange fittings: NPS 1/2 through NPS 24 metric/inch standard
ANSI/NACE MR 0175/ISO 15156:2009	Petroleum and natural gas industries – Materials for use in H ₂ S-containing environments in oil and gas production
API Spec 4F:2013	Specification for drilling and well servicing structures
API RP 5A3:2011	Recommended practice on thread compounds for casing, tubing, line pipe and drill stem elements
API RP 5C1 (R 2010)	Recommended practice for care and use of casing and tubing
API TR 5C3:2008	Technical report on equations and calculations for casing, tubing, and line pipe used as casing or tubing; and performance properties tables for casing and tubing
API Spec 5CT:2011	Specification for casing and tubing
API Spec 5DP:2009	Specification for drill pipe
API Spec 5L:2012	Specification for line pipe
API Spec 6A:2013	Specification for wellhead and Christmas tree equipment
API Spec 6D:2012	Specification for pipeline valves
API Spec 7-1 (R 2012)	Specification for rotary drill stem elements
API Spec 7-2:2010	Specification for threading and gauging of rotary shouldered thread connections
API Spec 7F:2010	Specification for oil field chain and sprockets
API RP 7G (R 2010)	Recommended practice for drill stem design and operation limits
API Spec 7K:2010	Specification for drilling and well servicing equipment
API Spec 8A:2001	Specification for drilling and production hoisting equipment
API RP 8B:2014	Inspections, maintenance, repair, and remanufacture of hoisting equipment
API Spec 8C:2014	Specification for drilling and production hoisting equipment (PSL 1 and PSL 2)
API Spec 9A:2012	Specification for wire rope
API RP 9B:2012	Application, care, and use of wire rope for oil field service
API Spec 13A:2010	Specification for drilling fluid materials
API RP 13B-1:2009	Recommended practice for field testing water-based drilling fluids

Drilling Proposal dan Drilling Program



Drilling Proposal

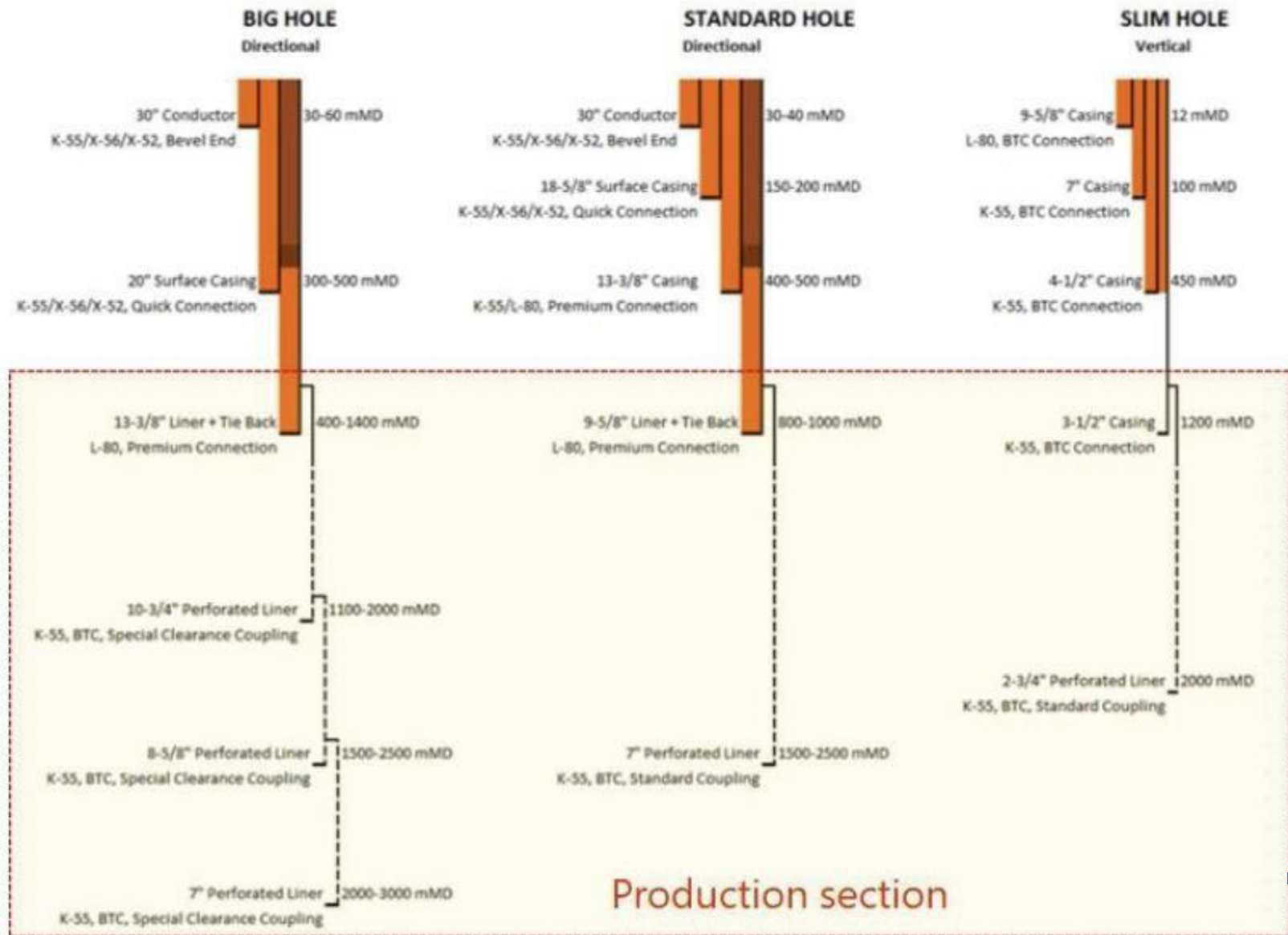
- Objective of the well
- Depth and location of target
- Geological cross section
- Pore pressure profile prediction

Drilling Program

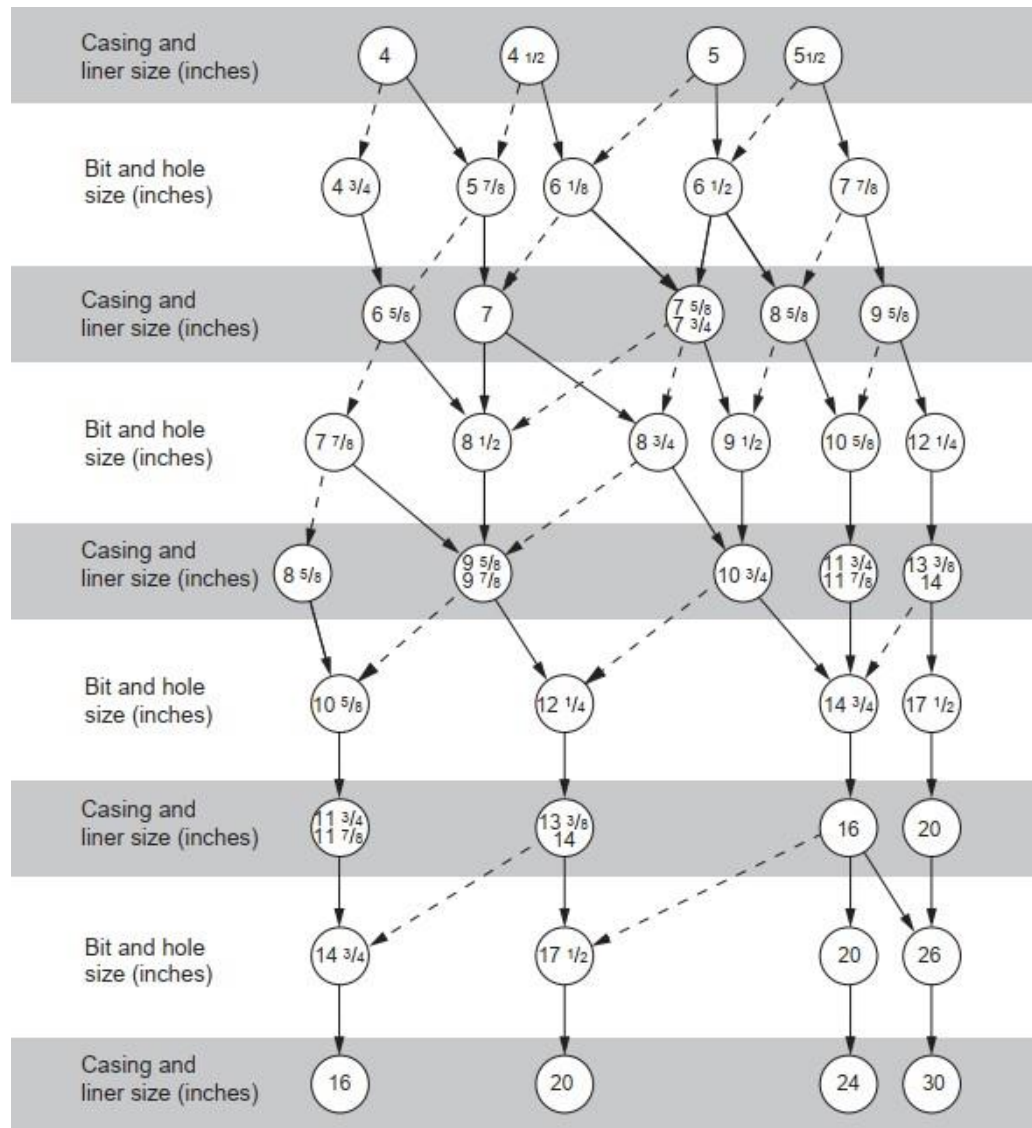


- Drilling Rig to be used for the well
- Proposed Location for the Drilling Rig
- Hole Sizes and Depths
- Casing Sizes and Depths
- Drilling Fluid Specification
- Directional Drilling Information
- Well Control Equipment and Procedures
- Bits and Hydraulics Program

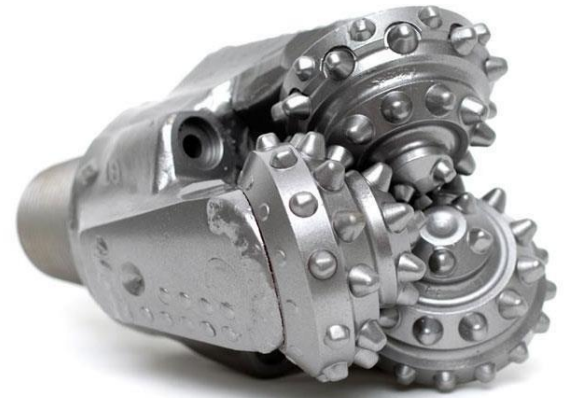
Jenis-Jenis Sumur Geotermal Berdasarkan Ukuran Casing



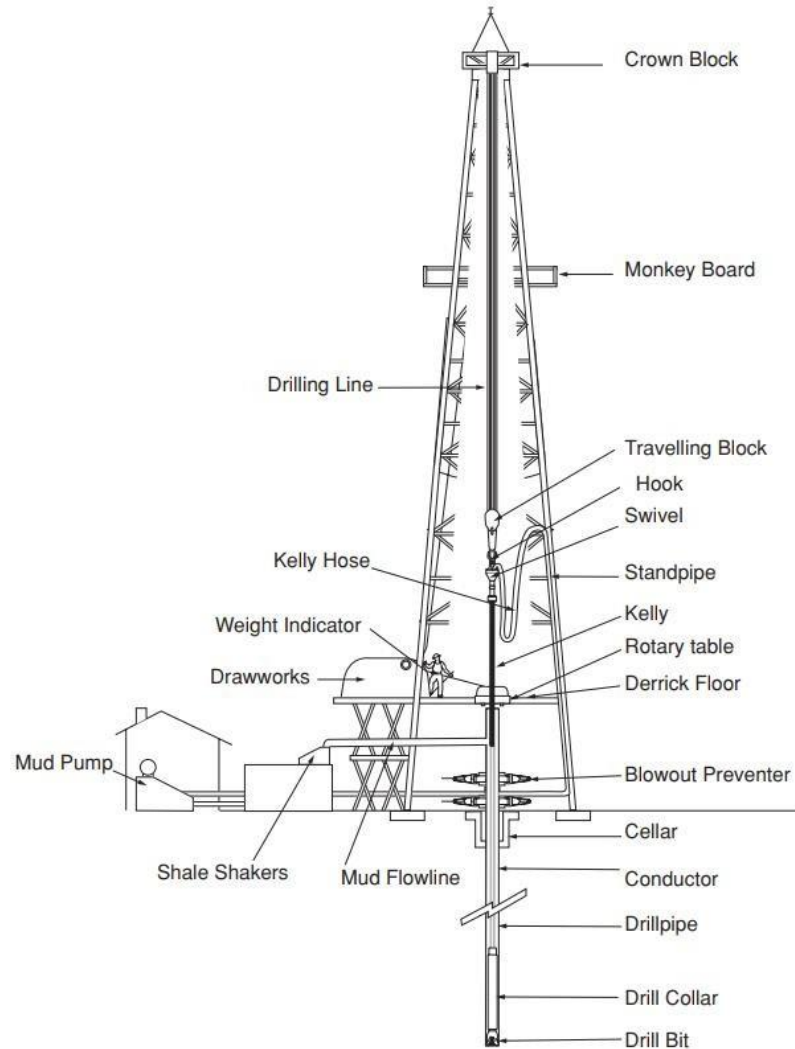
Casing Size Selection



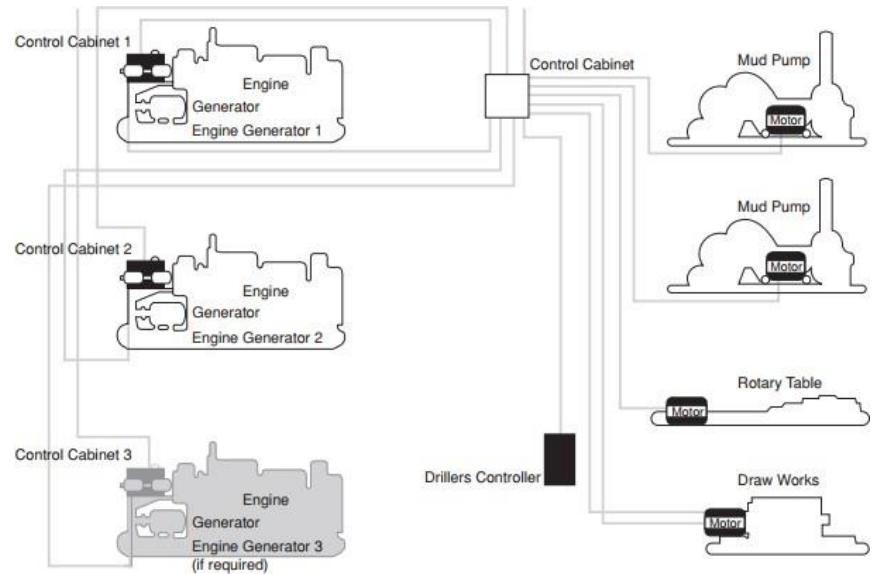
Drilling Equipment



Drilling Equipment

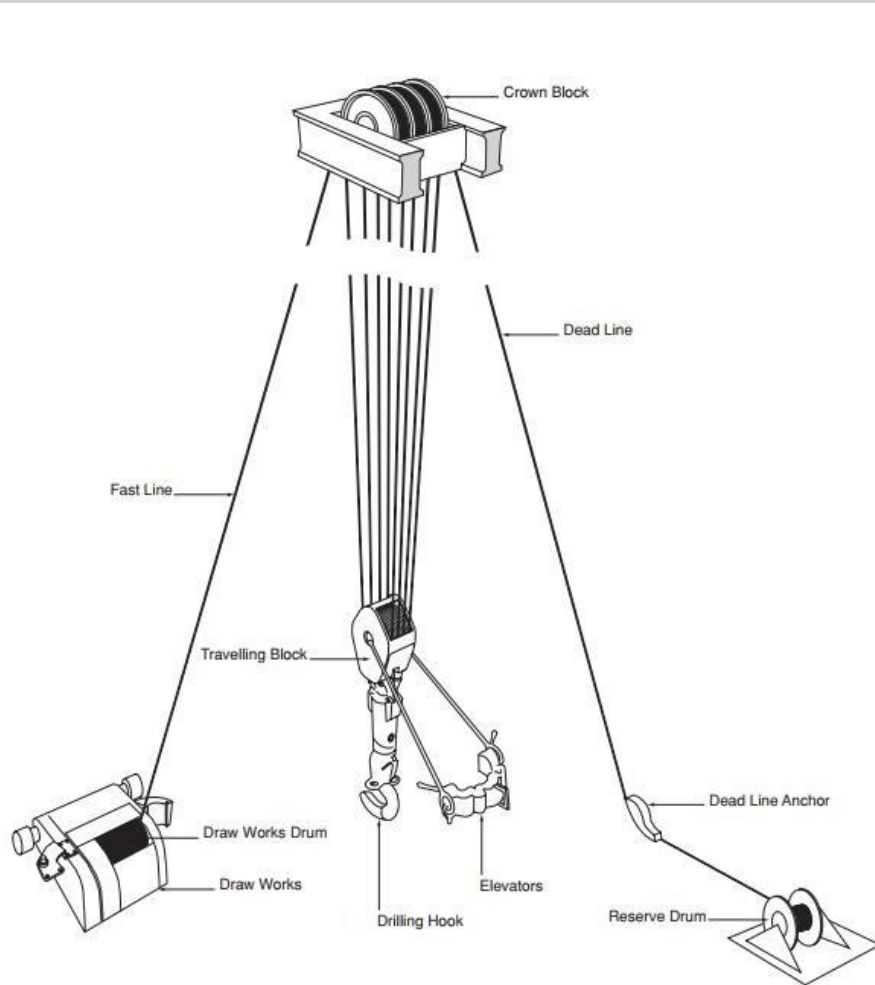


Rotary System

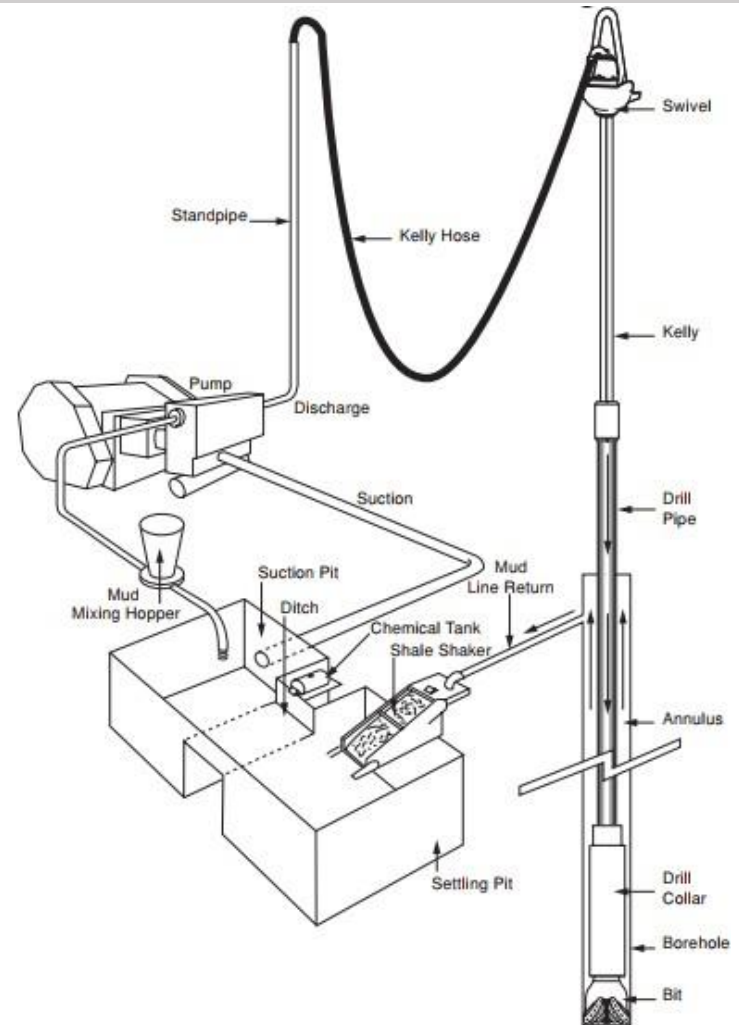


Power System

Drilling Equipment

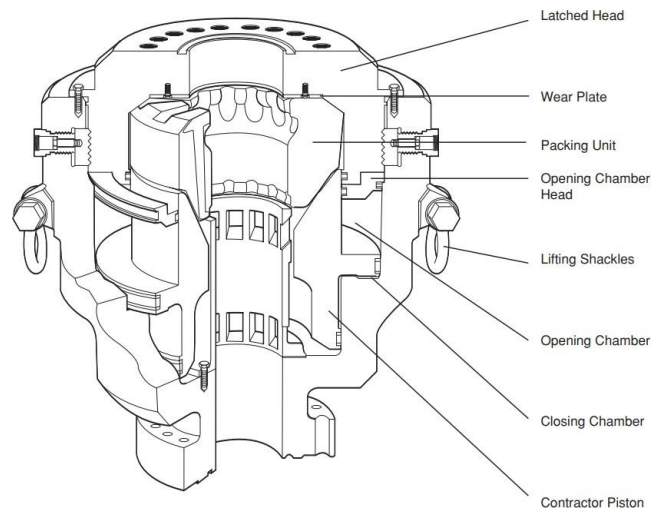


Hoisting System

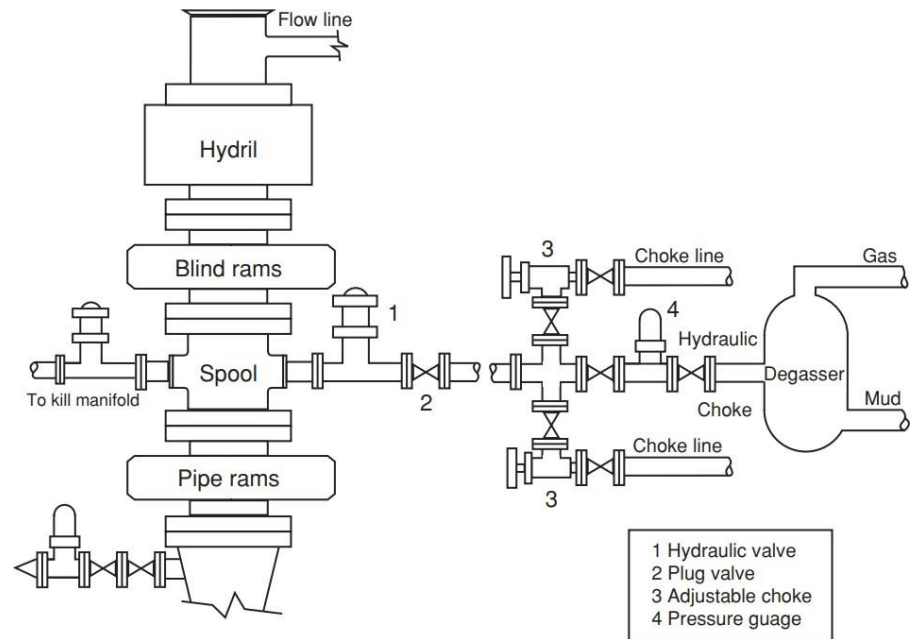


Circulating System

Drilling Equipment

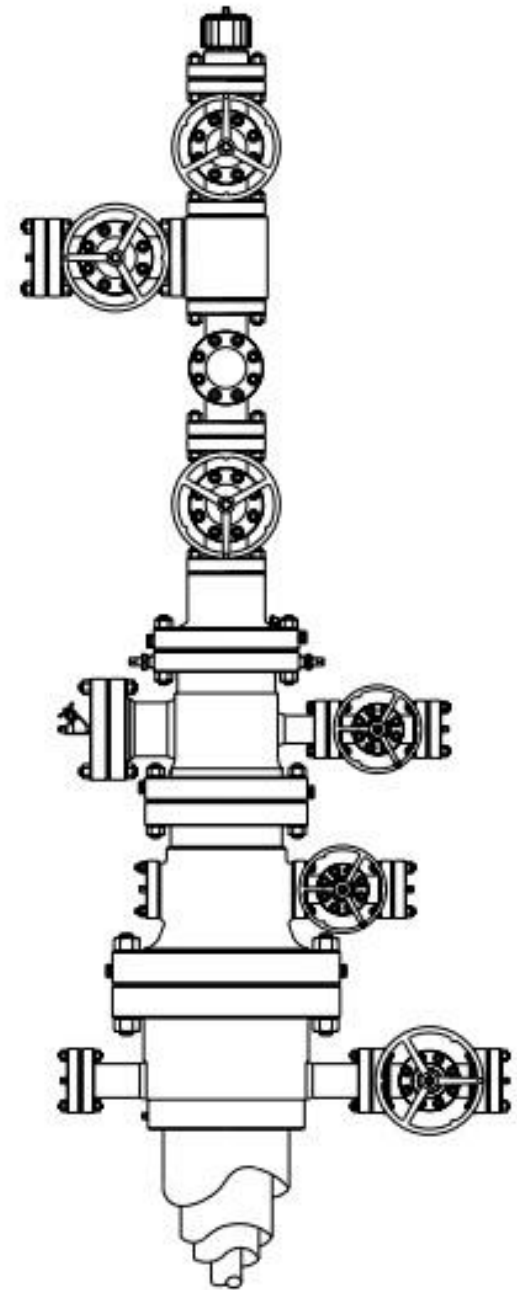


BOP



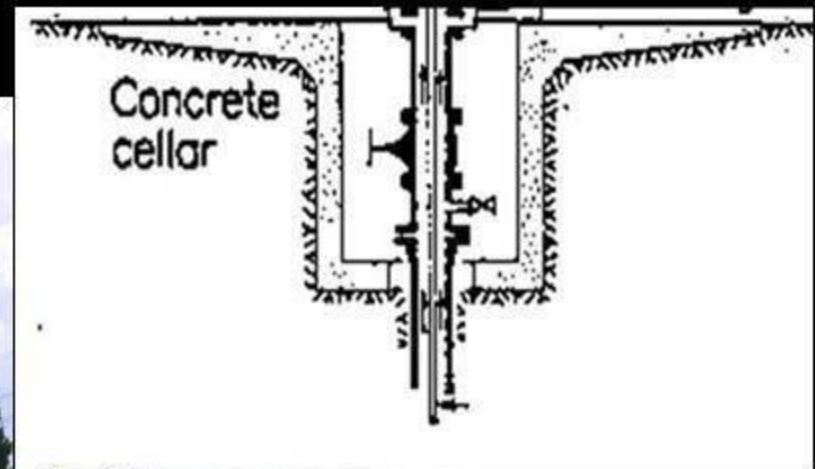
Well Control System

Well components



Tempat *Wellhead* (Kepala Sumur) dibeton (*concrete cellar*)

Lapangan Panas Bumi
Kamojang



VALVE DI KEPALA SUMUR



Wing Valve:
mengatur
aliran uap
ke steam
line (PLTP)

Service
Valve

Wing Valve:
mengatur aliran
uap ke rock
muffler
(saat uji sumur)

Master
Valve

Bleeding
Valve



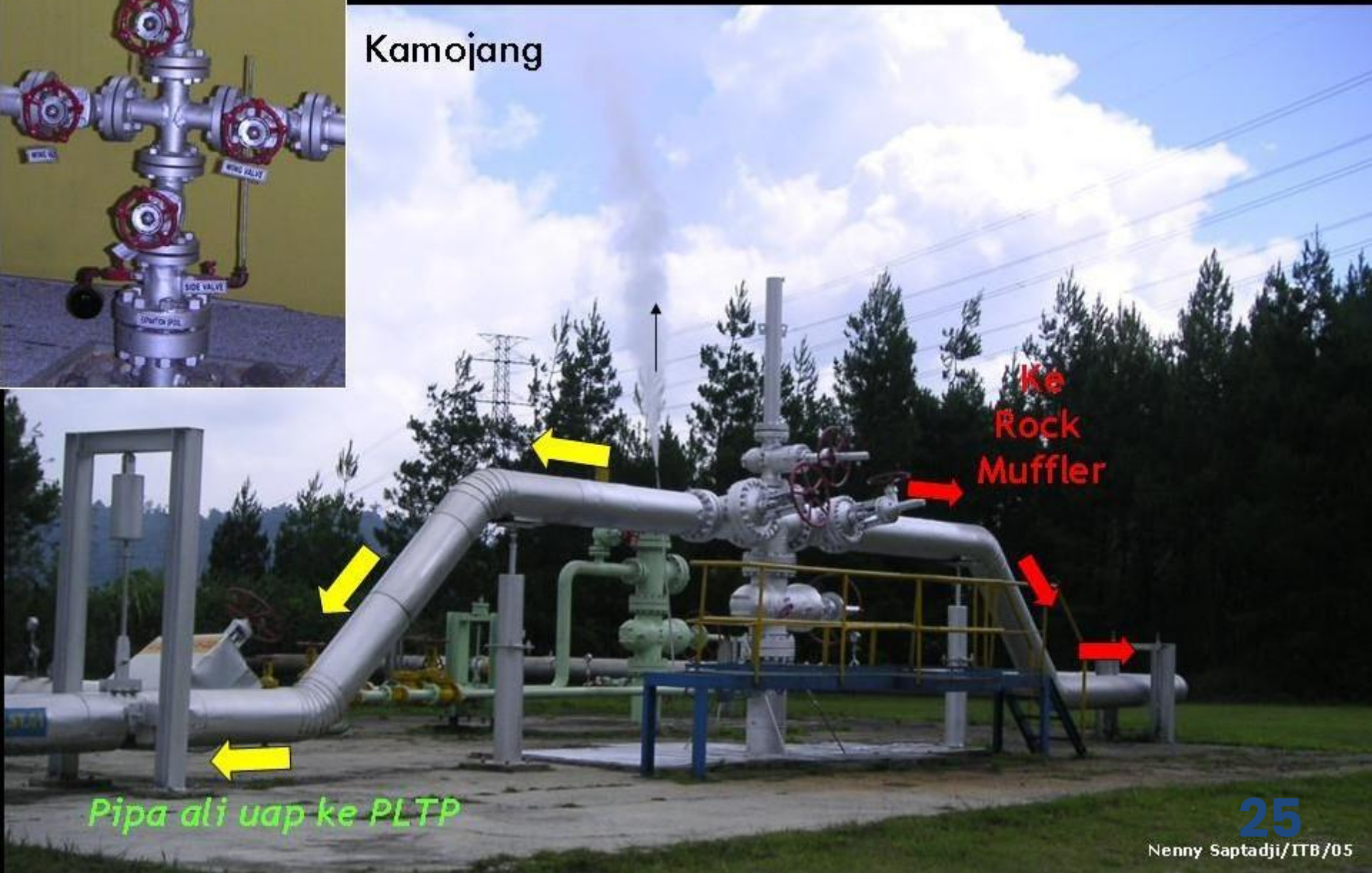
Master Valve



Wing Valves



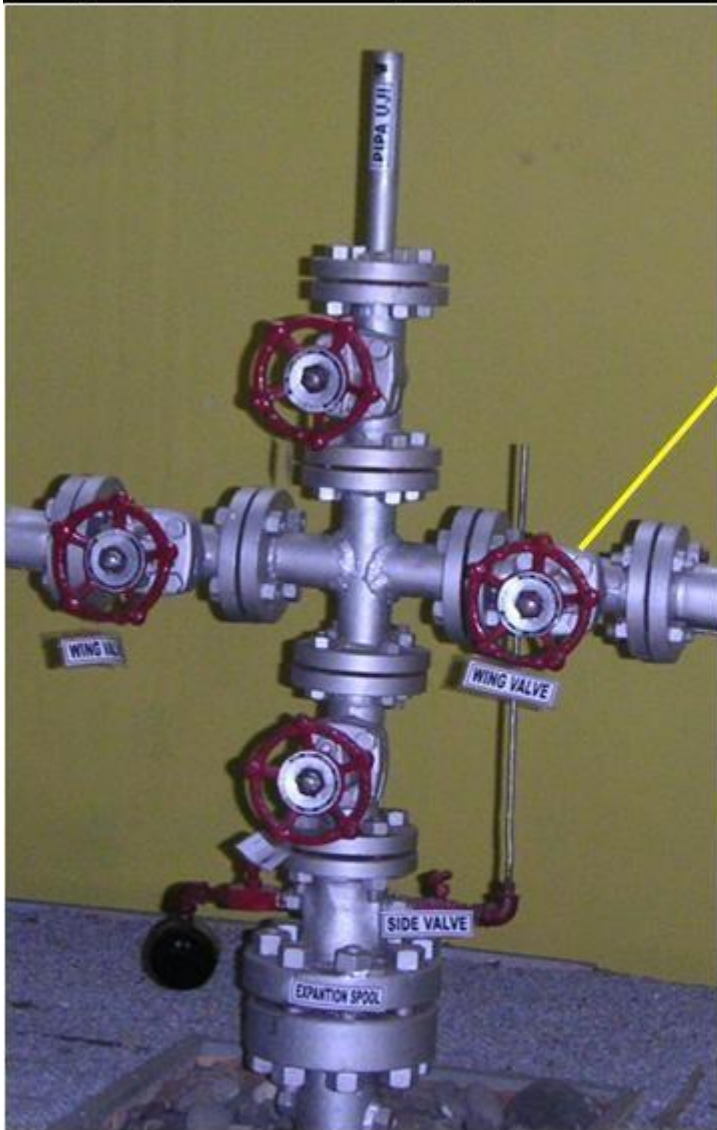
Kamojang



Ke
Rock
Muffler

Pipa ali uap ke PLTP

Lapangan Kamojang



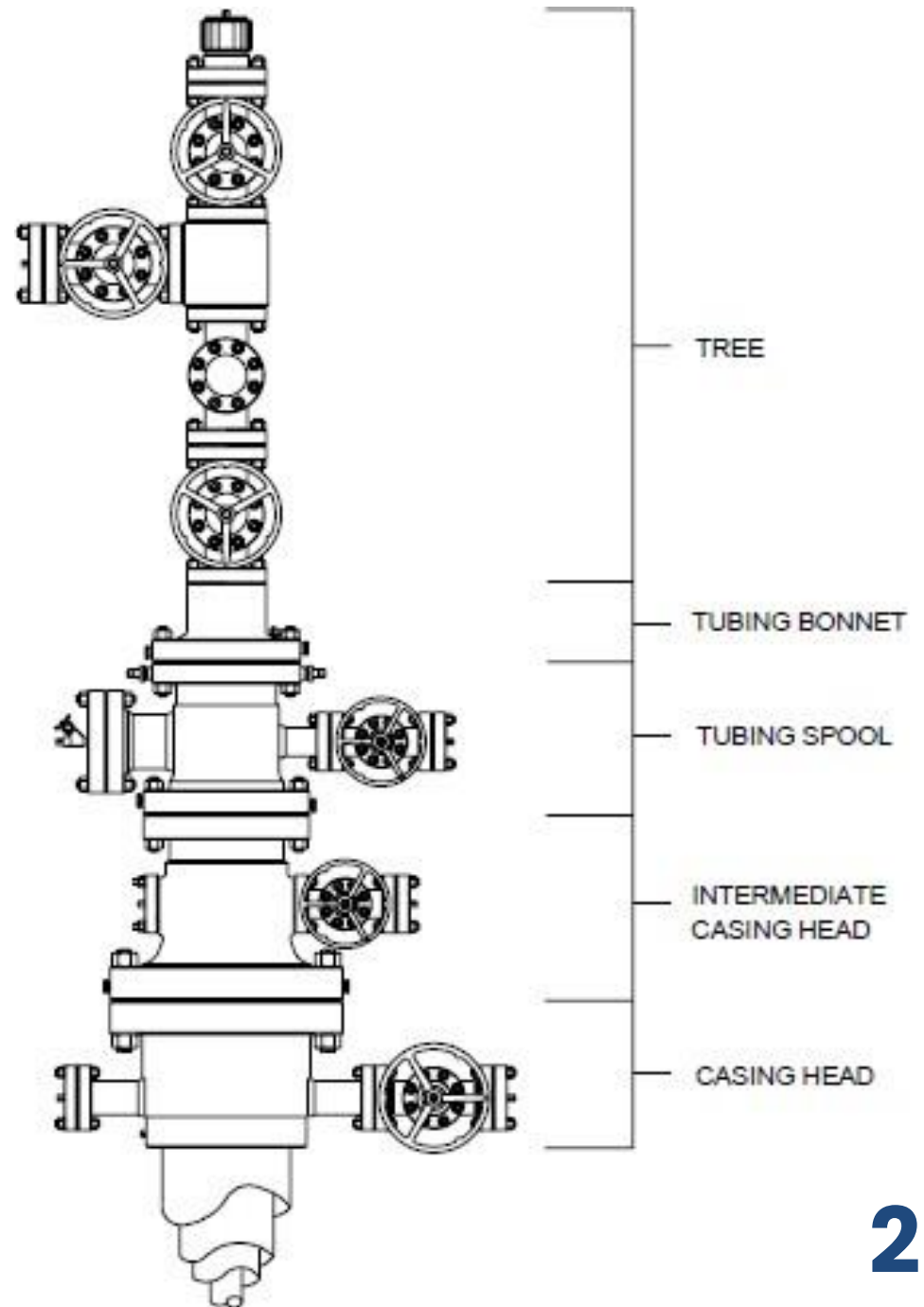
Wing Valve: mengatur aliran uap ke rock muffler(saat uji sumur)





Rock Muffler

Wellhead part



Casing



Introduction: Casing

Kegunaan masing-masing jenis casing

❑ Conductor casing:

- Memberikan arah pemboran pertama kali
- Tempat kedudukan bagi seksi casing berikutnya

❑ Surface casing:

- Tempat duduknya BOP
- Menjaga kestabilan lubang sumur
- Melindungi air tanah dari pencemaran oleh fluida pemboran

❑ Intermediate casing:

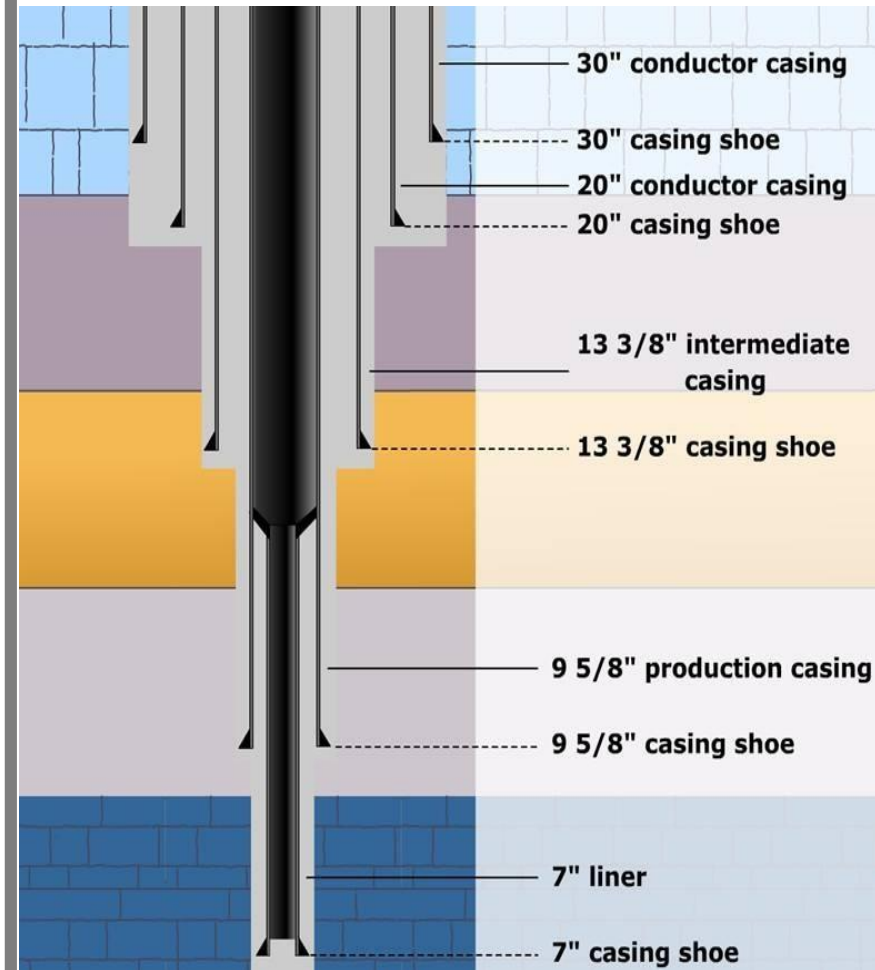
- Isolasi zona abnormal
- Isolasi zona lost circulation

❑ Production casing:

- Memisahkan zona produksi satu dengan zona produksi lainnya
- Menjaga kondisi lubang sumur yang dapat ditolerir subsurface completion equipment

❑ Liners:

- *sama dengan production casing





Well Drilling Types

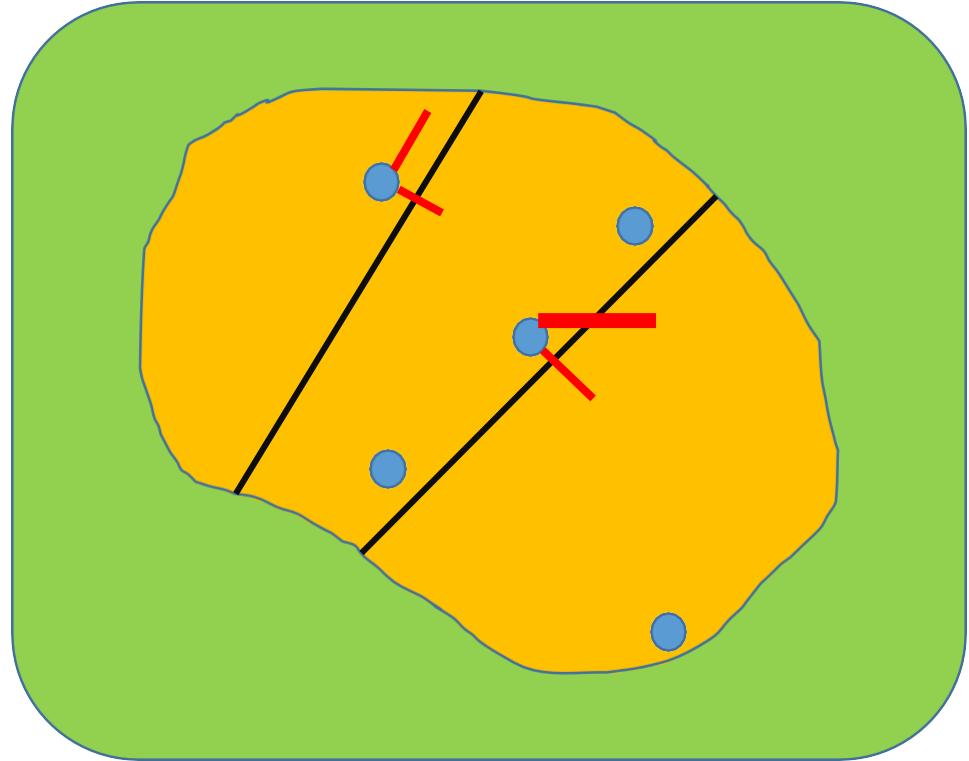
Berdasarkan
Tujuan

Berdasarkan
ukuran lubang
sumur



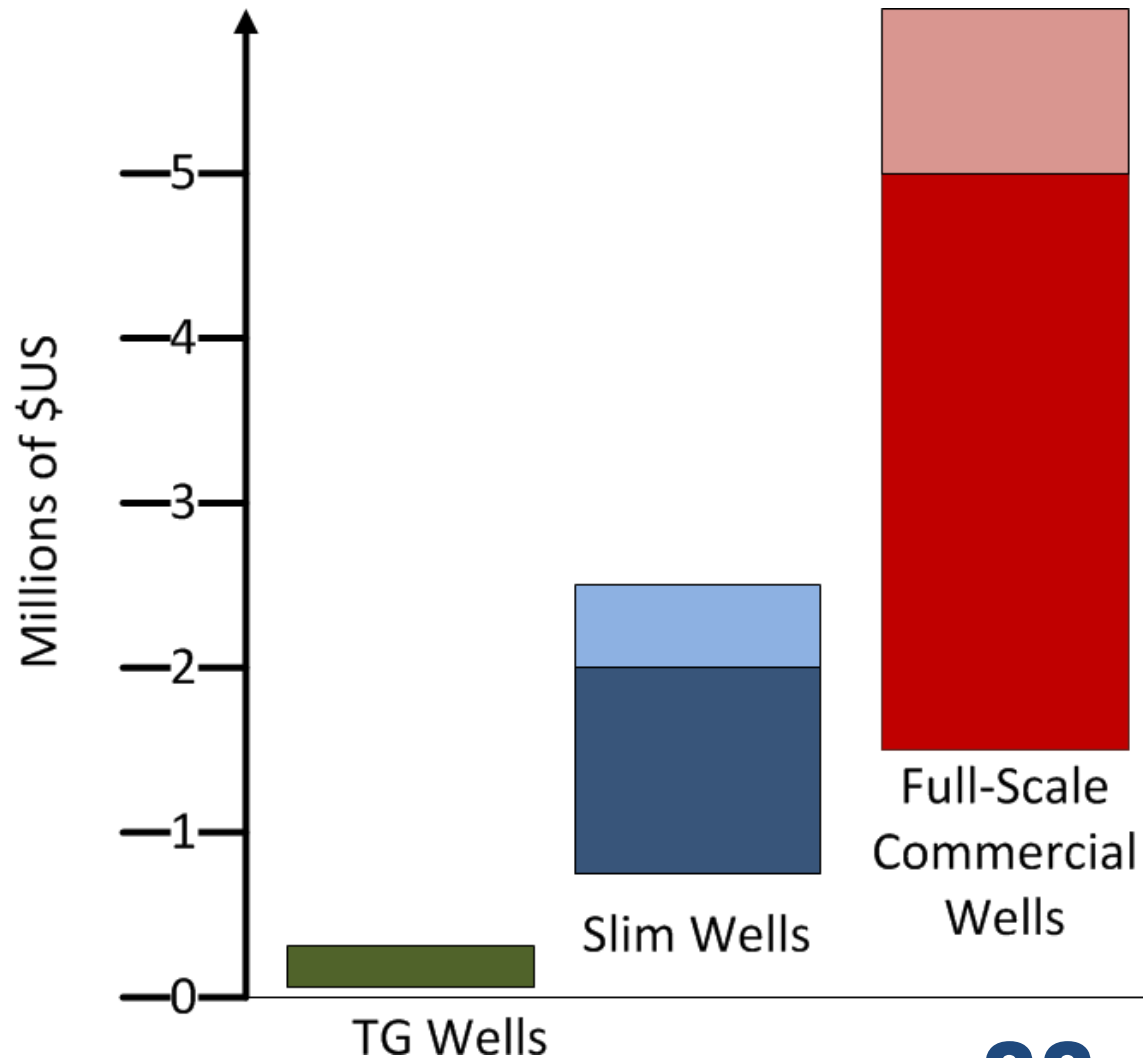
Berdasarkan Tujuan

- Exploration
- Development
- Production and Injection
- Make-up
- Workover



Berdasarkan Ukuran Lubang Sumur

- Temperature Gradient/Delienasi (TG)
- Slim Hole
- Standard Hole
- Big Hole



“Temperature Gradient” Well



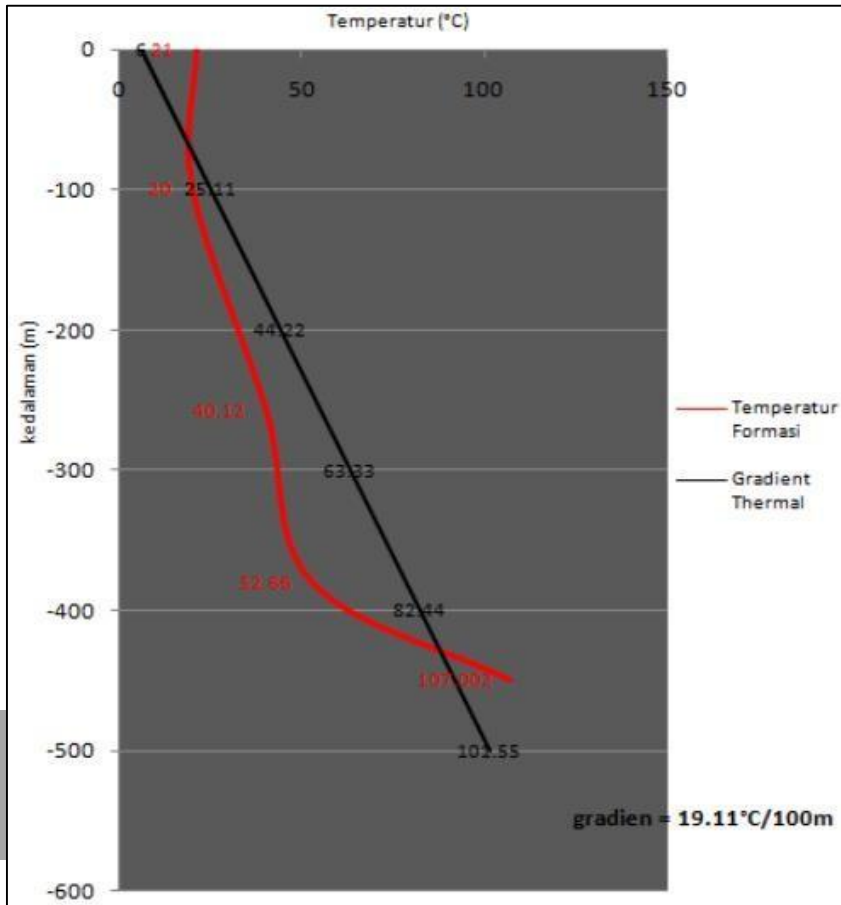
- Sumur awal
- Untuk mengetahui gradien temperatur
- Tidak untuk diproduksi

Karakteristik Sumur Landaian Suhu

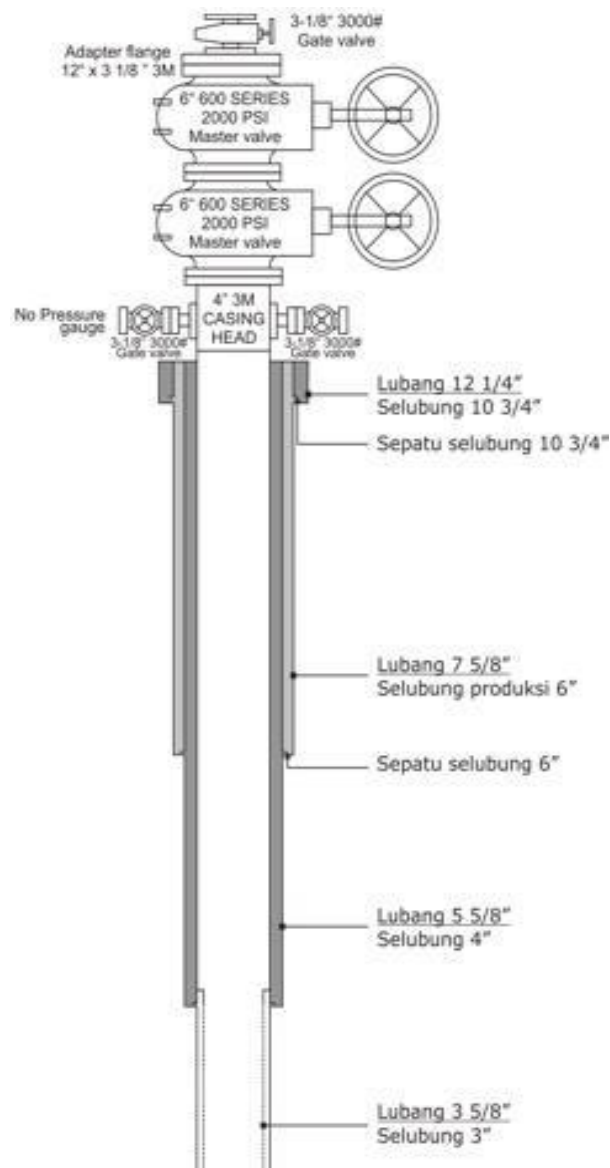
- Dangkal (< 500 m)
- Diameter kecil
- Truk Rig
- Dibor tanpa BOP



- Data geologi dangkal
- Data temperatur
 - Data jangka pendek
 - Data jangka panjang
 - Monitoring



Data obtained
36

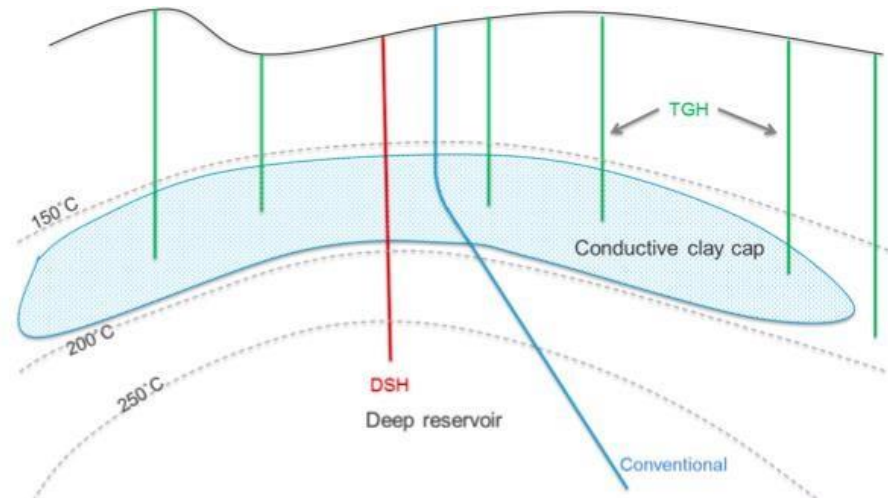


- Diameter lebih besar dari sumur TG
- TD (Total kedalaman) hingga reservoir
- Dibor dengan menggunakan BOP kelas ringan

“Slim Hole” Well Drilling

“Slim Hole” Well (cont.)

- Dibor pada tahap eksplorasi lanjut
- Tujuan pemboran untuk memastikan cadangan (dapat juga dijadikan sumur produksi?)
- Data yang diperoleh
 - - Kimia reservoir
 - - Data struktur
 - - Litologi
 - - Gradien temperature



Keunggulan Sumur Slim Hole

- Lebih murah → 30 – 50 % dari sumur standard
- Well pad kecil dan dapat dilakukan di remote area
- Dapat mendapatkan data yang sama dengan data pada sumur standard
- Dapat dijadikan sumur produksi/ injeksi atau sumur monitoring

Item	Slimhole Drilling	Conventional Drilling
Rig size for at least 1,000 m deep well	Small truck mounted	Large (more horsepower)
Truckloads for well	6 to 10 15%	40 to 80
Production hole diameter	2.98" (NQ), or 3.78" (HQ)	8-1/2" (standard), or 12-1/4" (large)
Approximate time to complete 1,000 m	40 days	25 days
Approximate rig pad size	40 m x 40 m 20%	100 m x 70 m
Storage area	40 m x 40 m 4%	200 m x 200 m
Access road	Single lane track or helicopter	2 lane with no steep grades
Highest structure	Mast up to 15 m	Mast up to 45 m
Waste sump volume	250 cu m approximately	1,000 cu m approximately
Noise	Not a significant issue	Significant in close proximity to site
Peak water supply requirements	60 cu m per day for coring HQ holes 2%	3,500 cu m per day
Approx. Cost/meter	1,000 USD 25%	4,000 USD

Kekurangan Sumur Slim Hole



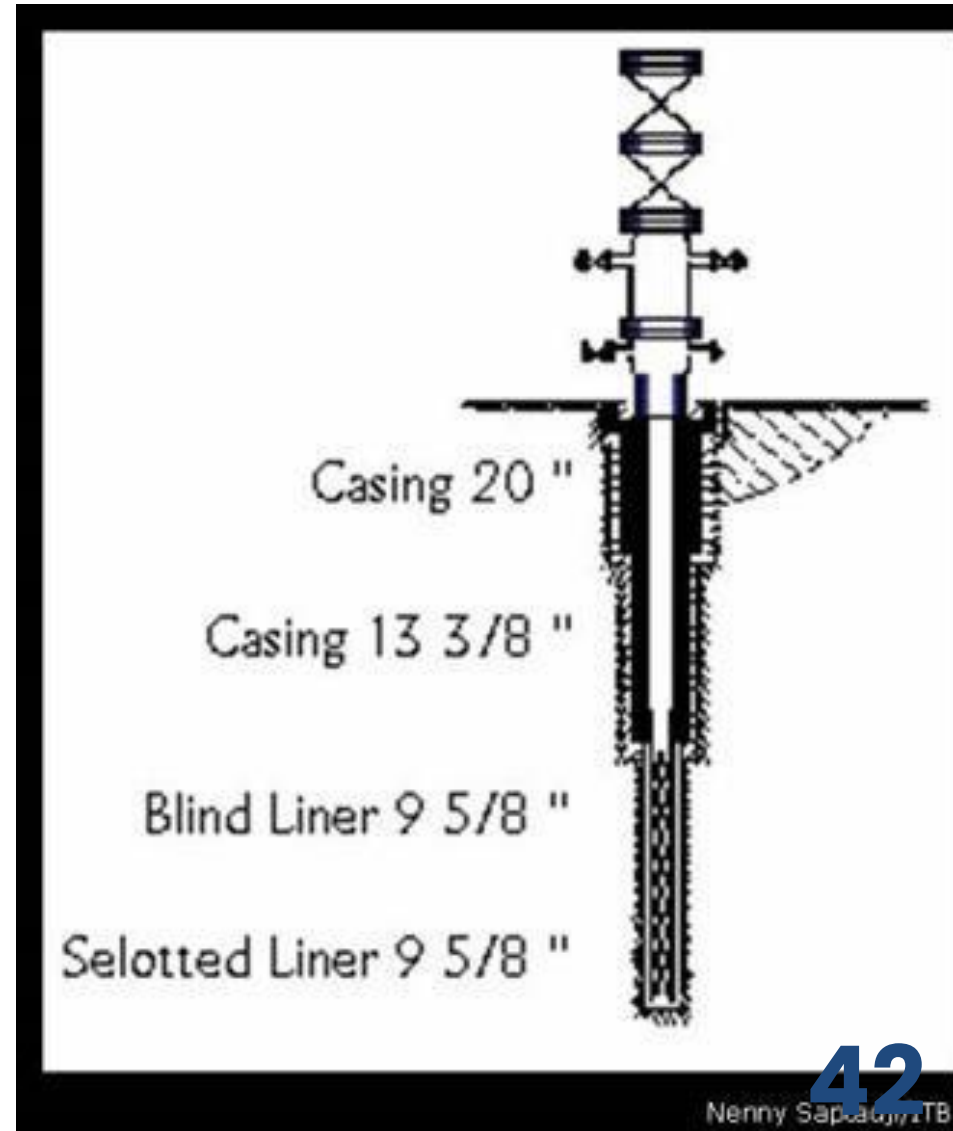
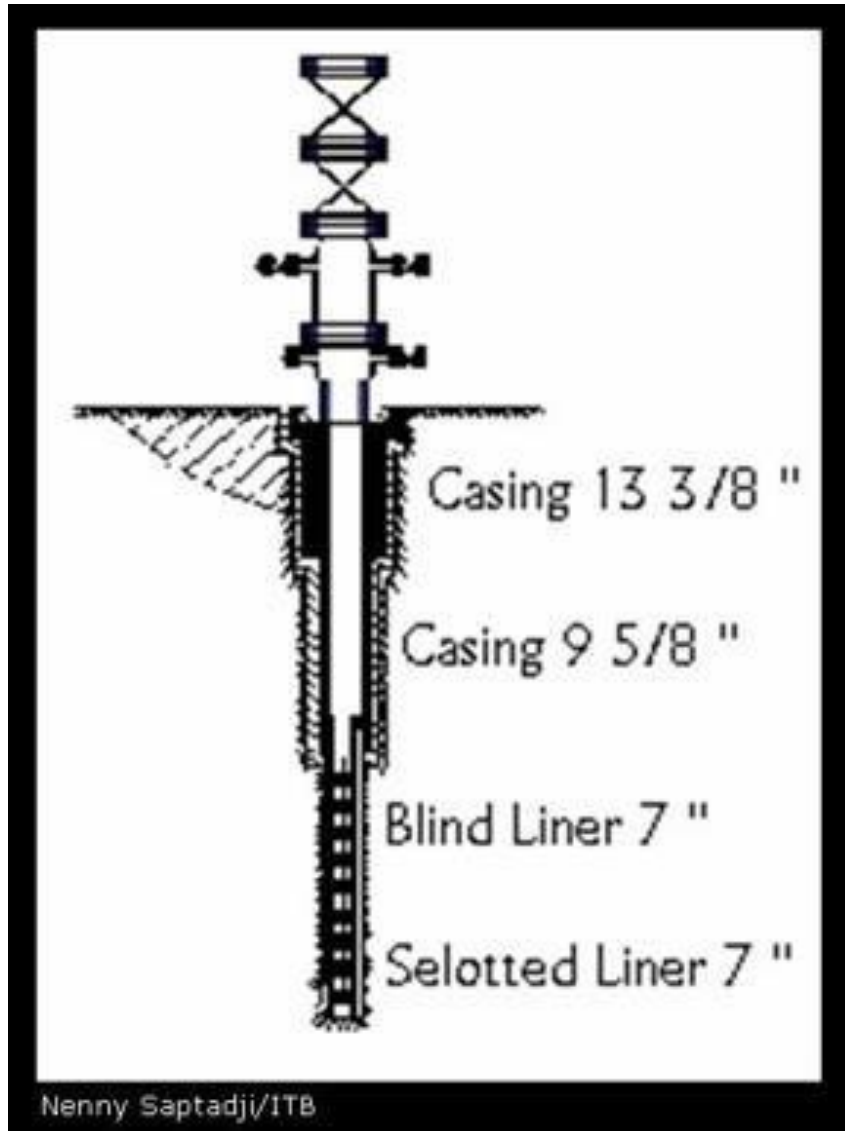
- Resiko pemboran lebih besar (aspek rig, aspek kestabilan sumur, dll)
- Produktivitas atau injektivitas rendah *
- Resiko produksi lebih besar (scaling, korosi, dll) *

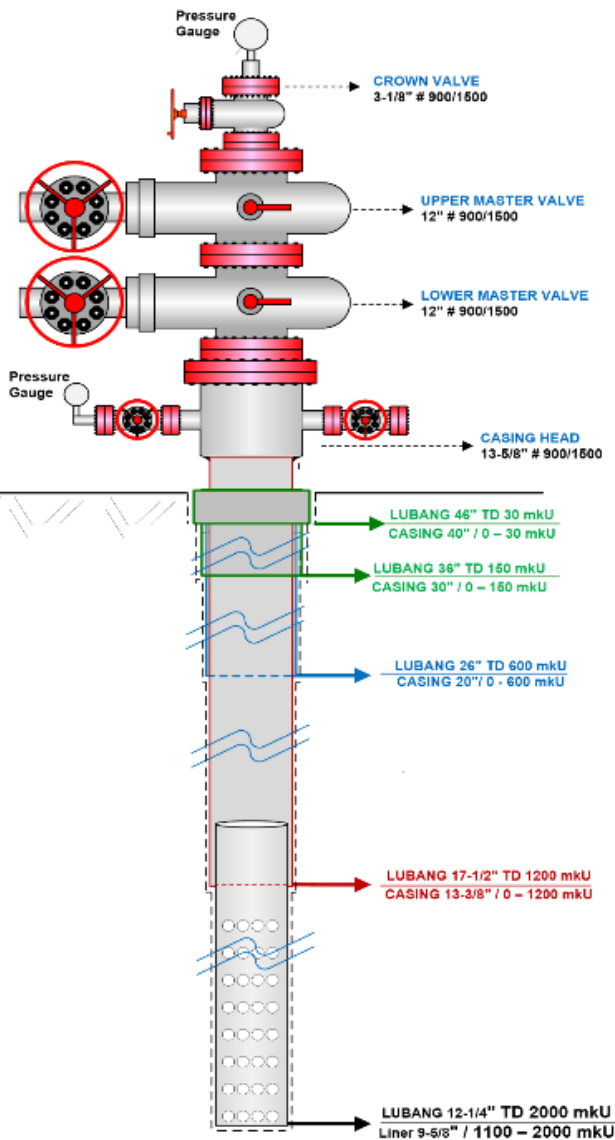
- ✓ Untuk Produksi atau Injeksi
- ✓ Didesain kuat dan tahan lama



Standard and Big Hole Well Driling₄₁

Standard vs Big Hole





- Lantai Rig tinggi untuk memberi ruang untuk BOP
- Kapasitas Pompa Lumpur yang besar
- Kolam lumpur yang besar
- Memiliki HP yang besar

Rig Pemboran Sumur Standard

Perencanaan dan Desain Sumur Panas Bumi



Drilling design process

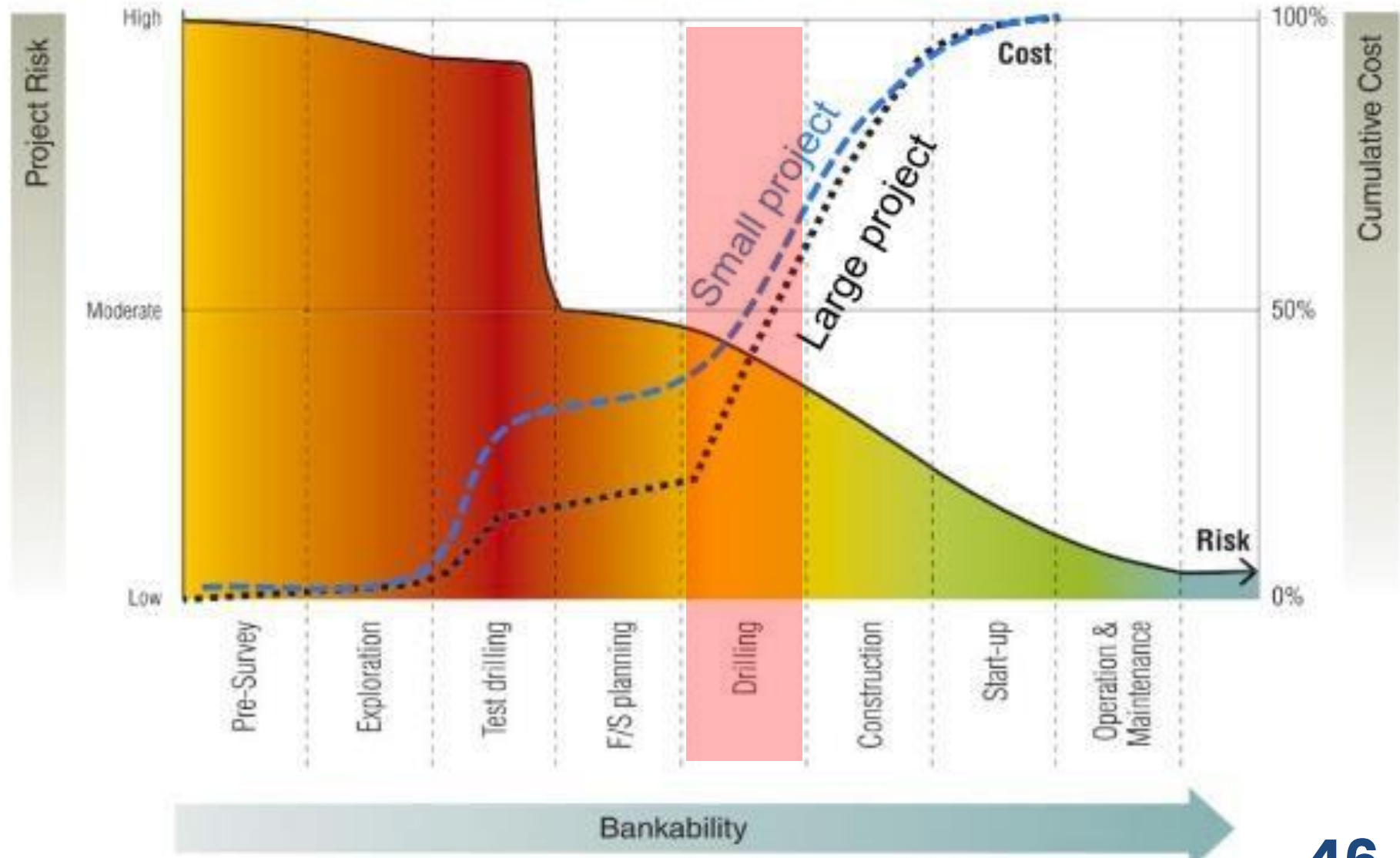
Table 2 – The well design process, step by step

Step	Action
1	Determine an intended vertical well depth, wellhead location, and well target.
2	Collect all available data and assessments on expected subsurface conditions between the wellhead and the well target. This includes but is not restricted to: (a) Temperature versus depth; (b) Pressure versus depth; (c) Geological formations expected; (d) Assessment on competency of geological formation and intervals of lost circulation; and (e) Assessment of any anticipated problem zones.
3	Assess the Maximum Design Pressure over the depth of the well.
4	Assess the Effective Containment Pressure of the formations over the depth of the well.
5	Determine safe casing shoe depths for each casing string back to surface. To do this, use: (a) Maximum Design Pressure; (b) Effective Containment Pressure; (c) Knowledge of formation integrity; and (d) Knowledge of anticipated problem zones.
6	Determine a desired production casing (or production liner) and open hole diameter. Consider the well's final use and desired production/reinjection performance when making this assessment.
7	Determine desired size, weight, grade, and connections of casing strings, and associated hole sizes for each casing string back to the surface.
8	Check each casing string for strength against running and cementing loads. Revise design if necessary.
9	Check each casing string for safe pressure containment of Maximum Design Pressure during well construction. Revise design if necessary.
10	Review each casing string for loadings under anticipated well service conditions, and revise design if necessary.

Development Risk

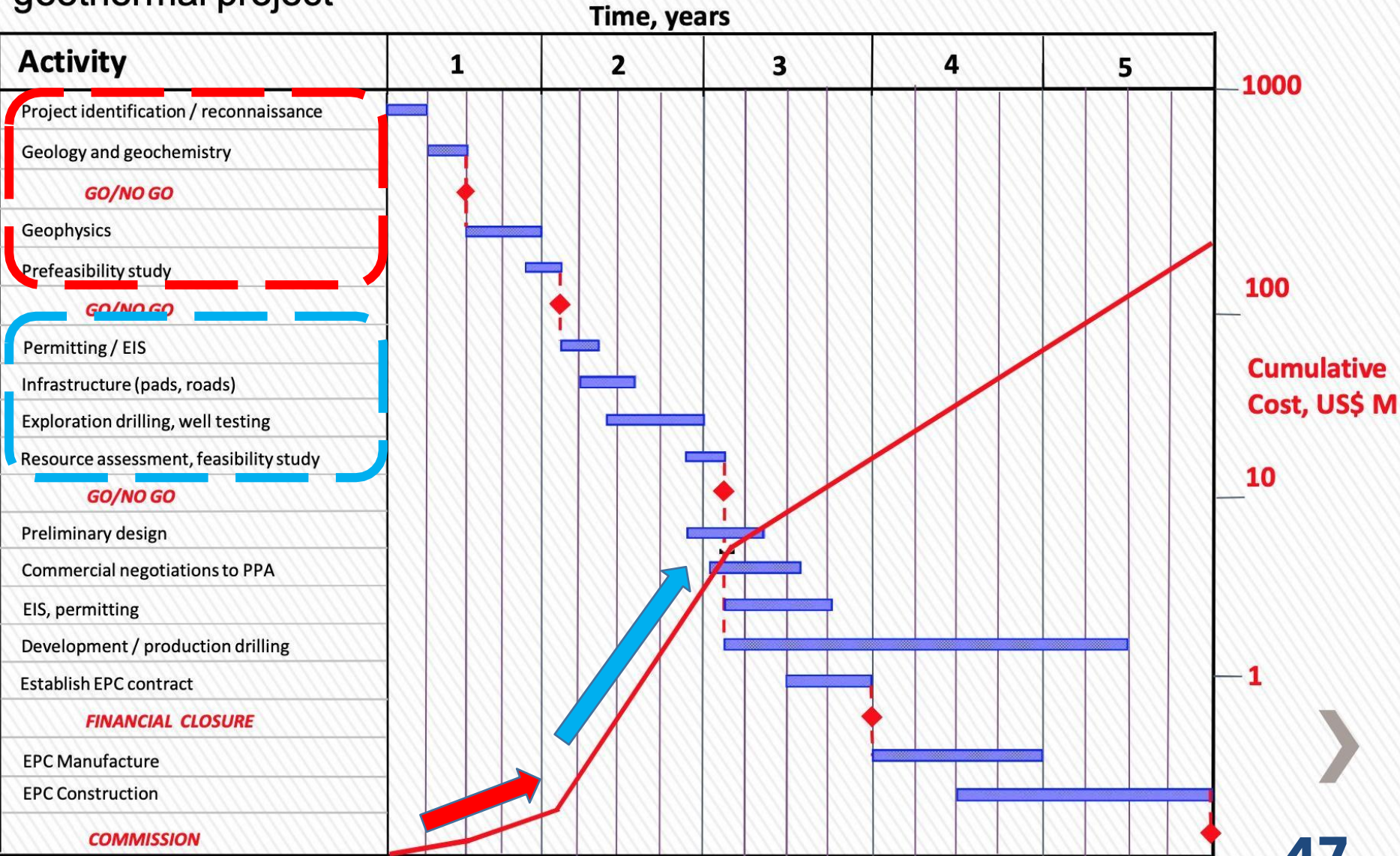
Gehring, M. and
Loksha, V.C. -
**Geothermal Handbook:
Planning and Financing
Power Generation.**
Washington DC: World
Bank Group, ESMAF:
2012.

Geothermal Project Risk and Cumulative Investment Cost



Staged Geothermal Exploration & Development

Nominal timing and capital cost requirements for a 100 MWe “Greenfield” geothermal project



Geothermal Drilling Technology Challenges

HIGH TEMPERATURE

Caused the tools temperature limitation (esp: directional tools), extreme thermal stress, and corrosion rate increases.

GEOLOGY COMPLEXITY

Sticking to the well design is difficult due to the poor predictability of geothermal geology.

LOST CIRCULATION

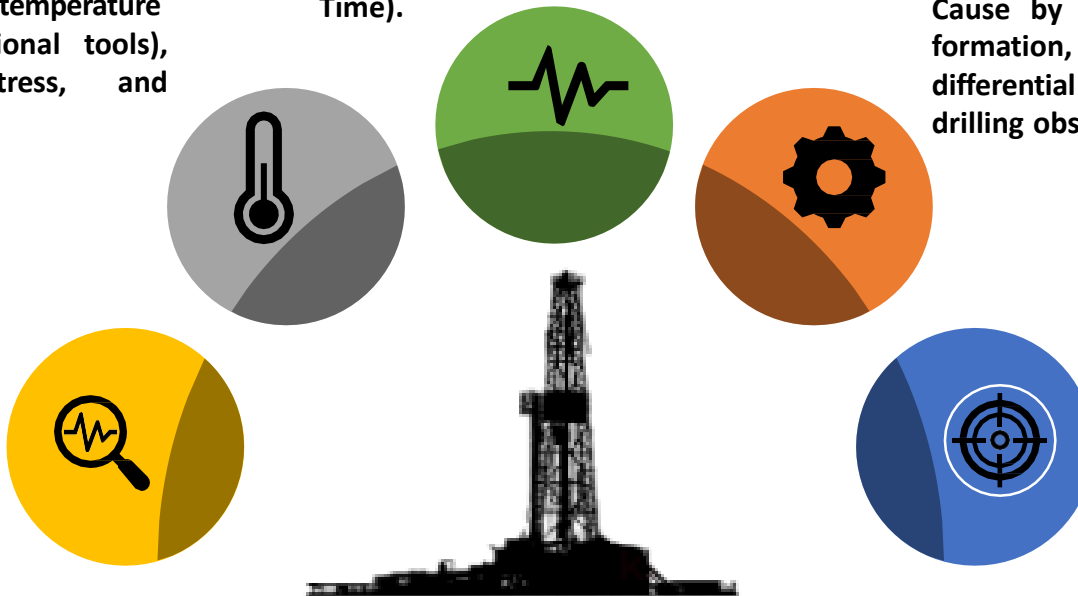
Around 10% of total cost of geothermal project is due to lost circulation mitigation which can caused stuck pipe (Non-Productive Time).

WELLBORE INSTABILITY

Cause by mechanically unstable formation, swelling clay, and differential stress that may lead to drilling obstruction.

CEMENTING AND CASING

Lost circulation during cementing can lead to poor cementing that can cause trapped water, which can affect the casing collapse.



Drilling Criteria



Reach the Target



Low Risk



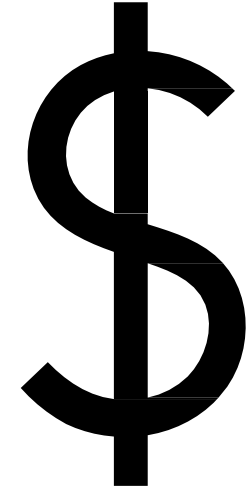
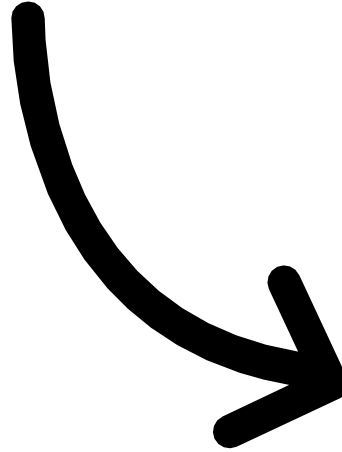
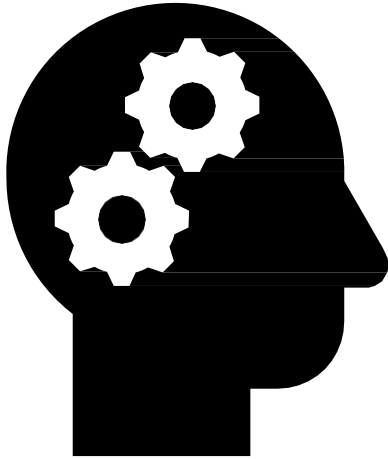
Efficiently Cost



High MW

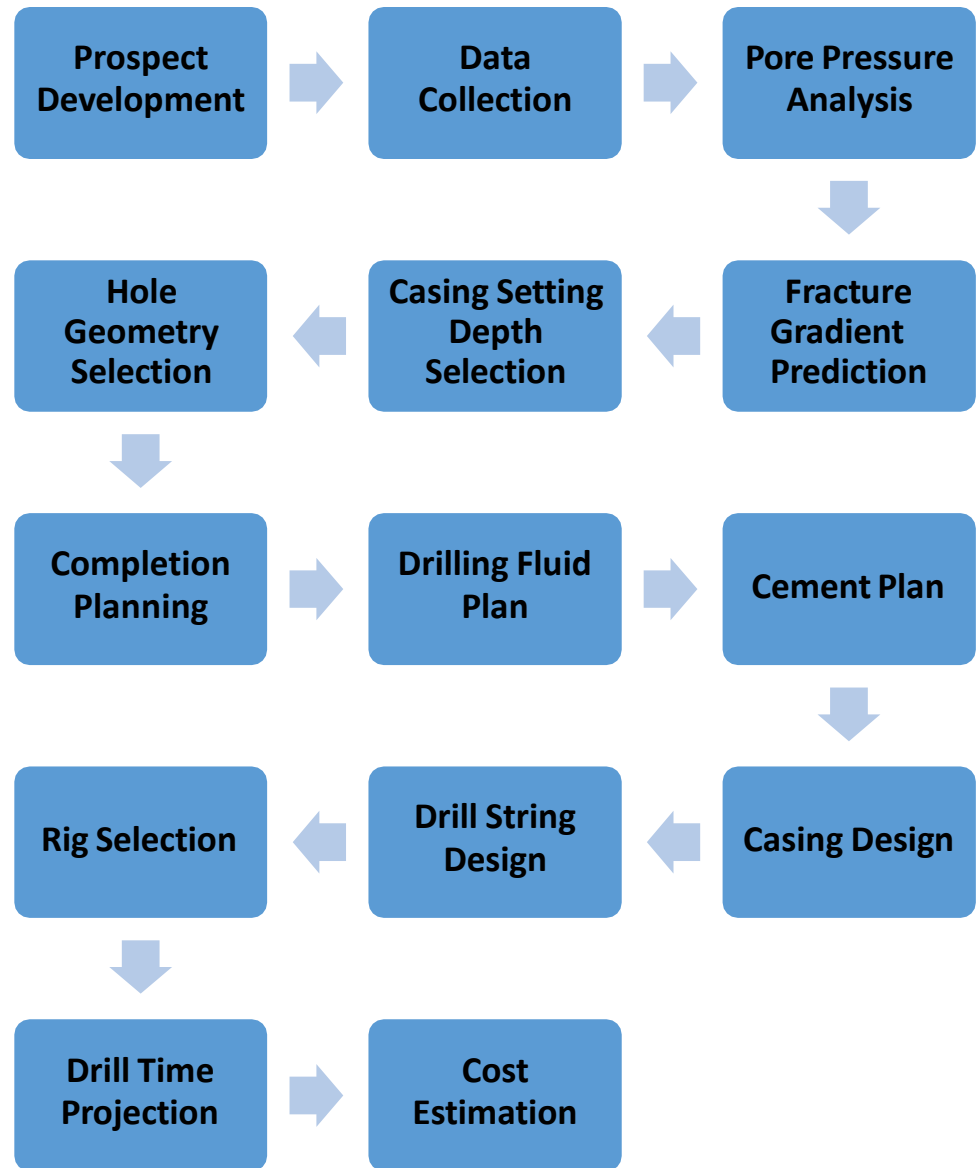


Low Maintenance



Well Planning Effort reduces Well Cost

Well Planning Workflow



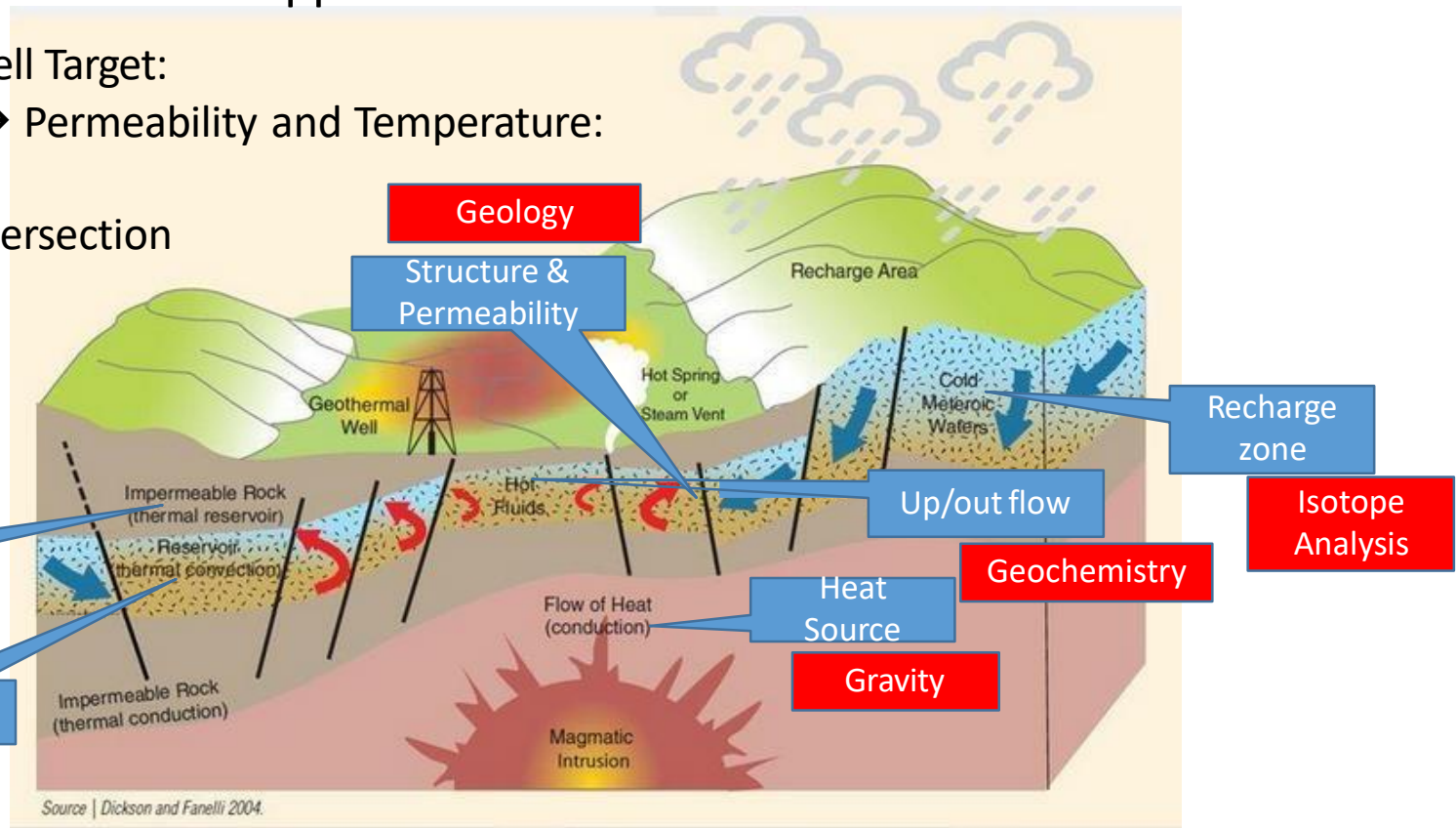
Project Development: Geothermal Well Targeting

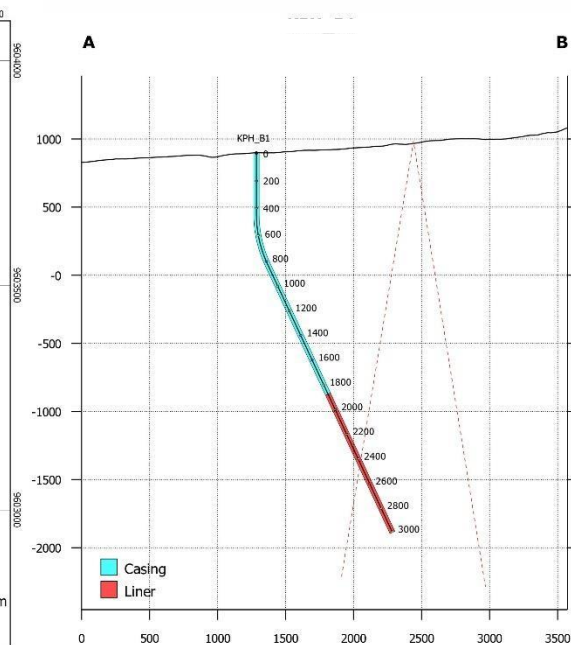
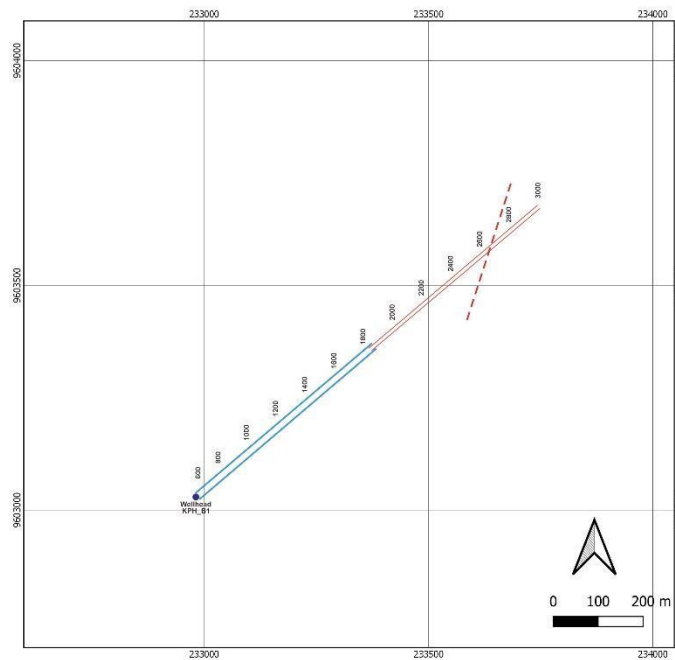
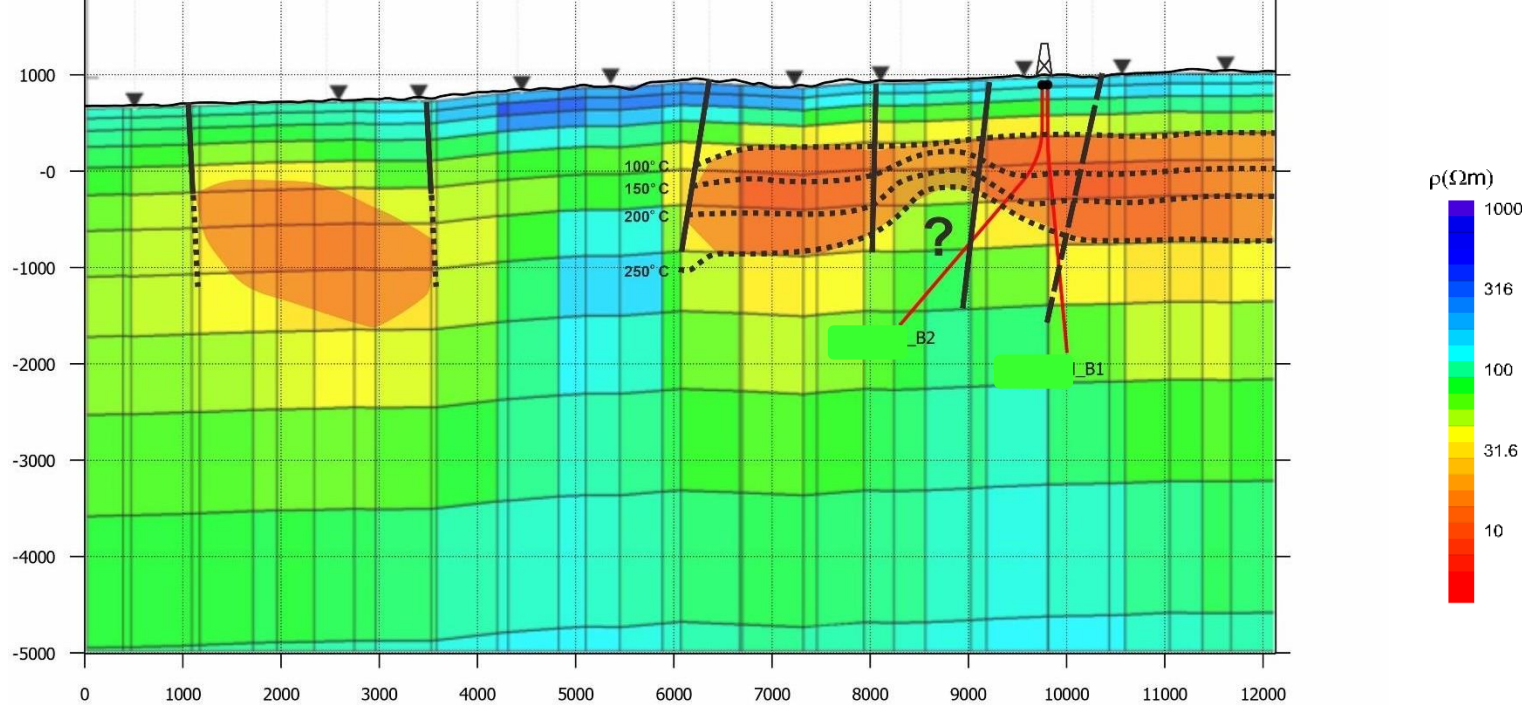
Conceptual Model Approach

Exploration Well Target:

❑ Reservoir → Permeability and Temperature:

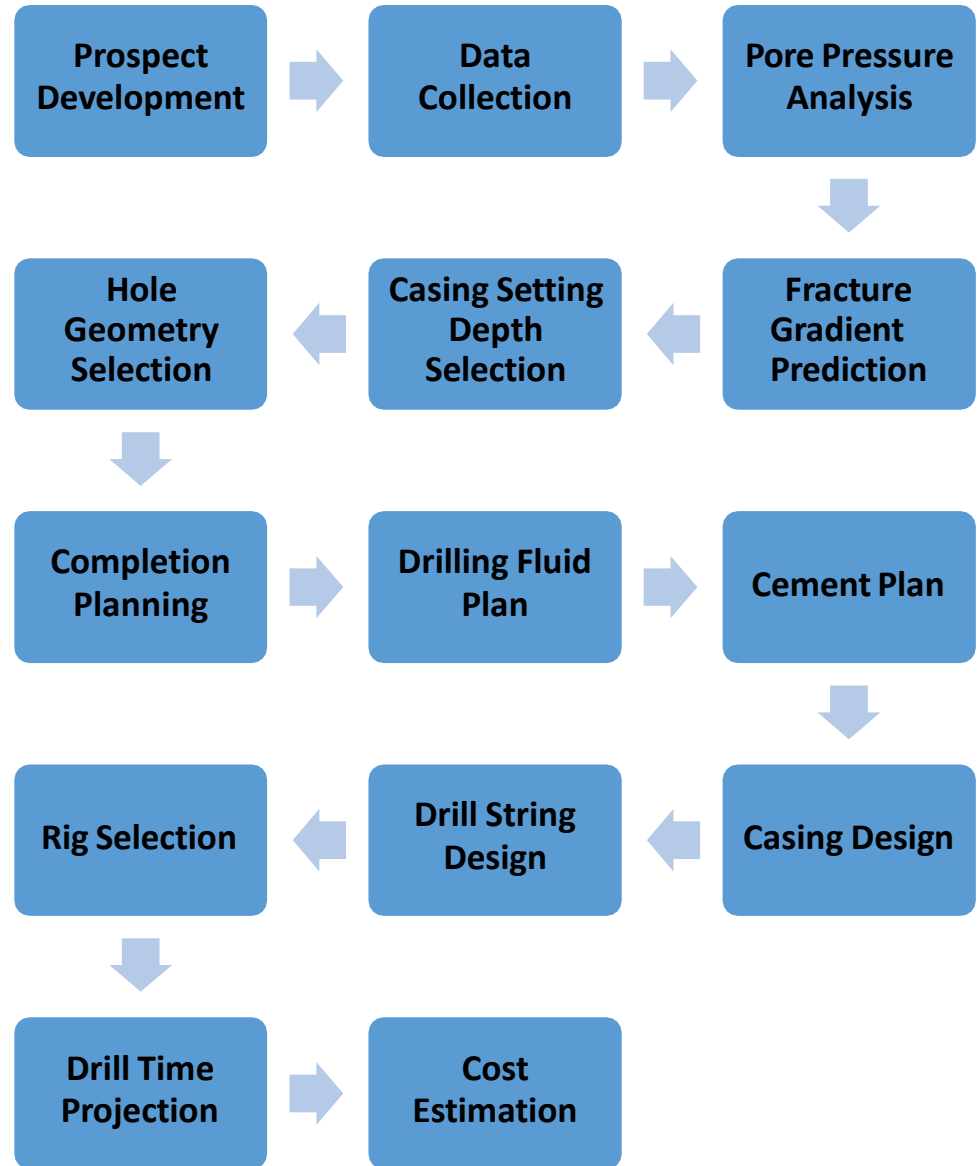
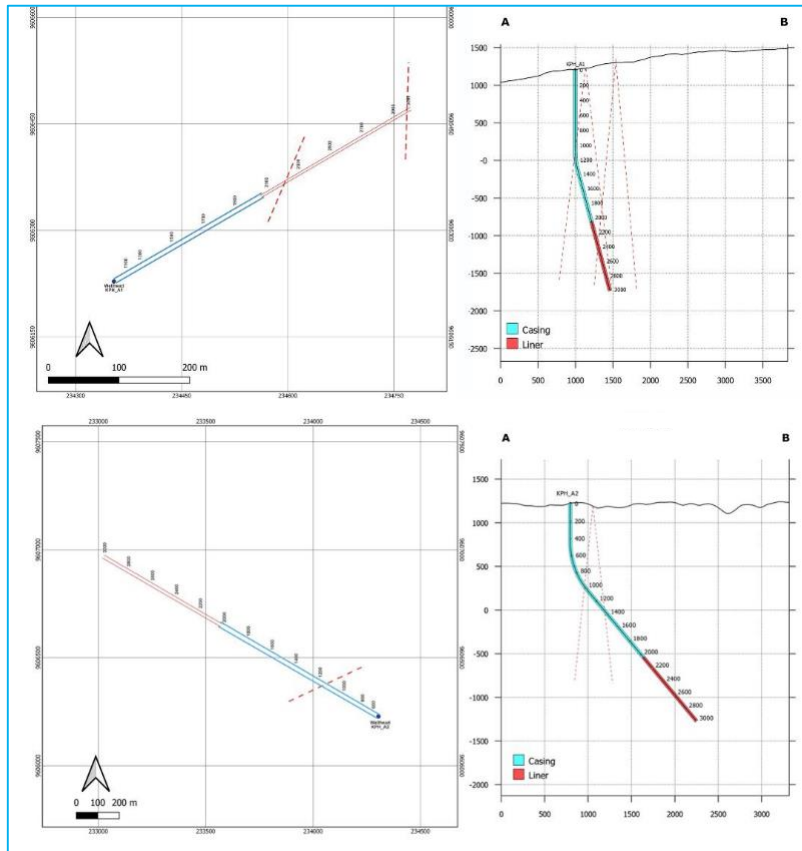
- Upflow
- Fault intersection





Contoh
Well Trajectory

Well Planning Workflow



Rock Characteristics ?



Drilling Engineer



**Expected Drilling
Geology**

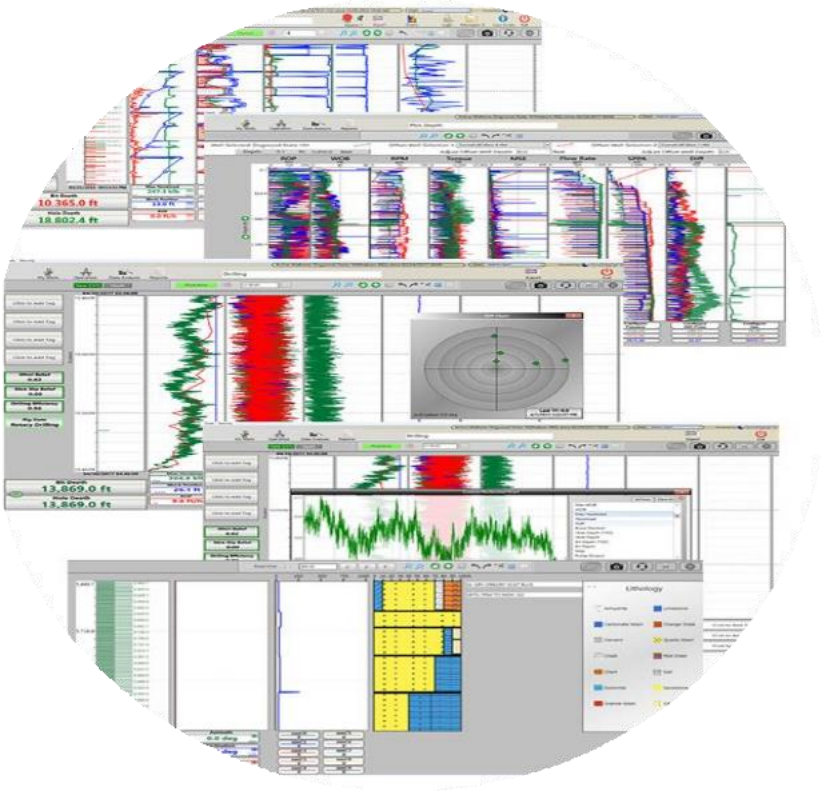
**Define Geological
Structure**

**Identify Geological
Anomaly**

**Determine
Formation Pressure
& Fracture
Gradient**

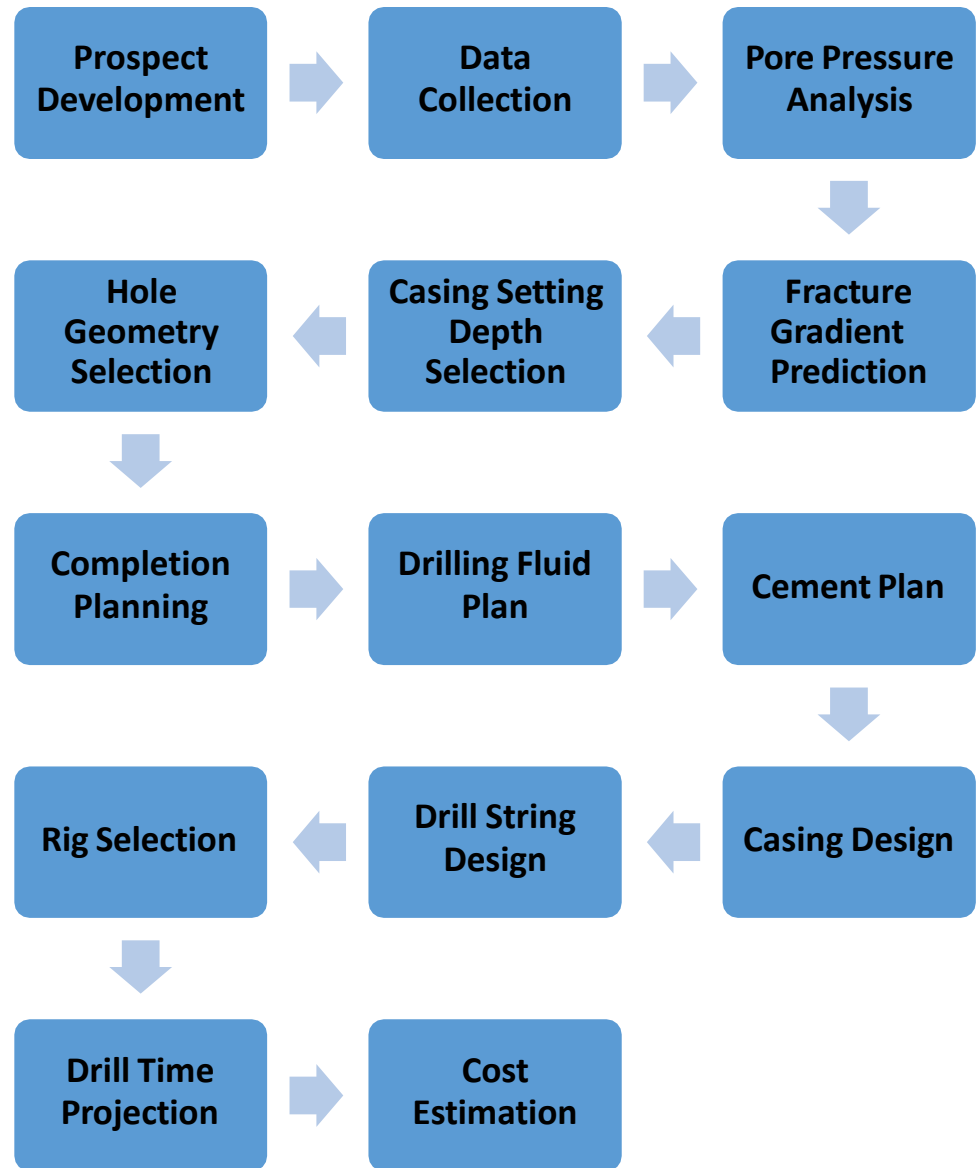


Common Data Types

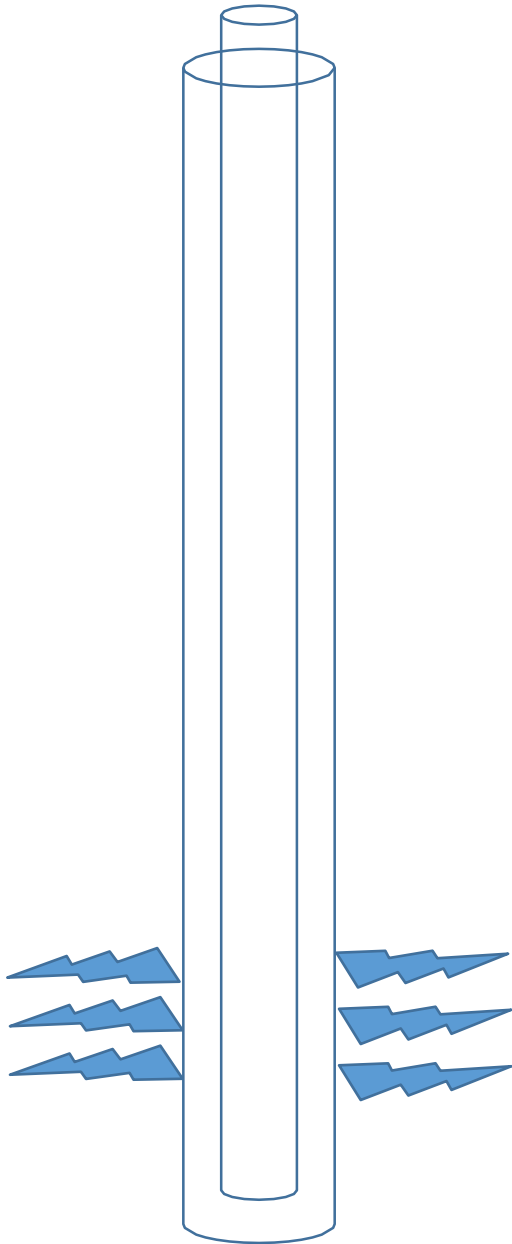


- Bit records
- Mud records
- Mud logging records
- IADC drilling reports
- Scout tickets
- Log header
- Production history
- Seismic studies
- Well surveys
- Geological contour
- Databases or service company files

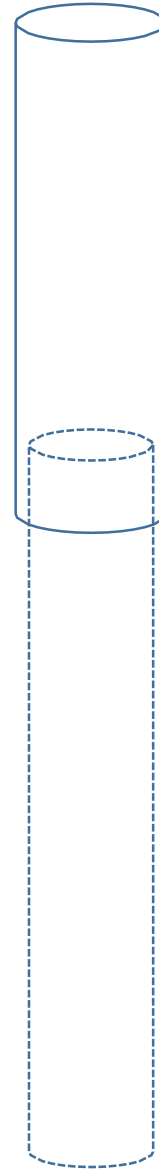
Well Planning Workflow



Oil vs Geothermal Casing Setting

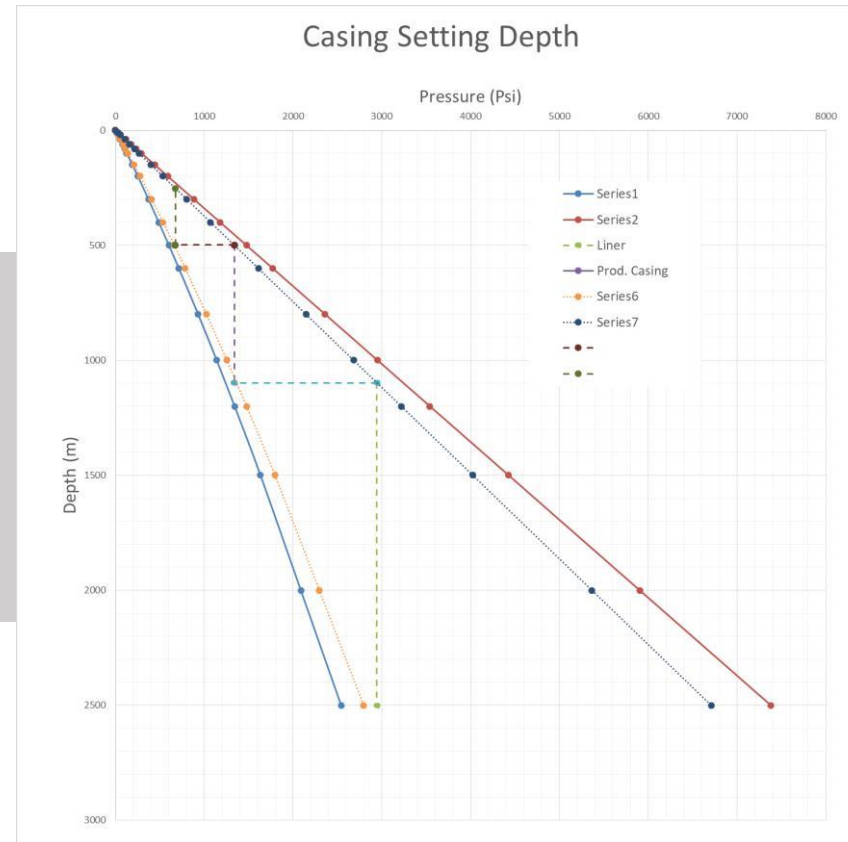


Oil Well

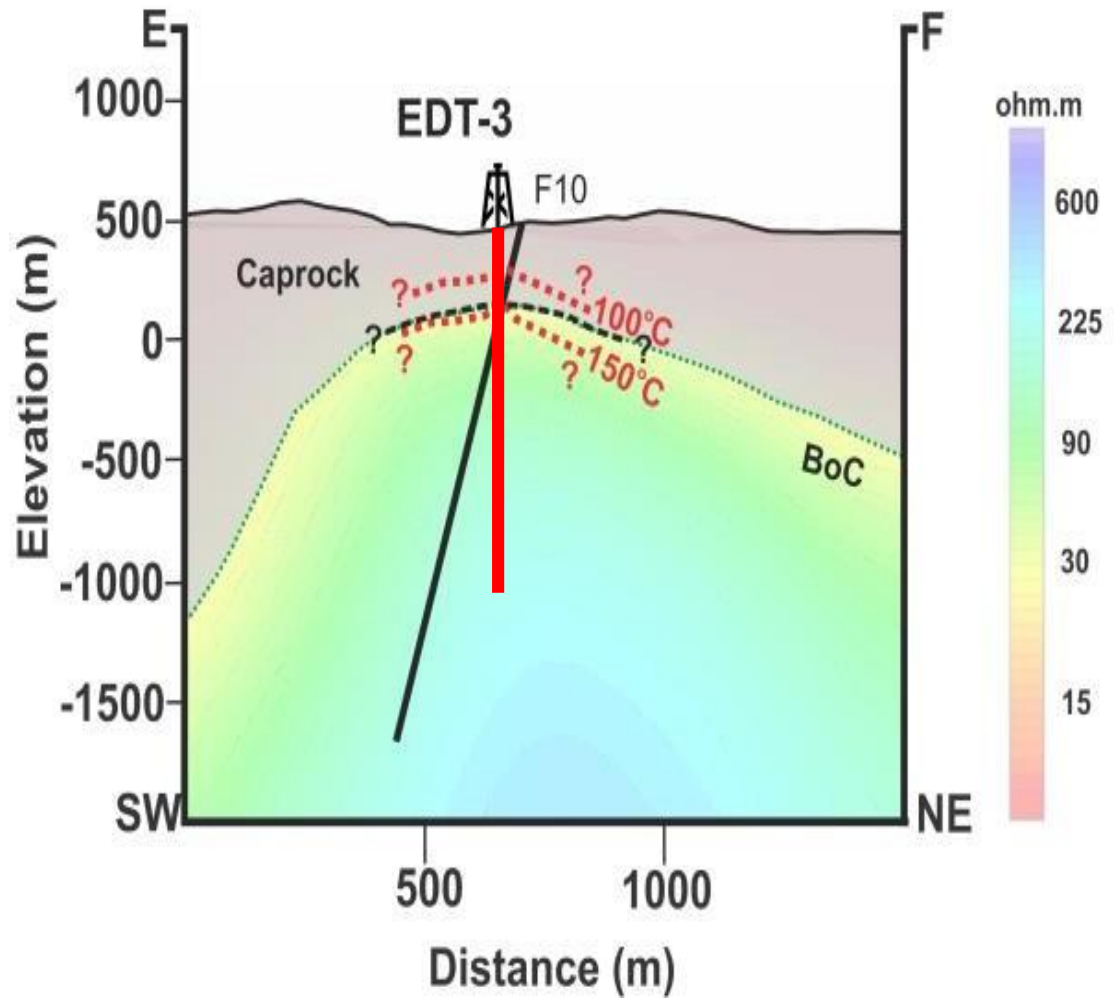


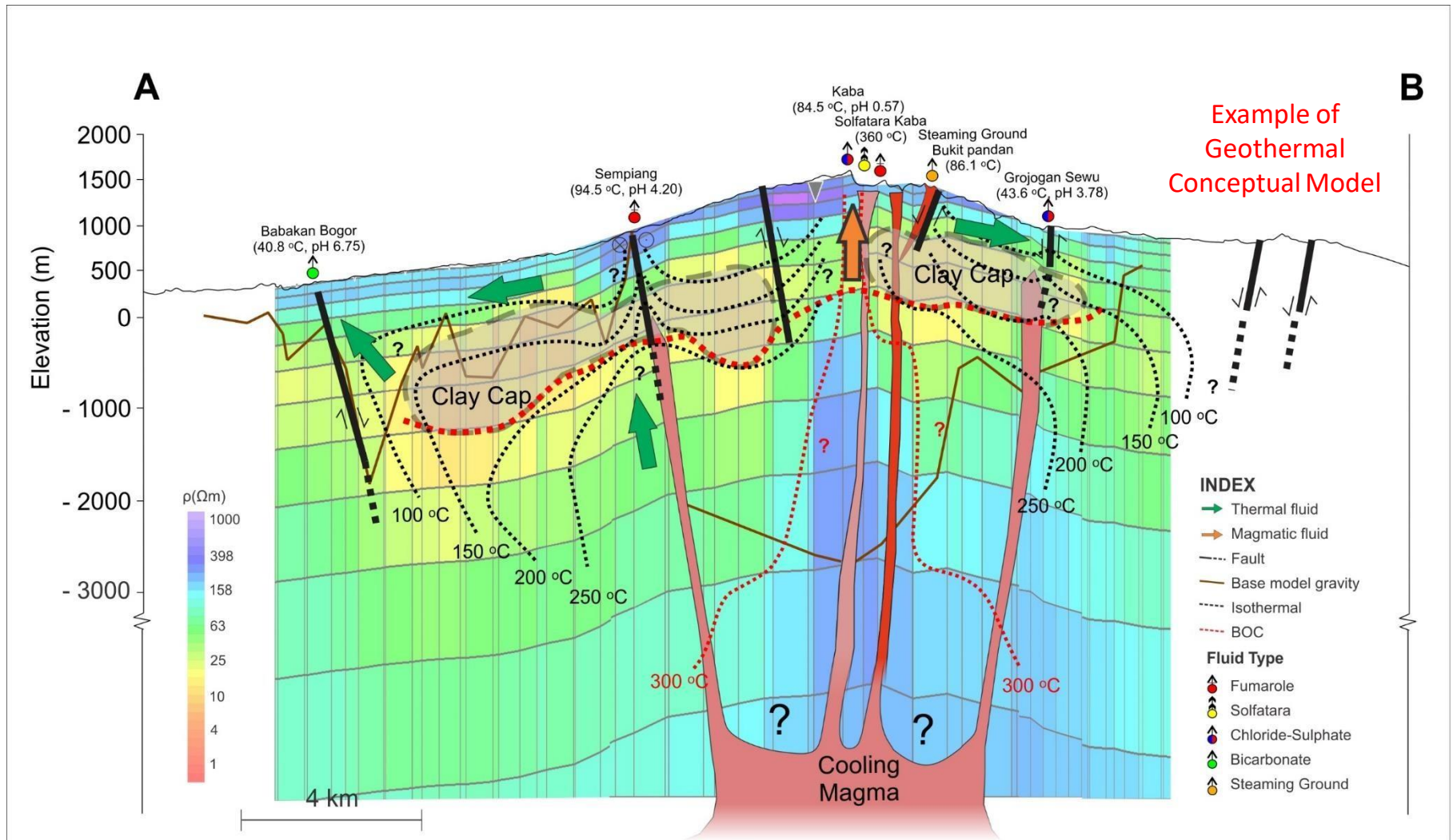
Gas* and Geothermal Well

Casing Setting Depth



WELL DEPTH DETERMINATION





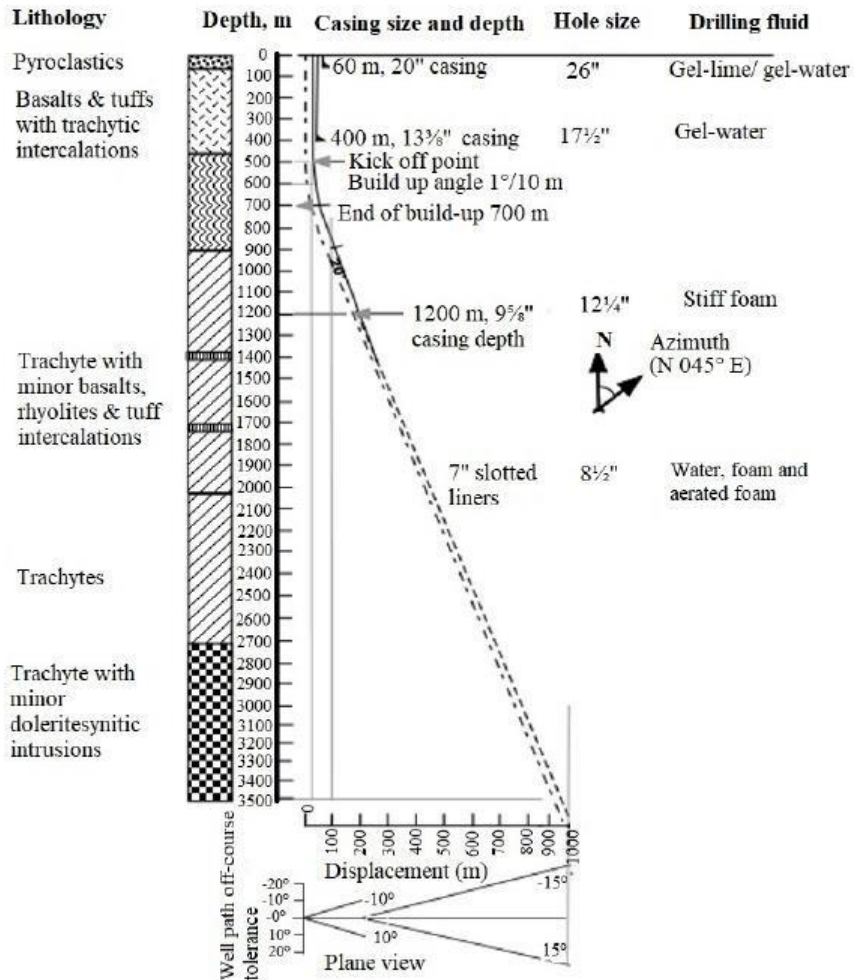


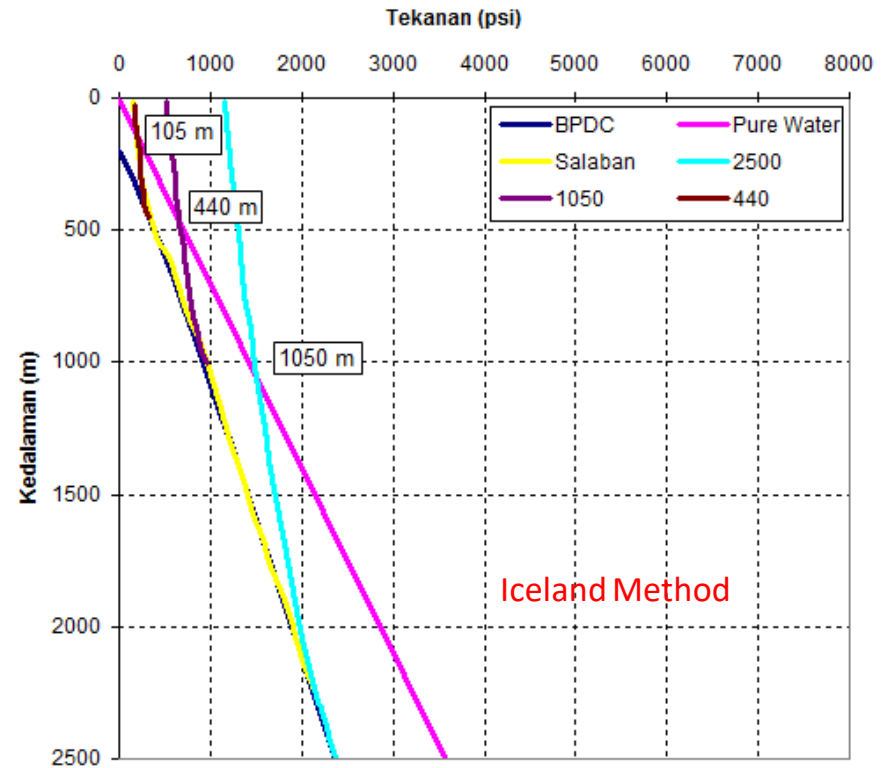
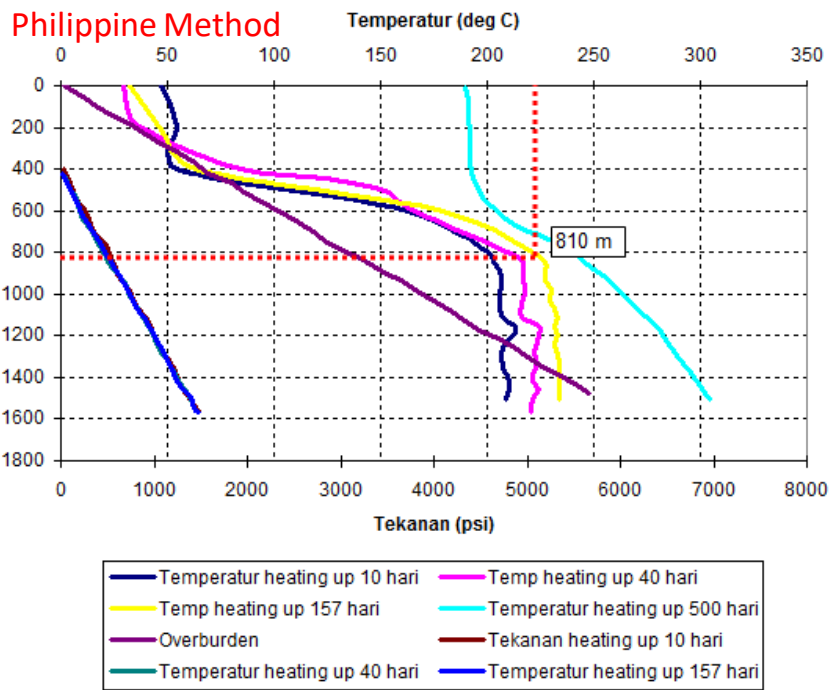
FIGURE 1: Well OW-736A design and geological cross-section (KenGen, 2014)

Casing Setting depth consideration

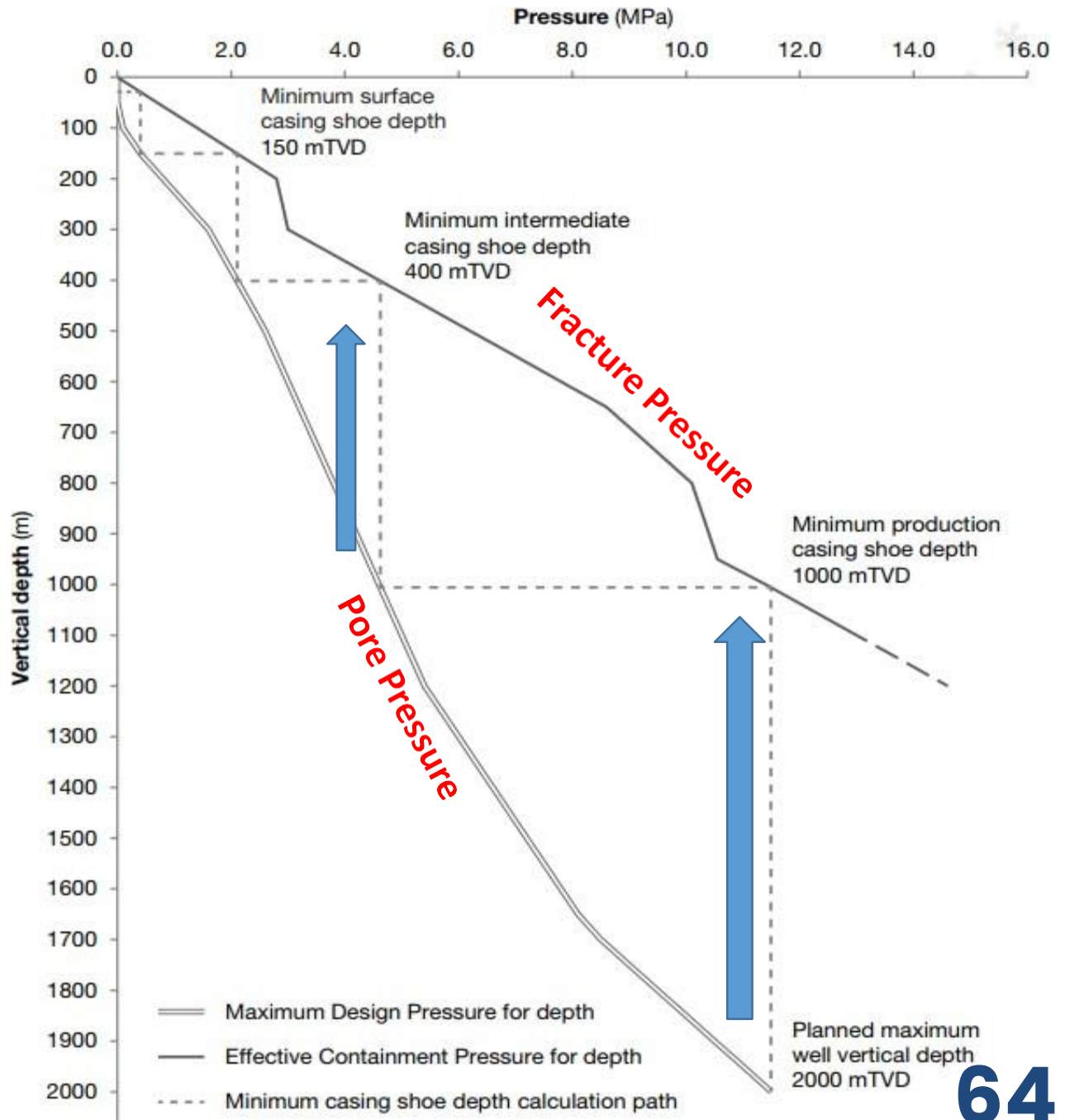
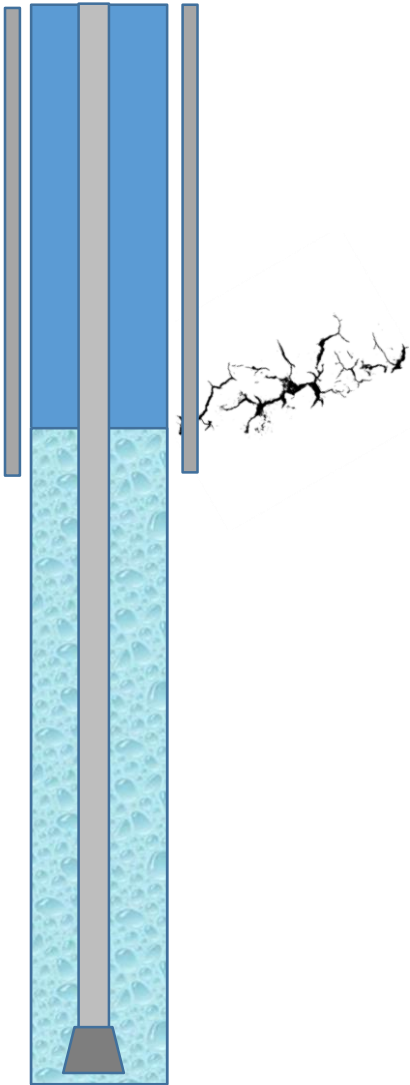
- Competent formation
- Case out problem zone
- Provide prevented Zone isolation
- Provide additional corrosion protection
- To prevent interference

Casing setting depth method

- Philippine Method
- Iceland Method
- New Zealand Method

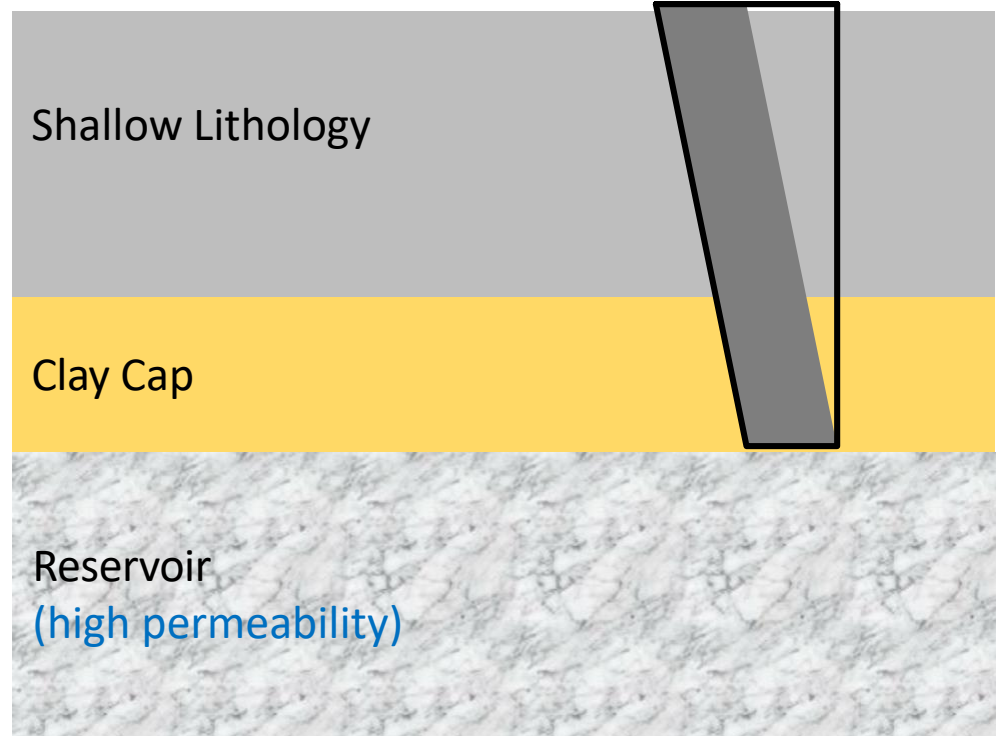


Casing Setting Depth



Production Casing Setting Depth

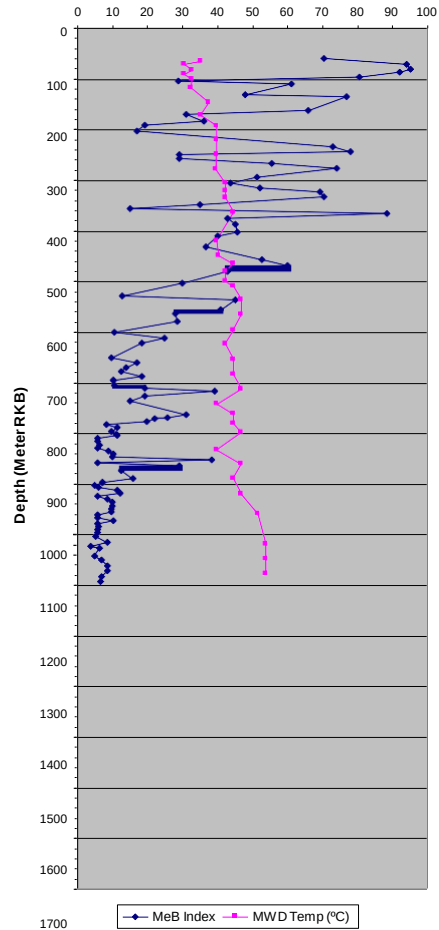
Common lithology layer



XXX-1

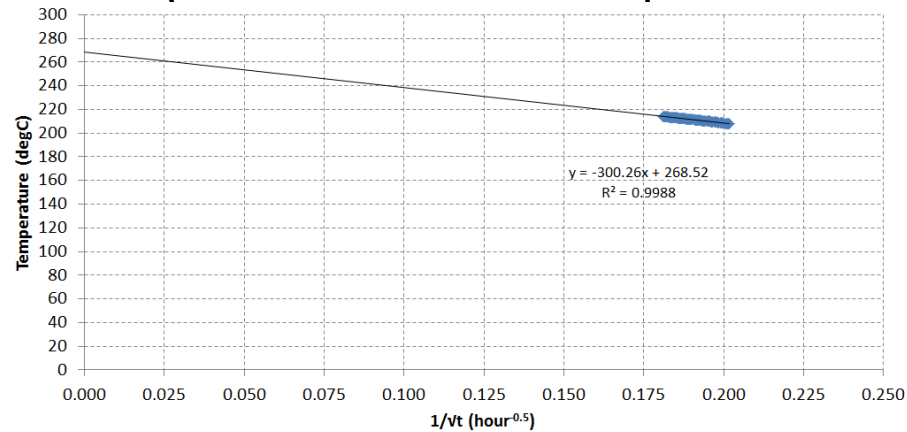
MeB Index Variation with Depth VS Temp. MWD

MeB & °C Index

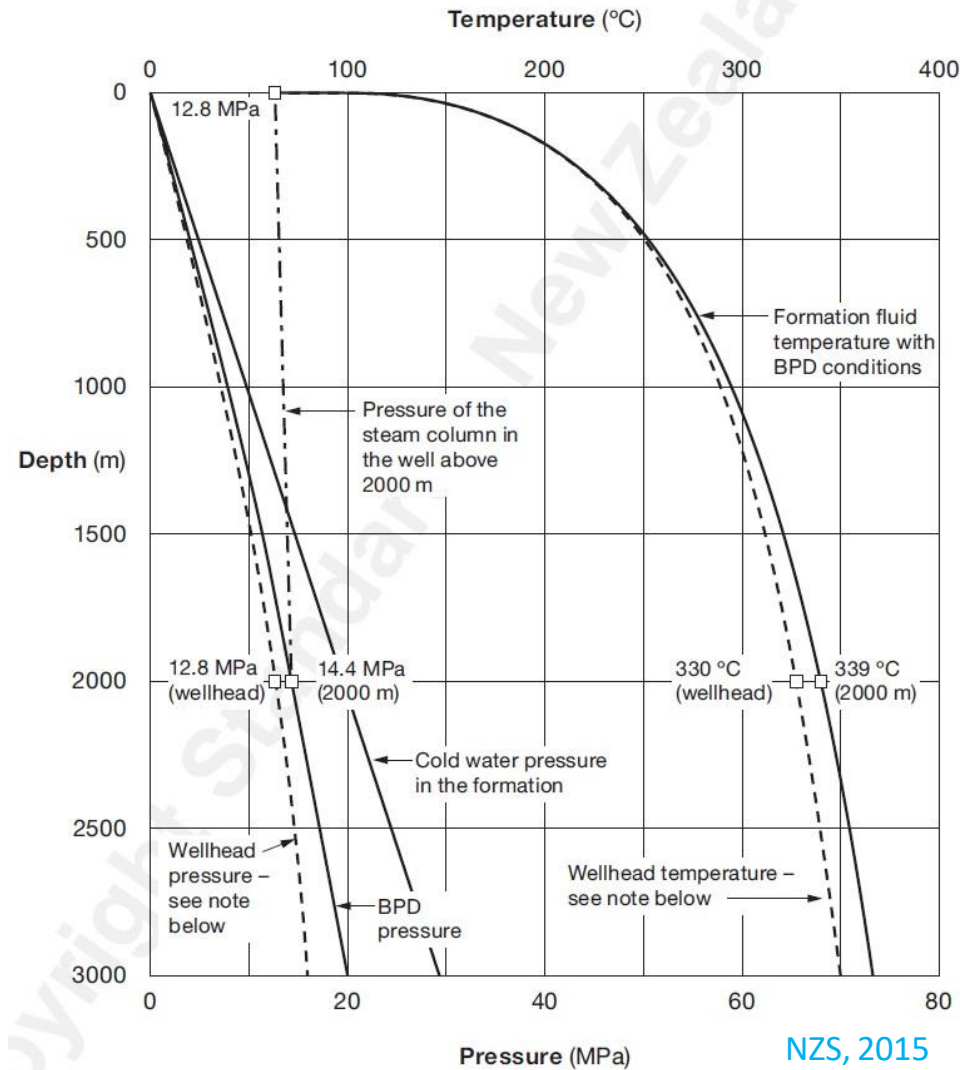
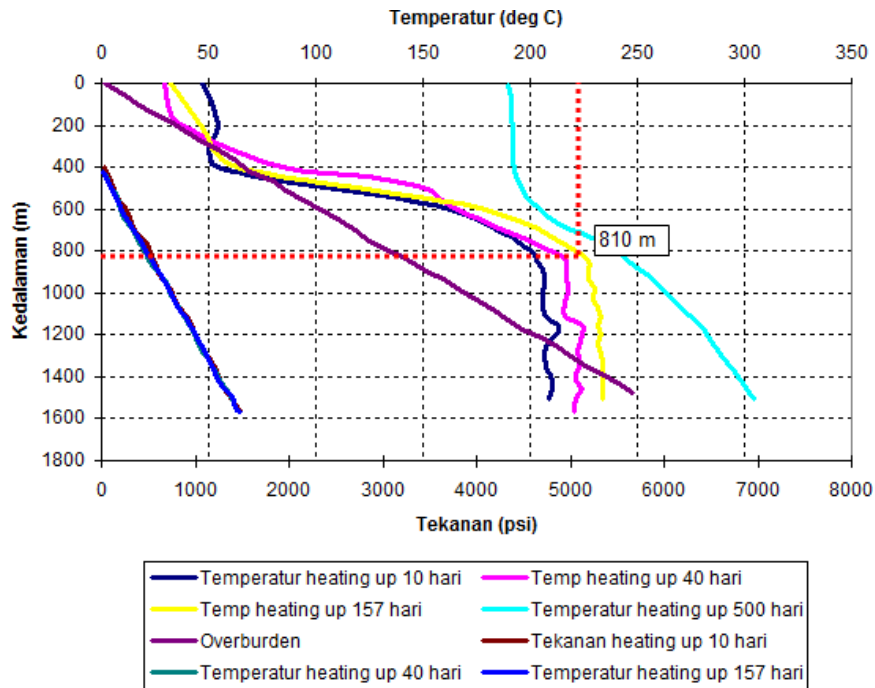


Top of reservoir determination

- High indication mineral appear consistently (epidote, adularia)
- MeB consistently < 10
- T mud out > 72 deg-C (Mud cooler On)
- PLC & TLC
- SFTT (Static formation temperature test)



Formation pressure



NZS, 2015

Fracture gradient

$$F = \frac{S - P}{D} \left(\frac{\nu}{1 - \nu} \right) + \frac{P}{D}$$

Where:

P = wellbore pressure, psi

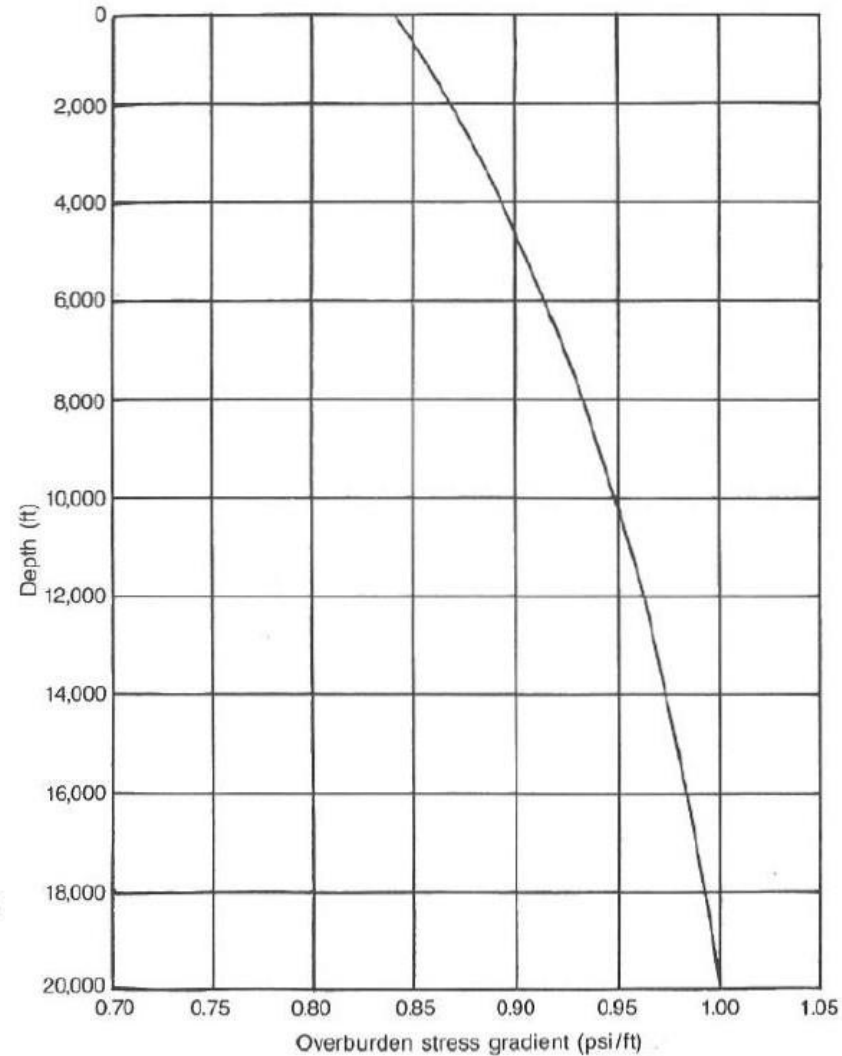
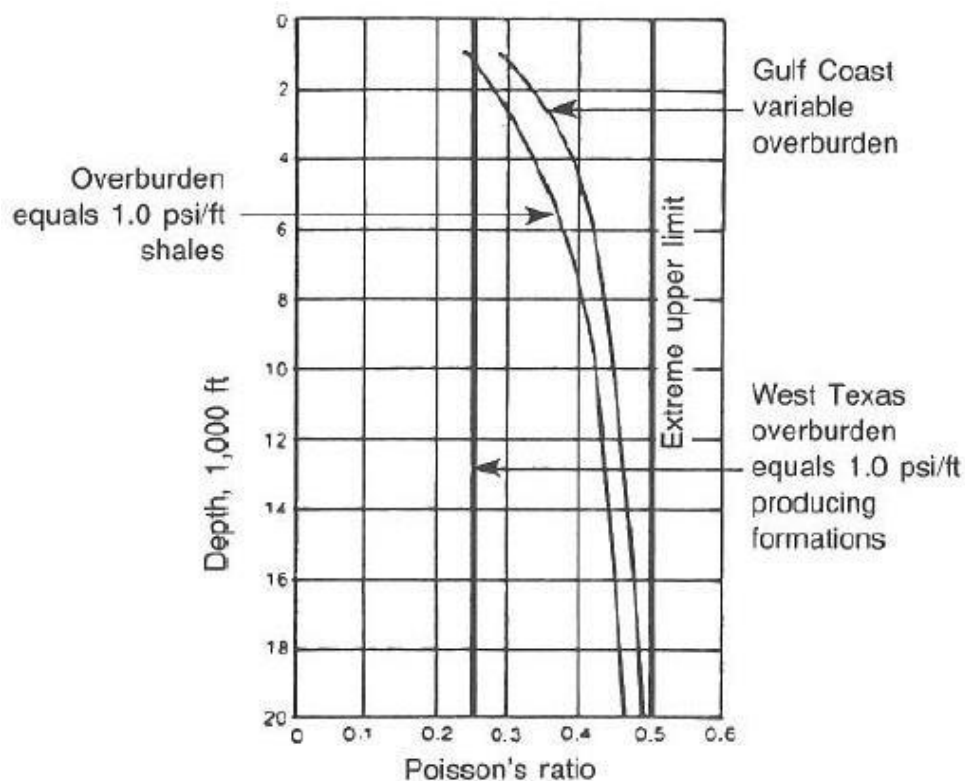
D = depth, ft

S = overburden stress, psi

ν = Poisson's ratio

F = fracture gradient, psi/ft

Jika poisson ratio
mendekati 0.5, maka
Fracture gradient
mendekati S/D

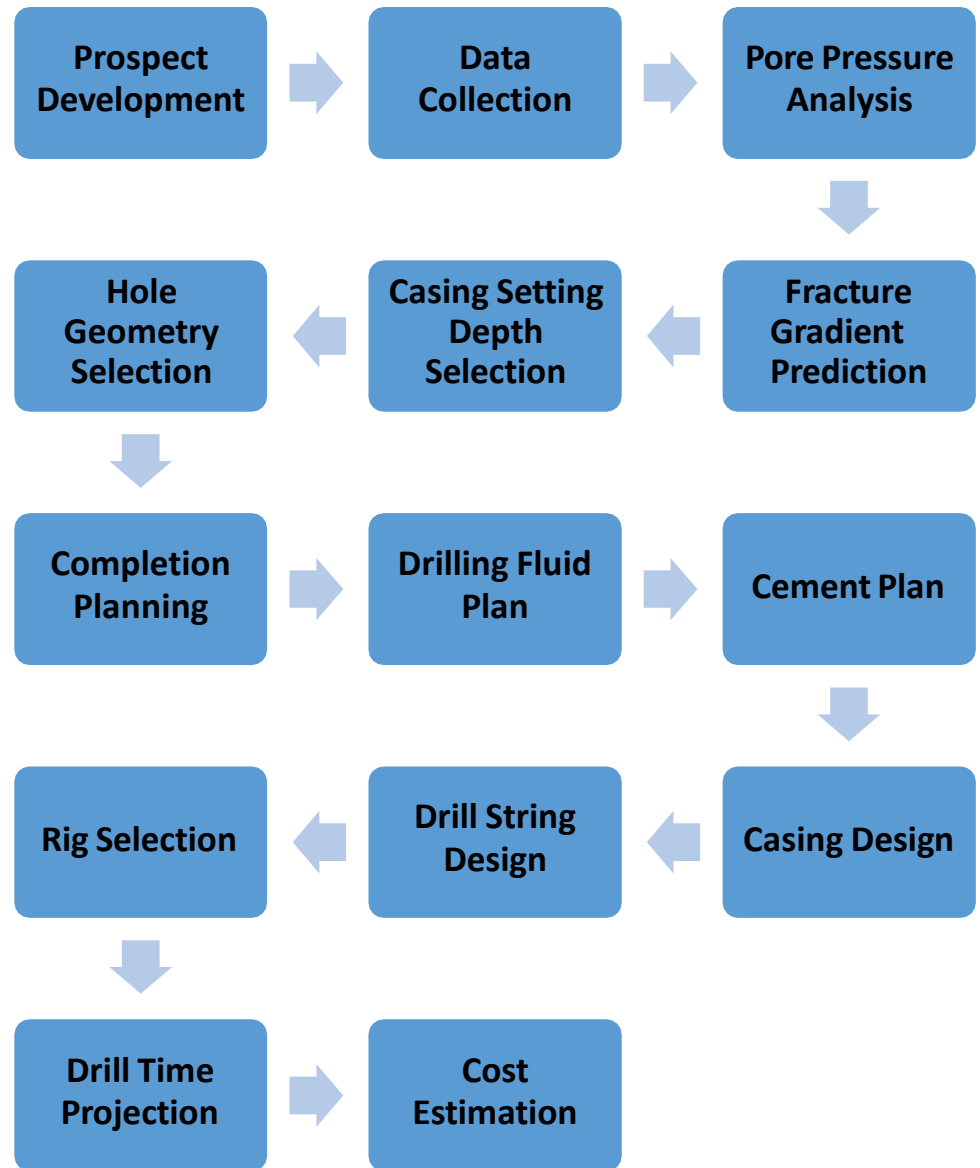


Adams, 1985

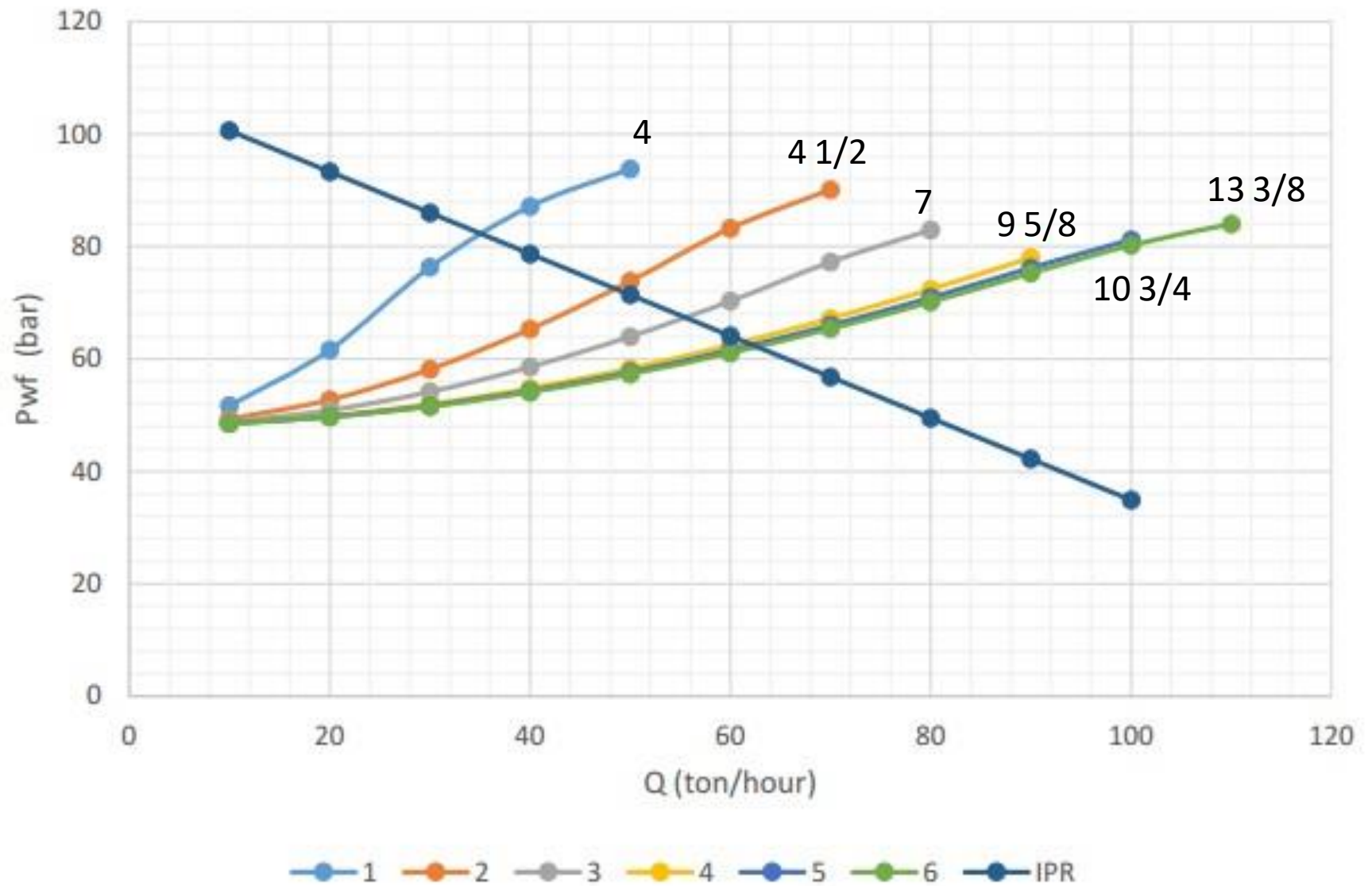
Two-stage liner

Casing Program	Section Chart	Programs	Objective
30" Conductor 20" Casing Max angle 45° 13 3/8" Casing 10 3/4" Casing 8 5/8" Casing		▪ Rock cuttings description per 3m ▪ MeB analysis per 6 to 9 m interval ▪ Gas, temperature, and chloride monitoring ▪ Drilling parameter and water loss monitoring	▪ Top of reservoir (TOR) determination ▪ Production casing set (PCS) ▪ Reservoir rock characterization ▪ Real time well database
		XRMI + GR ± Sonic ± Density ±BHPT	Permeability Assessment and reservoir rock characterization
		Coring	Reservoir rock characterization
		Rig Test	Well permeability & Productivity

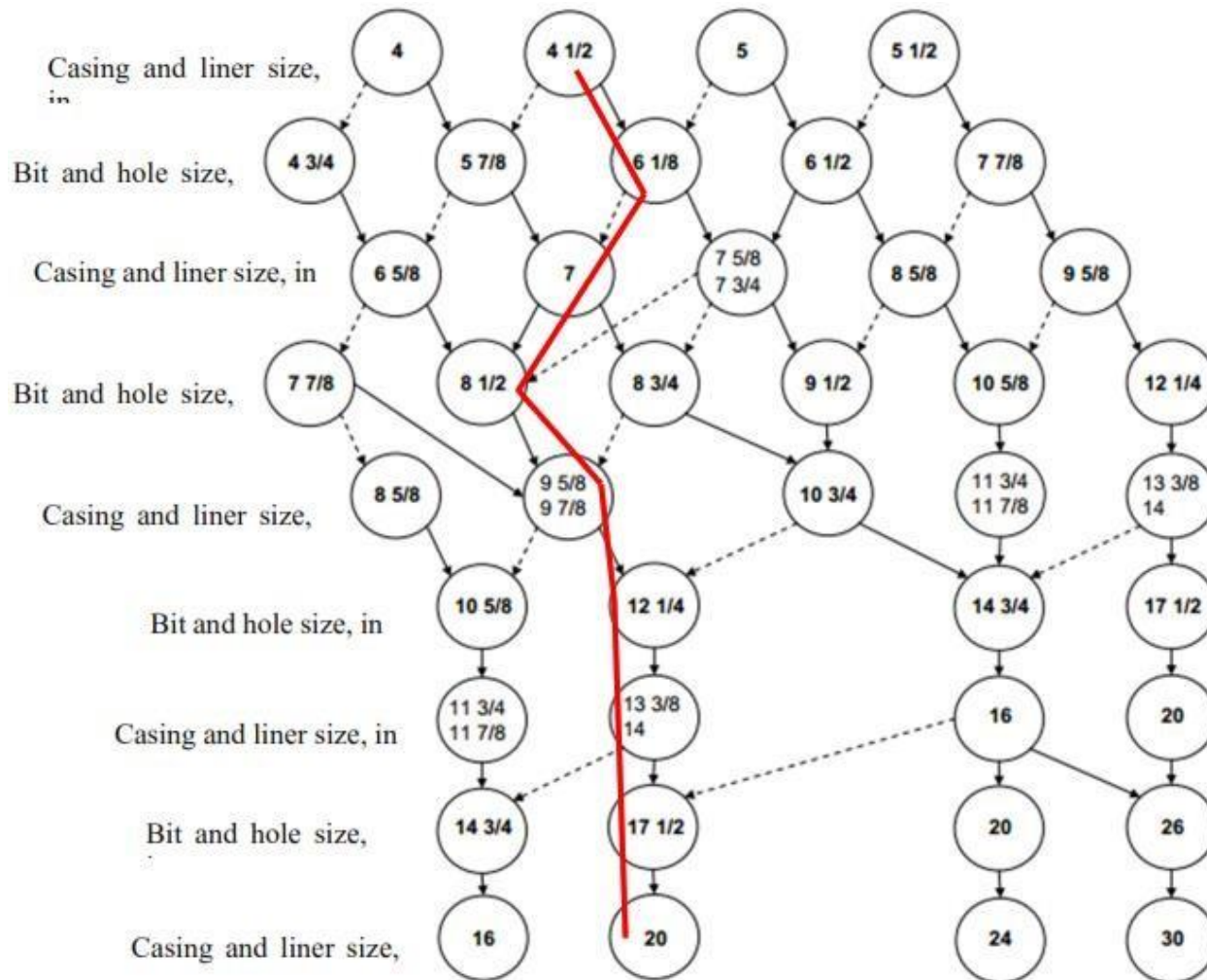
Well Planning Workflow



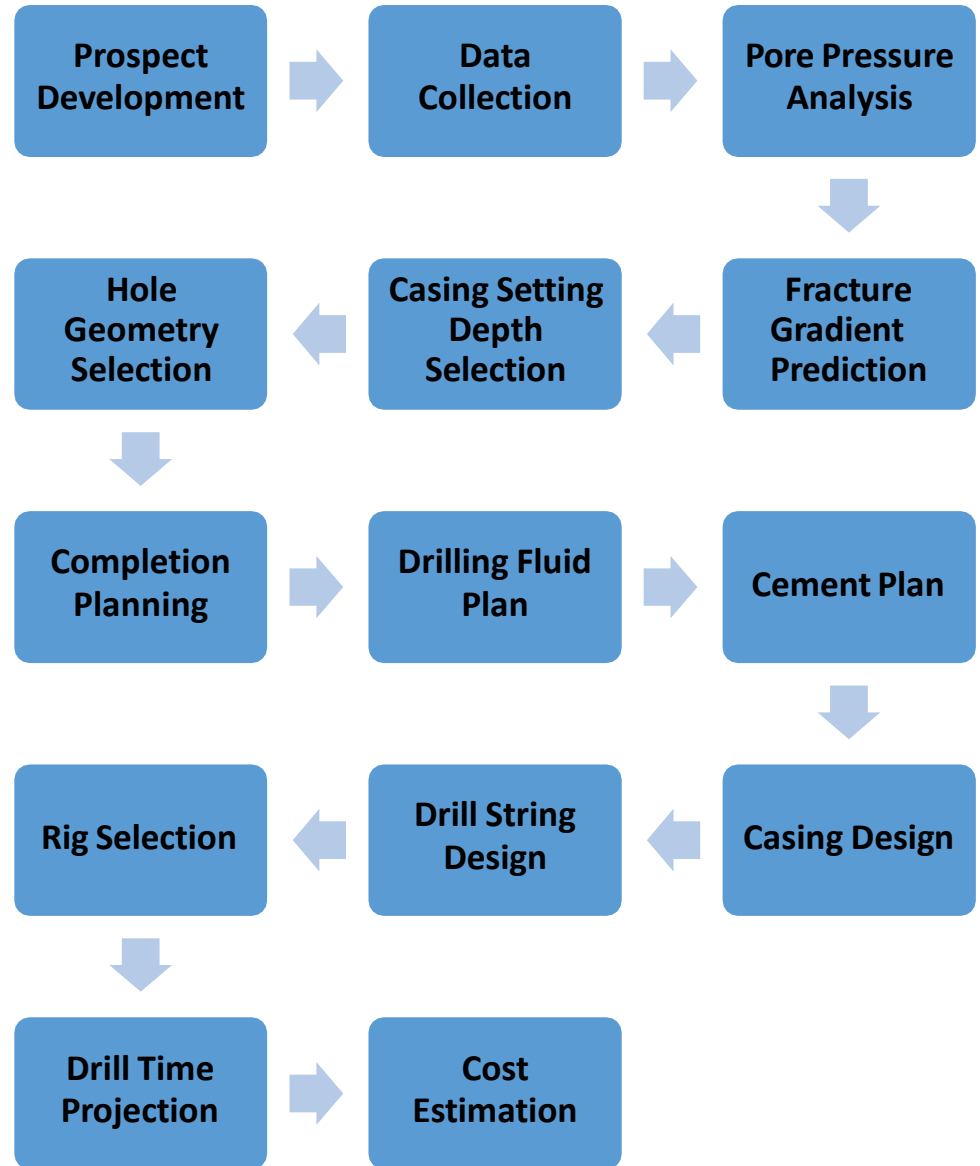
IPR vs. TPR



Hole and Casing Size



Well Planning Workflow

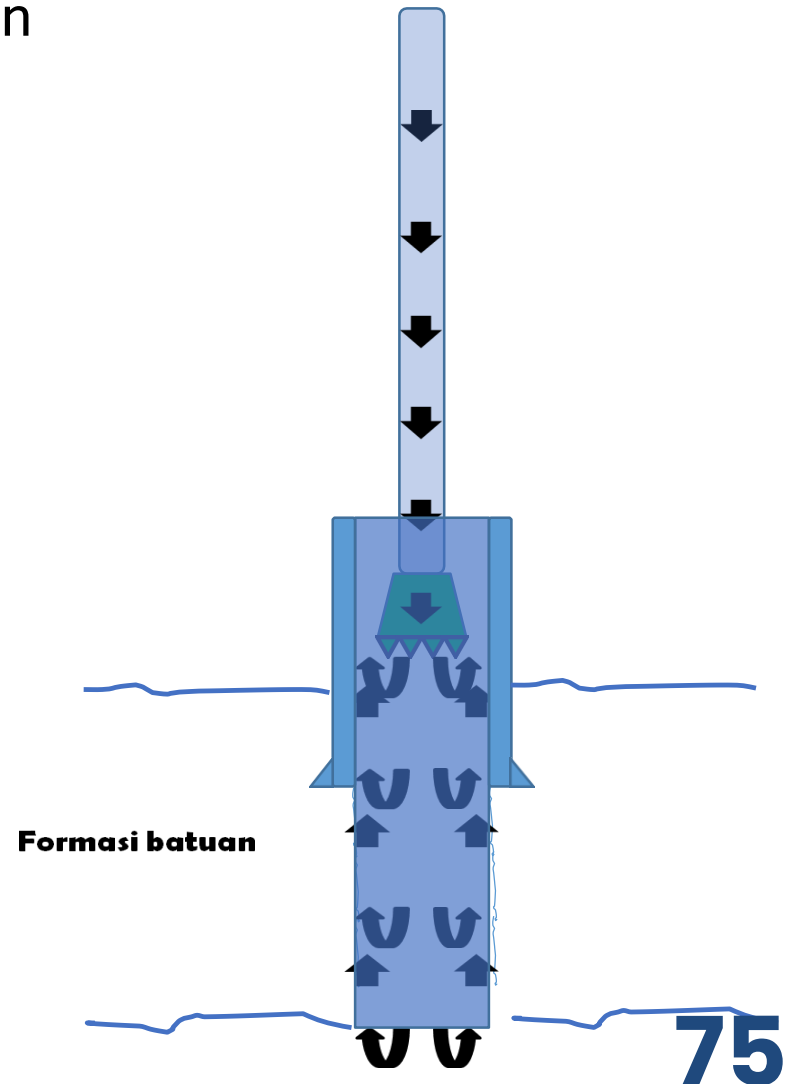


Program Lumpur Pemboran



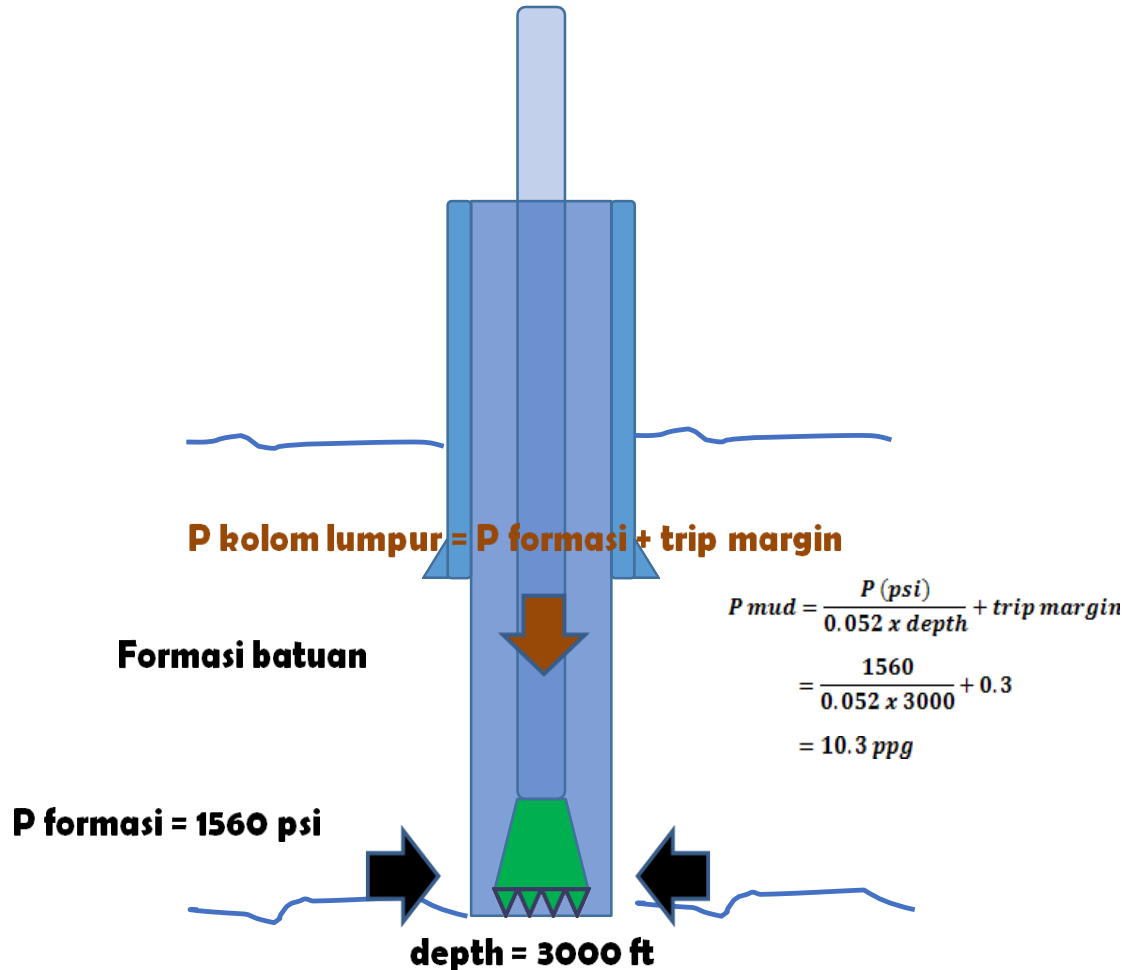
Fungsi lumpur pemboran

- Mengangkat cutting ke permukaan
kecepatan aliran umumnya
100 - 120 ft/min
- Pendingin dan pelumas
- Membentuk mud cake
- Menahan P formasi
- Suspensi cutting
- Buoyancy effect



Sifat-sifat lumpur pemboran

1. Densitas: sebagai penahan tekanan formasi.



Sifat-sifat lumpur pemboran

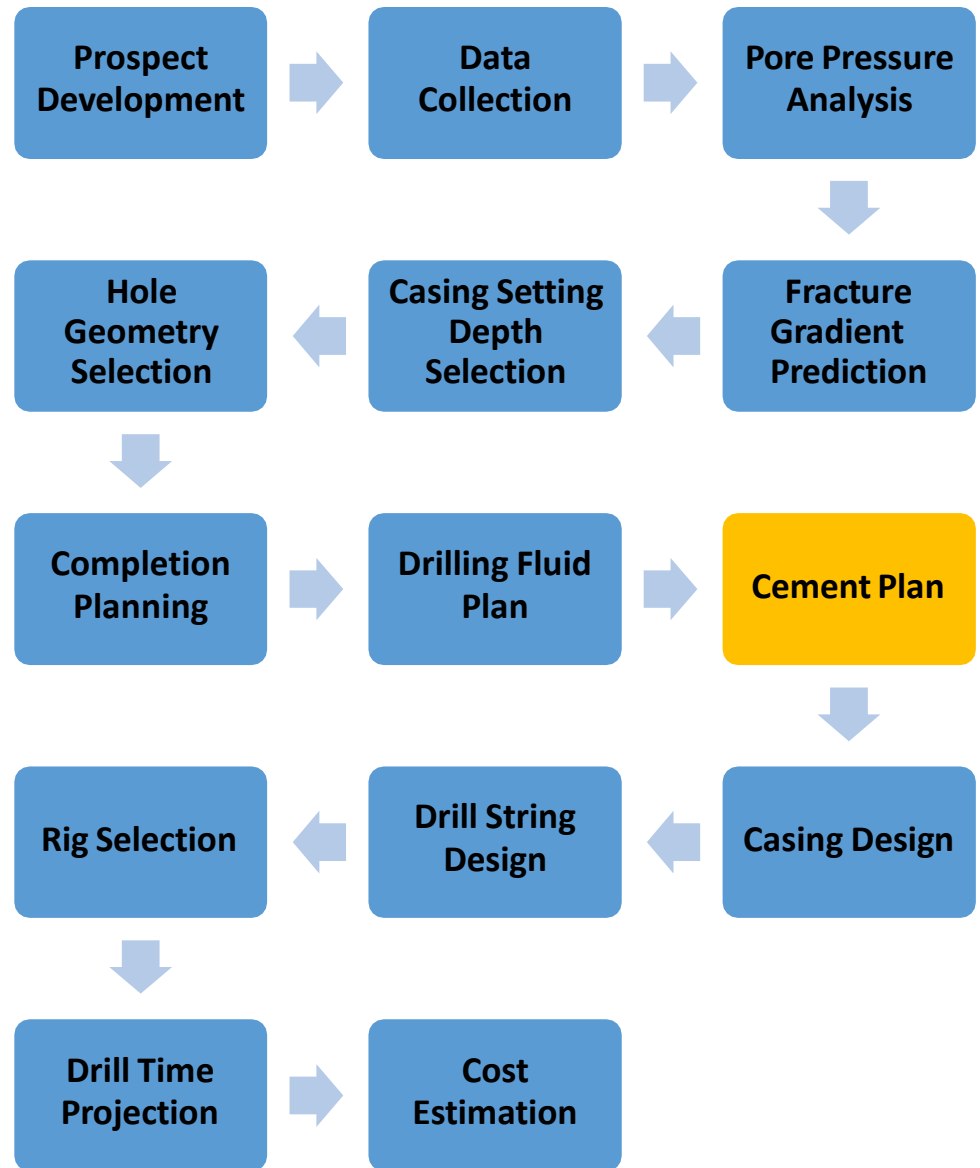
2. Viskositas & gel strength:

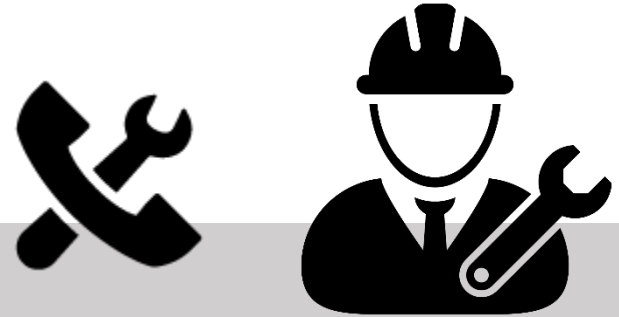
Viskositas retensi fluida terhadap perubahan bentuk di bawah tekanan shear.

- Plastic viscosity: resistensi mengalir yang disebabkan oleh friksi mekanik.
- Apparent viscosity: viskositas fluida yang terukur di alat (misal: Fann VG viscometer).



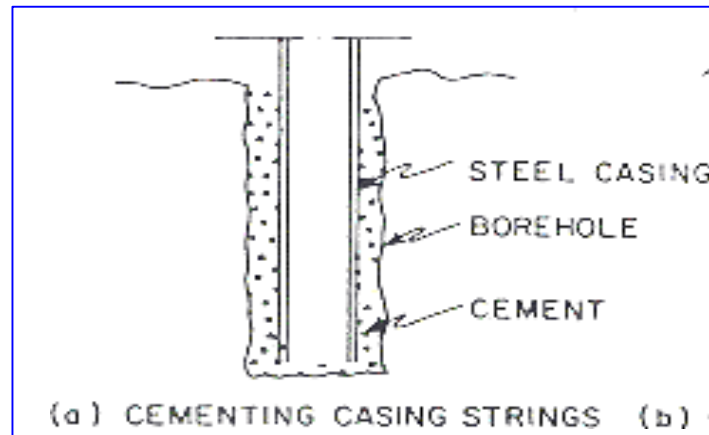
Well Planning Workflow



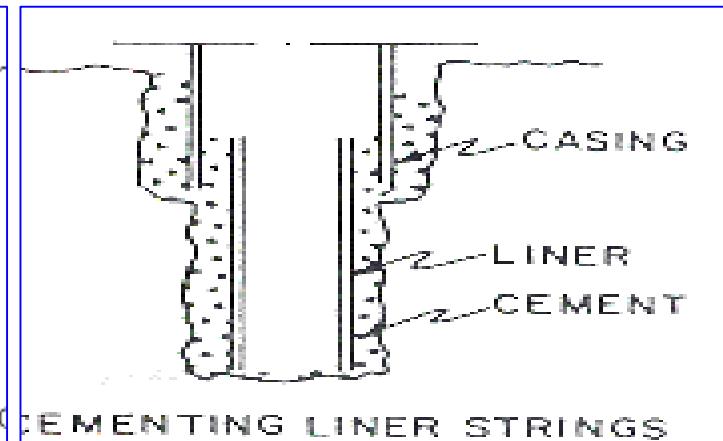


Operasi Penyemenan Sumur

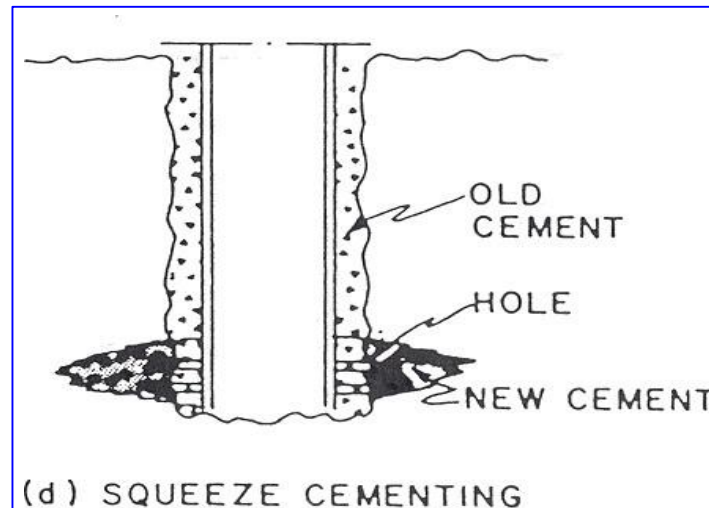
Fungsi semen



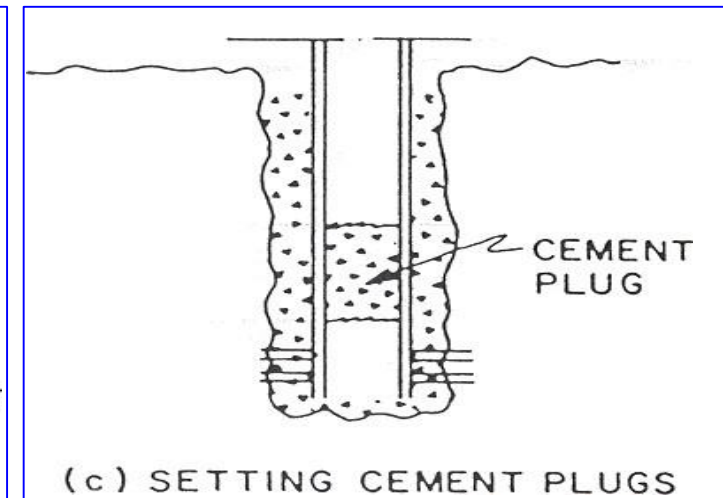
Melekatkan casing dengan dinding lubang sumur



Melekatkan liner dengan dinding lubang sumur



Menambal kebocoran di casing



Cement plug untuk well abandonment

Tie-back Cementing

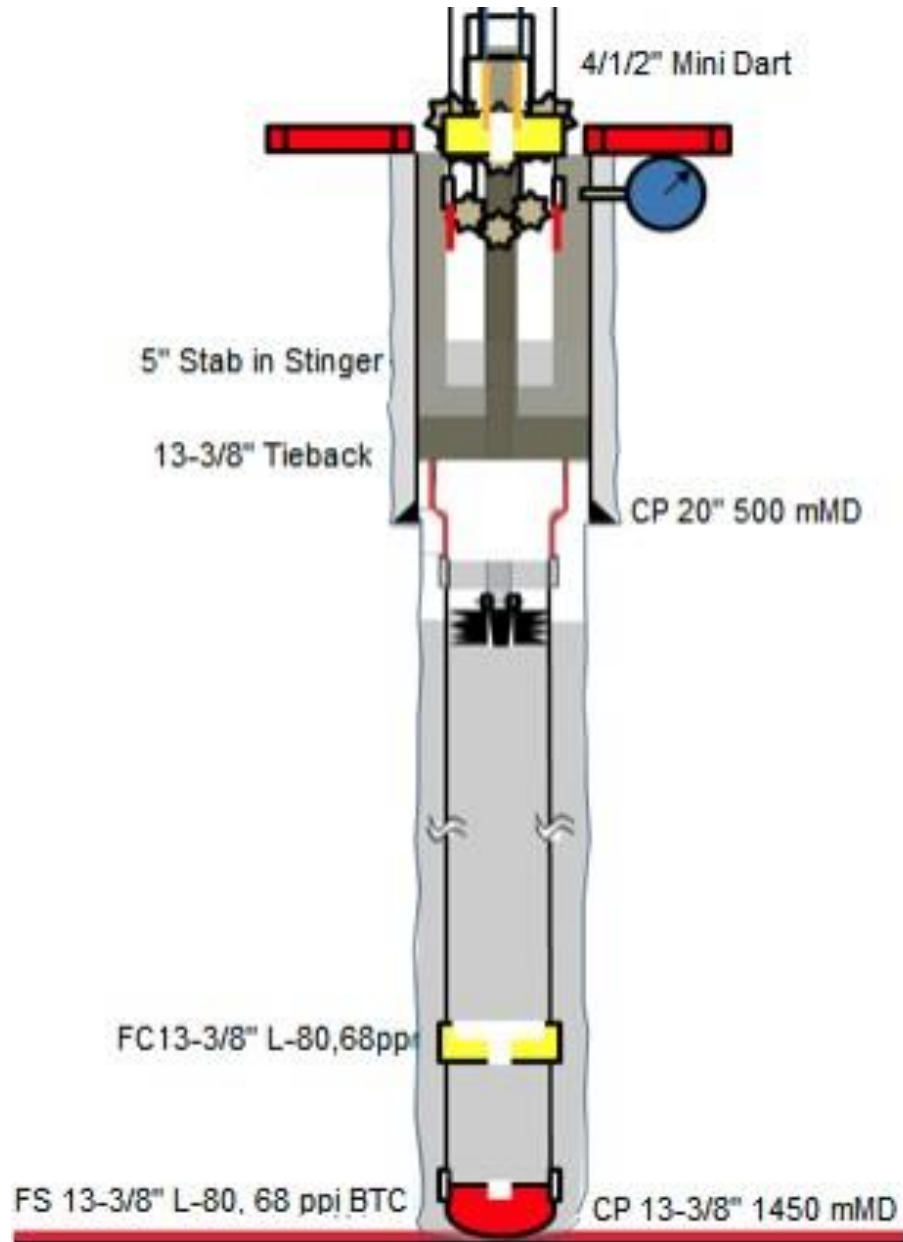


Figure 1: Tie-back cementing (Pertamina Geothermal Energy Team, 2016).

Fungsi semen

Primary cementing

Umum:

Melekatkan casing pada dinding lubang sumur.

Conductor casing, surface casing:

Mencegah kontaminasi air tanah oleh lumpur pemboran.

Intermediate casing:

Menutup zona abnormal, menutup zona lost circulation.

Production casing:

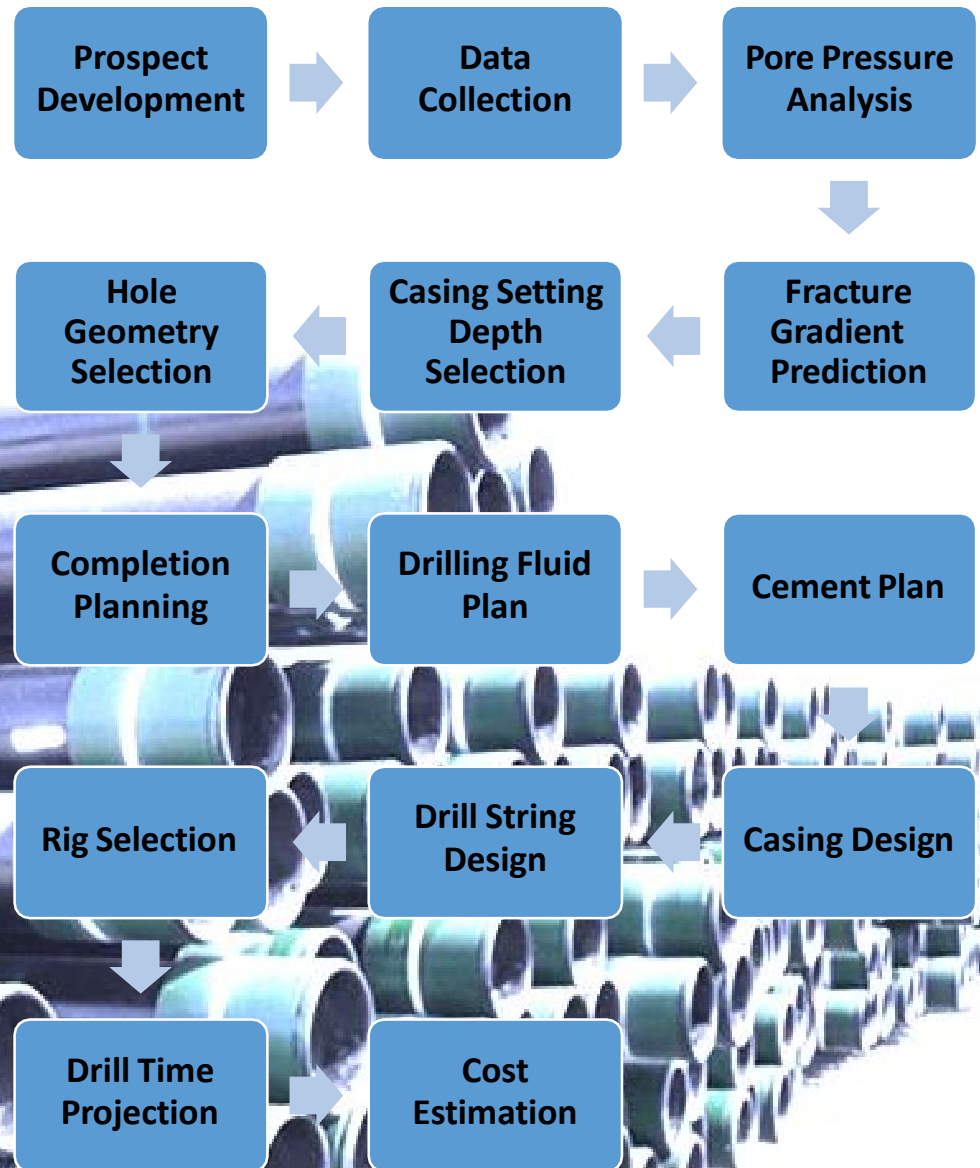
Isolasi zona produktif.

Klasifikasi semen

■ Kelas-kelas semen & spesifikasinya (API standard):

Kelas semen	Range kedalaman	Keterangan
A	0 – 6000 ft	Ordinary type only
B	0 – 6000 ft	Sulphate resistant
C	0 – 6000 ft	Early strength
D	6000 – 12000 ft	HPHT
E	6000 – 14000 ft	HPHT
F	10000 – 16000 ft	Very HPHT
G	0 – 8000 ft	Standard cement
H	0 – 8000 ft	Standard cement

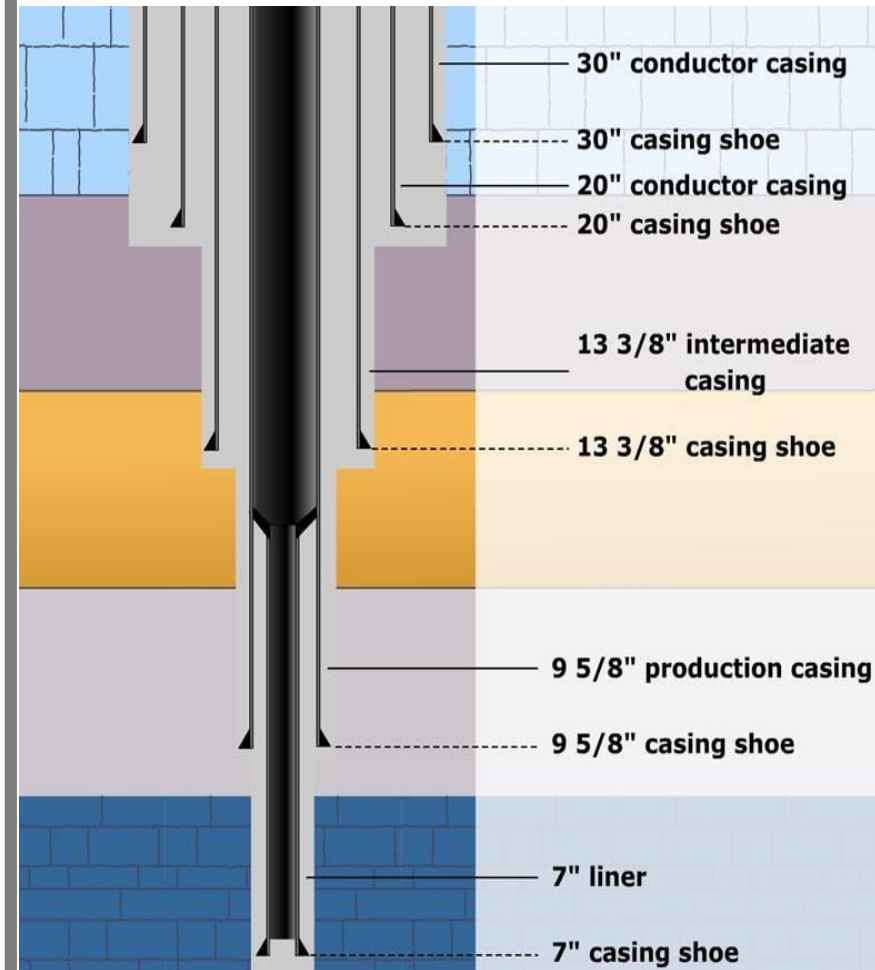
Well Planning Workflow



Introduction: Casing

Kegunaan masing-masing jenis casing

- ❑ Conductor casing:
 - Memberikan arah pemboran pertama kali
 - Tempat dudukkan bagi seksi casing berikutnya
- ❑ Surface casing:
 - Tempat duduknya BOP
 - Menjaga kestabilan lubang sumur
 - Melindungi air tanah dari pencemaran oleh fluida pemboran
- ❑ Intermediate casing:
 - Isolasi zona abnormal
 - Isolasi zona lost circulation
- ❑ Production casing:
 - Memisahkan zona produksi satu dengan zona produksi lainnya
 - Menjaga kondisi lubang sumur yang dapat ditolerir subsurface completion equipment
- ❑ Liners:
 - *sama dengan production casing



Introduction: Casing

□ Tipe casing

Type	Diameter (in)
Conductor	16 – 60
Surface	16 – 48
Intermediate	8 5/8 – 20
Production	7 5/8 – 13 3/8
Liner	4 ½ - 9 5/8

□ Klasifikasi casing

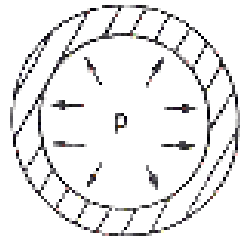
Parameter	Definisi
OD (Outside diameter)	casing body eksternal diameter
Pounder (lbf/ft)	weight per foot of casing
Wall thickness	OD – ID
ID (Inside Diameter)	Nominal ID, Drift diameter
Type of joint connection	API 8rd, Butress, Premium
Length range	Range 1 (16-25 ft), Range 2 (25-34 ft), Range 3 (> 34 ft)

Introduction: Casing

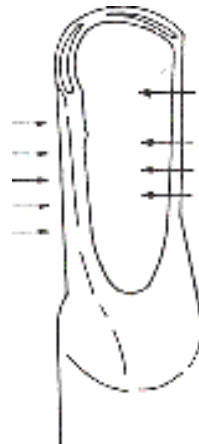
□ Grade casing

<u>Most Common Grades</u>	<u>Minimum Yield Strength (PSI)</u>	<u>Ultimate Tensile Strength (PSI)</u>
H-40	40,000	60,000
J-55	55,000	75,000
K-55	55,000	95,000
C-75	75,000	95,000
L-80	80,000	95,000
N-80	80,000	100,000
C-90	90,000	100,000
C-95	95,000	105,000
S-95	95,000	105,000
P-110	110,000	125,000
V-150	150,000	160,000

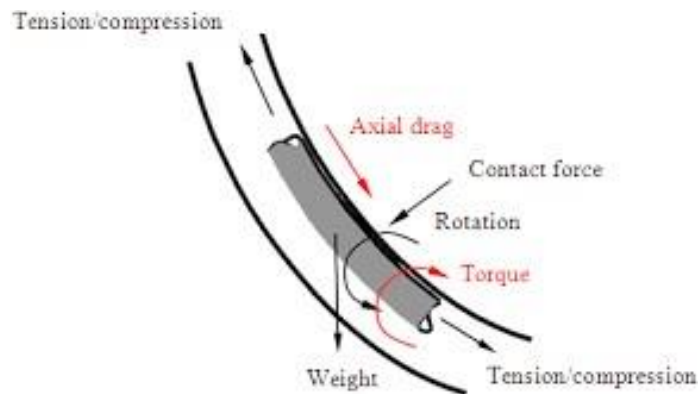
Kriteria Desain Casing



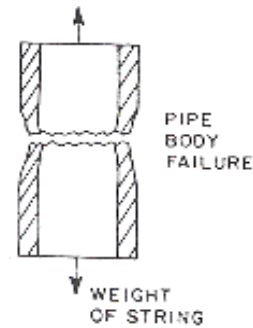
1. Menahan Beban Burst



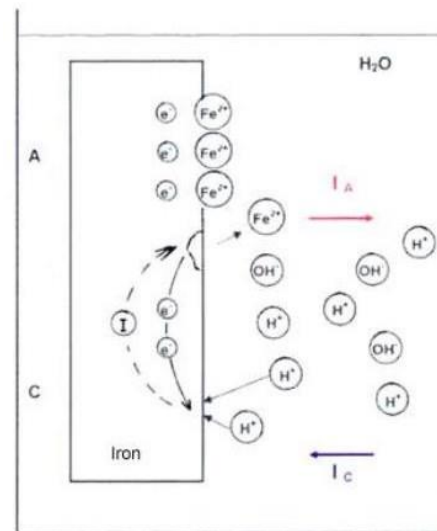
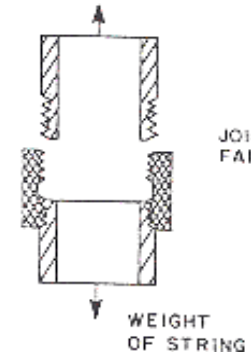
2. Menahan Beban Collapse



4. Menahan Beban Tension dan Torsi akibat kemiringan sumur



3. Menahan Beban Tension

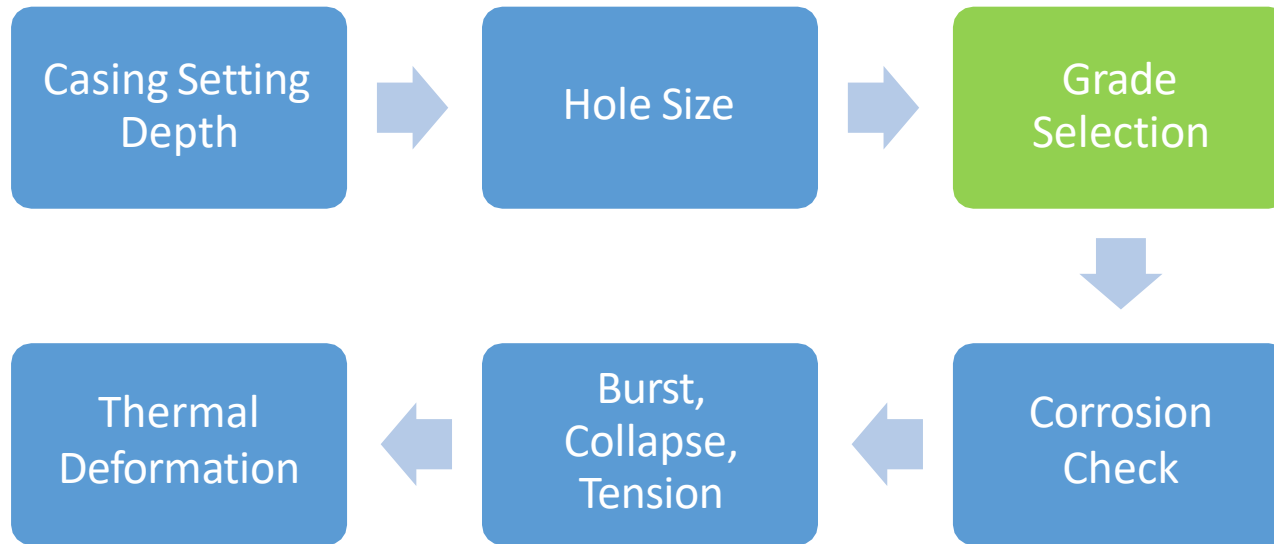


4. Menahan Korosi

Effect of Temperature

Grade	Temperature (°C)						
	20	100	150	200	250	300	350
	API yield strength (factor)						
J55/K55	1.00	0.94	0.90	0.90	0.85	0.80	0.70
L80/C90/T95	1.00	0.96	0.92	0.90	0.88	0.85	0.81
	Tensile strength (factor)						
All grades	1.00	0.96	0.92	0.90	0.88	0.86	0.84
	Modulus of elasticity (10^3 MPa)						
All grades	210	205	201	197	194	190	185
NOTE –							
(1) The tabulated values are generally within 5% of selected published figures.							
(2) Sufficient published test data on some of the preferred steels listed in 2.9.2 are not available for inclusion in this table.							
(3) This table was prepared based on published data collected from companies such as Tenaris, Sumitomo, Grant Prideco, and Nippon. Some variation exists between these data sets and this table represents a realistic lower bound of the data.							

Simple Casing Design



Corrosion

Ekasari & Marbun, 2015

$$Cr_{eq} = \frac{2.981 - 4.532(pH) - 25.052\left(\frac{1}{T}\right) - \log(CR)}{2.912}$$

Cr_{eq} = Chromium Equivalent

CR = Corrosion Rate, mm/year

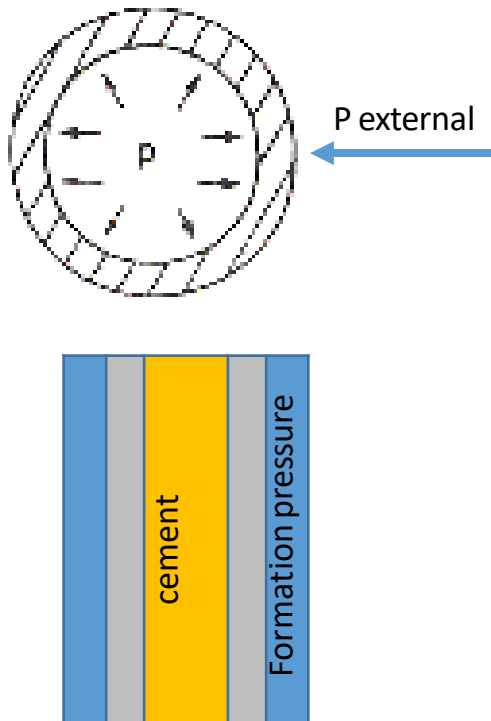
pH = Geothermal fluid pH

T = Temperature, deg-C

Relative to corrosion resistance	Corrosion rate	
	(mpy)	(mm/year)
Outstanding	< 1	<0.025
Excellent	1 - 5	0.025 - 0.1
Good	5 - 20	0.1 - 0.5
Fair	20 - 50	0.5 - 1.27
Poor	50 - 200	1.27 - 5.08
Unacceptable	200 ++	5.08 ++

Grade casing (API 5CT)		Minimum	Maksimum
Grup 1	H-40	-8,5	-7,5
	J-55	-6	-5,4
	K-55	-6	-5
	N-80-1	-2	-1,5
	N-80-Q	-2	-1,5
Grup 2	M-65	-5,5	-4,3
	L-80-1	-2,4	-1,8
	L-80-9Cr	10	11,5
	L-80-13Cr	11	12

Burst



2.10.6.2

During cementing the maximum differential internal pressure of the string occurs near the shoe or stage cementing ports when all of the following apply:

- (a) The casing is filled with cement slurry;
- (b) The annulus contains either a column of water or is subject to formation pressure; and
- (c) There is sufficient constriction within the casing (for example, a cementing plug) to hold the differential pressure.

$$\Delta P_{internal} = P_{int} - P_{ext}$$

P_{int} = cement column

P_{ext} = Formation Pressure/ Water/ Drilling Fluid

If Formation pressure not available yet, use BPD pressure

$$P_b = [0.052 \times h_c \times \rho_c] - P_{BPD}$$

P_b = P burst, Psi

h_c = Cement column high, feet

ρ_c = Cement Density, ppg

P_{BPD} = Formation saturation pressure, psi

Burst rate

- Internal pressure that the pipe can withstand prior to failure.

$$P_B = 0.875 \frac{2Y_P t}{OD}$$

Where:

P_B = burst pressure rounded to the nearest 10 psi

Y_P = specified minimum yield strength, psi

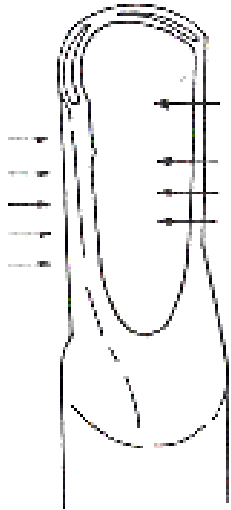
t = nominal wall thickness, in.

OD = nominal outside diameter, in.

Collapse Pressure

2.10.7.2

During the later stages of the casing cementing operation, the maximum differential external pressure occurs near the casing shoe when the casing annulus is filled with dense cement slurry, and the casing is filled with water. The maximum differential external pressure is:



$$\Delta P_{external} = P_{ext} - P_{int}$$

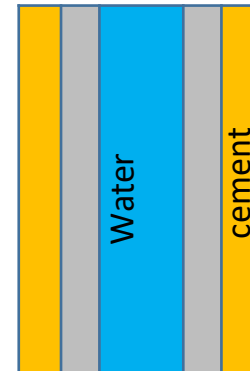
$$P_c = 0.052 \times h \times [\rho_c - \rho_w]$$

P_c = P collapse, Psi

h = fluid column high, feet

ρ_c = Cement Density, ppg

ρ_w = Water Density, ppg



Collapse Rate

Table 11-6 Formula Factors and D/t Ranges for Plastic Collapse

1	2	3	4	5
Grade*	Formula Factor			D/t Range
	A	B	C	
H-40	2.950	0.0465	754	16.44 to 27.01
-50	2.976	0.0515	1,056	15.24 to 25.63
J-K-55 & D	2.991	0.0541	1,206	14.81 to 25.01
-60	3.005	0.0566	1,356	14.44 to 24.42
-70	3.037	0.0617	1,656	13.85 to 23.38
C-75 & E	3.054	0.0642	1,806	13.60 to 22.91
L-80 & N-80	3.071	0.0667	1,955	13.38 to 22.47
-90	3.106	0.0718	2,254	13.01 to 21.69
C-95	3.124	0.0743	2,404	12.85 to 21.33
-100	3.143	0.0768	2,553	12.70 to 21.00
P-105	3.162	0.0794	2,702	12.57 to 20.70
P-110	3.181	0.0819	2,852	12.44 to 20.41
-120	3.219	0.0870	3,151	12.21 to 19.88
-125	3.239	0.0895	3,301	12.11 to 19.63
-130	3.258	0.0920	3,451	12.02 to 19.40
-135	3.278	0.0946	3,601	11.92 to 19.18
-140	3.297	0.0971	3,751	11.84 to 18.97
-150	3.336	0.1021	4,053	11.67 to 18.57
-155	3.356	0.1047	4,204	11.59 to 18.37
-160	3.375	0.1072	4,356	11.52 to 18.19
-170	3.412	0.1123	4,660	11.37 to 17.82
-180	3.449	0.1173	4,966	11.23 to 17.47

Table 11-7 Formula Factors and D/t Range for Transition Collapse

1	2	3	4
Grade*	Formula Factors		D/t Range
	F	G	
H-40	2.063	0.0325	27.01 to 42.64
-50	2.003	0.0347	25.63 to 38.83
J-K-55 & D	1.989	0.0360	25.01 to 37.21
-60	1.983	0.0373	24.42 to 35.73
-70	1.984	0.0403	23.38 to 33.17
C-75 & E	1.990	0.0418	22.91 to 32.05
L-80 & N-80	1.998	0.0434	22.47 to 31.02
-90	2.017	0.0466	21.69 to 29.18
C-95	2.029	0.0482	21.33 to 28.36
-100	2.040	0.0499	21.00 to 27.60
P-105	2.053	0.0515	20.70 to 26.89
P-110	2.066	0.0532	20.41 to 26.22
-120	2.092	0.0565	19.88 to 25.01
-125	2.106	0.0582	19.63 to 24.46
-130	2.119	0.0599	19.40 to 23.94
-135	2.133	0.0615	19.18 to 23.44
-140	2.146	0.0632	18.97 to 22.98
-150	2.174	0.0666	18.57 to 22.11
-155	2.188	0.06825	18.37 to 21.70
-160	2.202	0.0700	18.19 to 21.32
-170	2.231	0.0734	17.82 to 20.60
-180	2.261	0.0769	17.47 to 19.93

*Grades indicated without letter designation are not API grades but are grades in use or grades being considered for use and are shown for information purposes.
Courtesy American Petroleum Institute

Table 11-8 D/t Range for Elastic Collapse

1	2
Grade*	D/t Range
H-40	42.64 and greater
-50	38.83 and greater
J-K-55 & D	37.21 and greater
-60	35.73 and greater
-70	33.17 and greater
C-75 & E	32.05 and greater
L-80 & N-80	31.02 and greater
-90	29.18 and greater
C-95	28.36 and greater
-100	27.60 and greater
P-105	26.89 and greater
P-110	26.22 and greater
-120	25.01 and greater
-125	24.46 and greater
-130	23.93 and greater
-135	23.44 and greater
-140	22.98 and greater
-150	22.11 and greater
-155	21.70 and greater
-160	21.32 and greater
-170	20.60 and greater
-180	19.93 and greater

*Grades indicated without letter designation are not API grades but are grades in use or grades being considered for use and are shown for information purposes.
Courtesy American Petroleum Institute

$$P_p = Y_p \left[\frac{A}{D/t} - B \right] - C$$

$$P_t = Y_p \left[\frac{F}{D/t} - G \right]$$

$$P_e = \frac{46.95 \times 10^6}{(D/t) [(D/t) - 1]^2}$$

Tension

$$F_a = [W \times BF] + MOP$$

F_a = Tension, lbs

W = Casing weight at surface, lbs

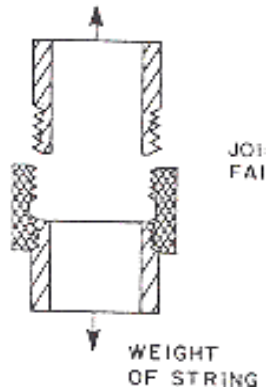
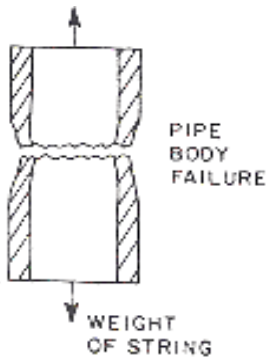
BF = Buoyancy factor, ratio

MOP = Margin of overpull, lbs (Commonly 100,000 lbs)

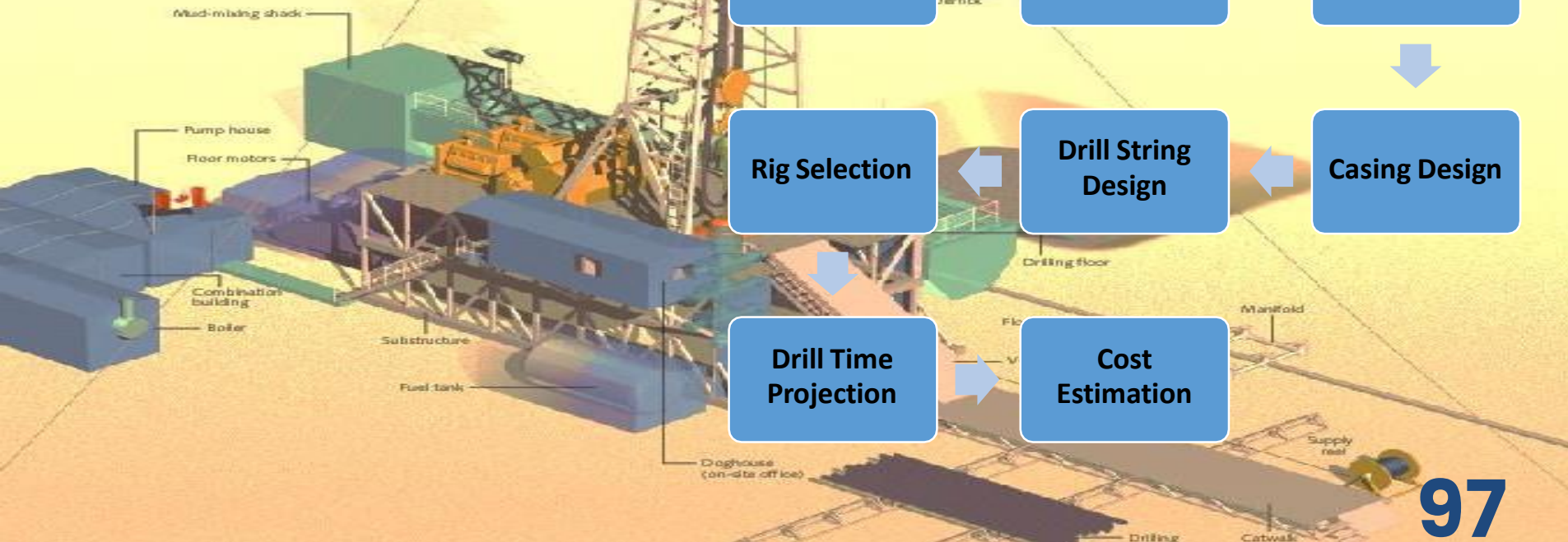
$$BF = 1 - \left(\frac{\rho_m}{\rho_s} \right)$$

ρ_m = Mud Density, ppg

ρ_s = casing material density, 65.5 ppg



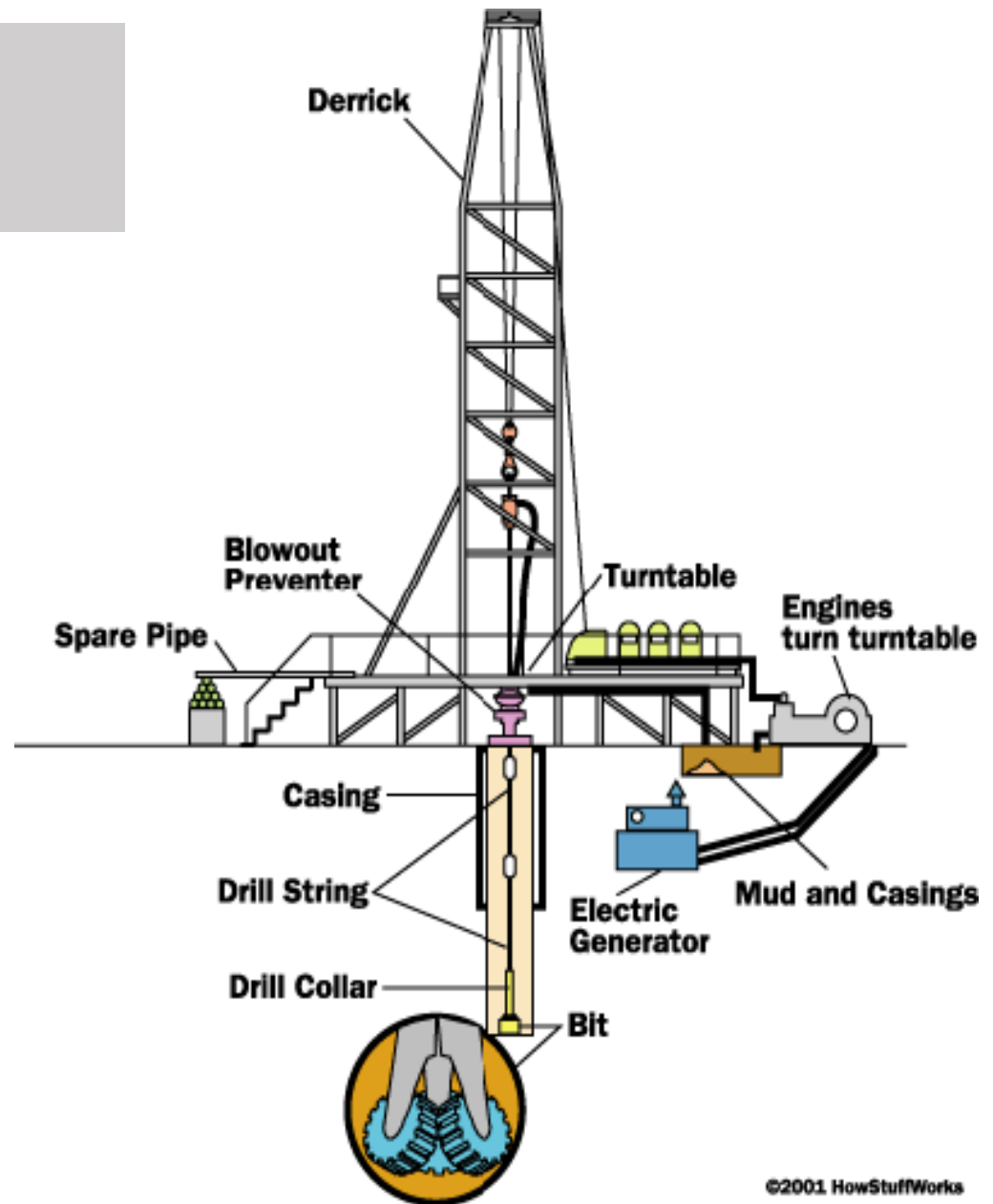
Well Planning Workflow



RIG COMPONENTS SELECTION

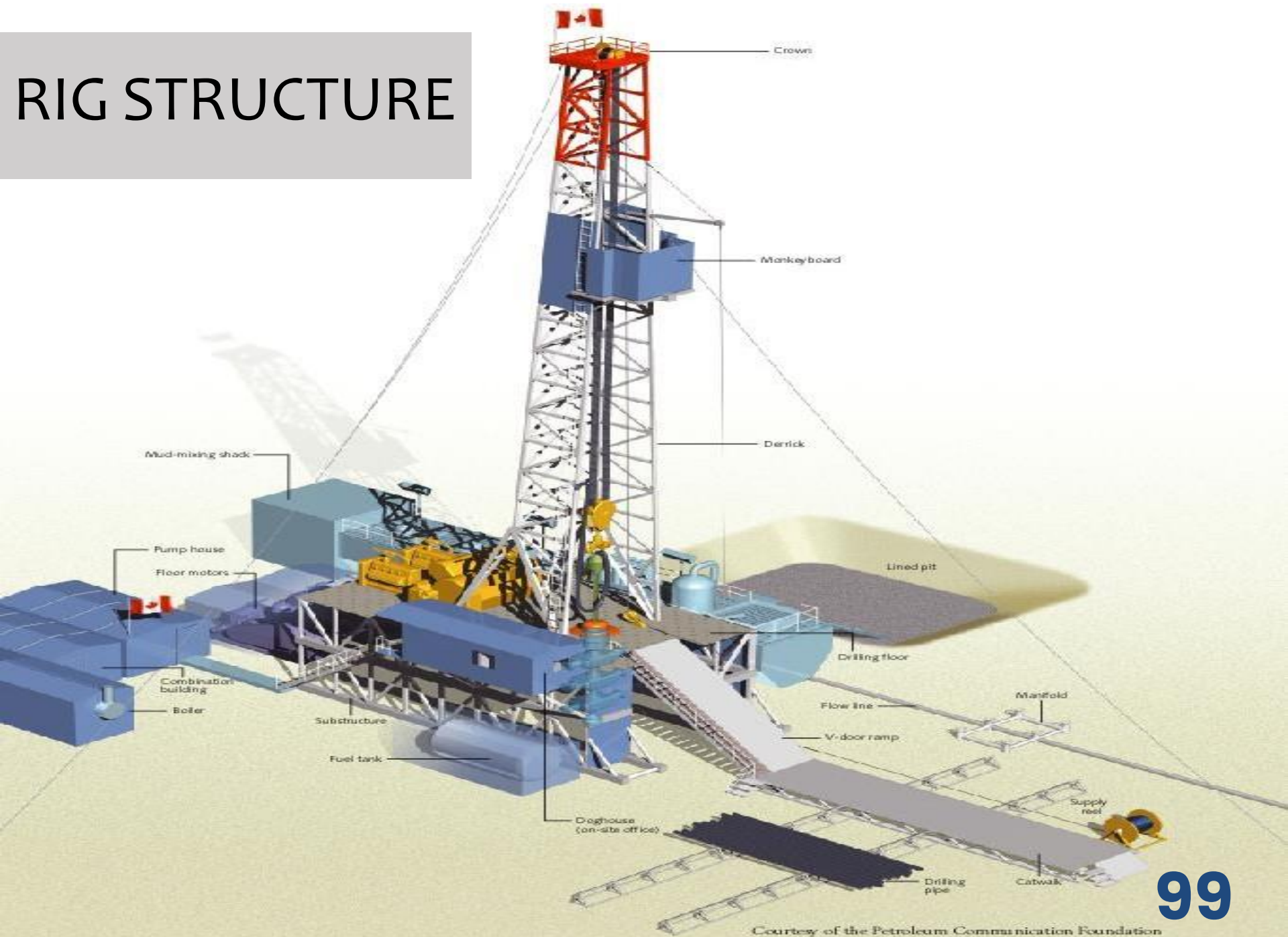
Five Main System of the Rig:

- Power system
- Hoisting system
- Rotating system
- Circulating system
- Control & Measurement system



©2001 HowStuffWorks

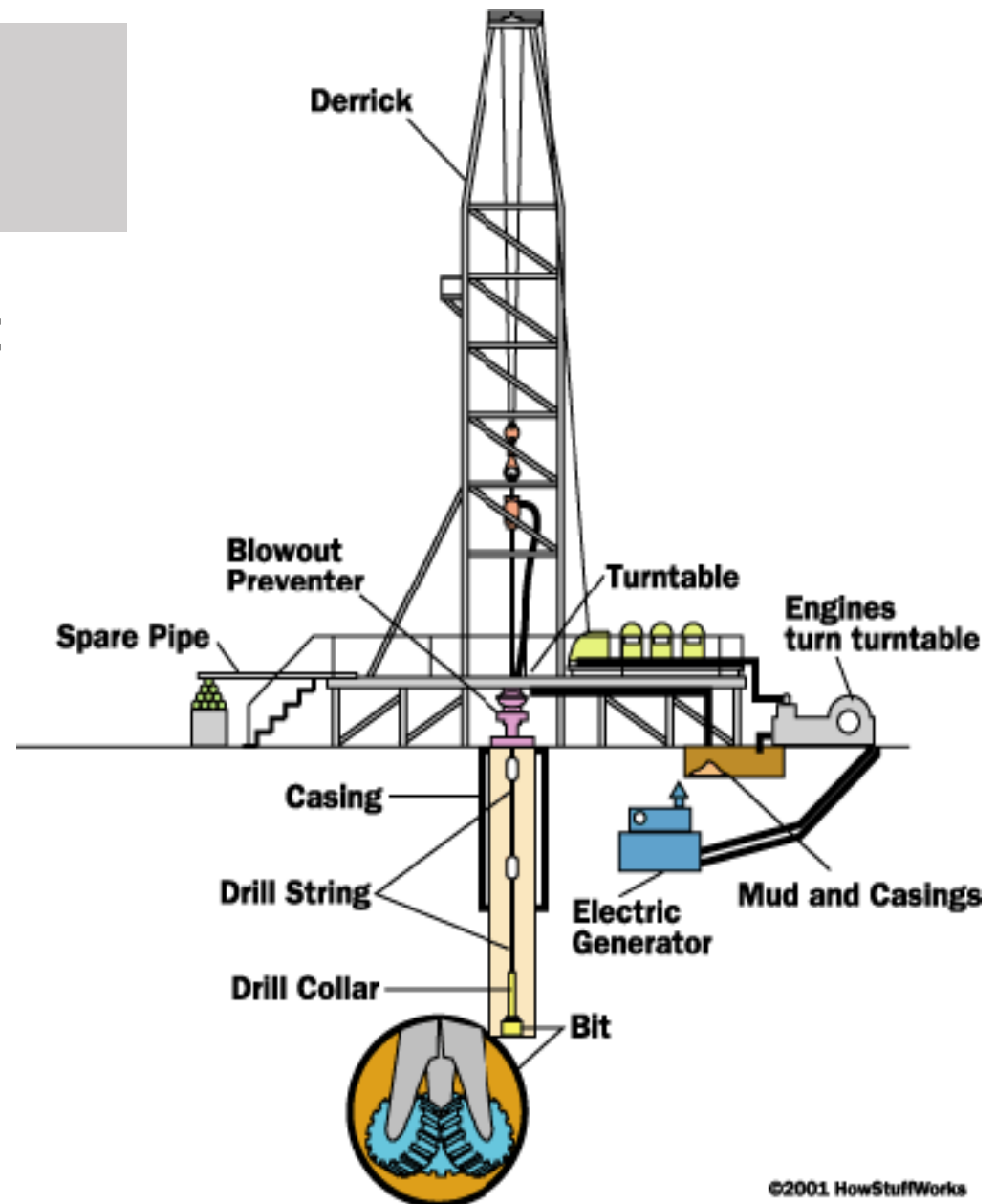
RIG STRUCTURE



RIG COMPONENTS

Five Main System of the Rig:

- Power system
- Hoisting system
- Rotating system
- Circulating system
- Control & Measurement system

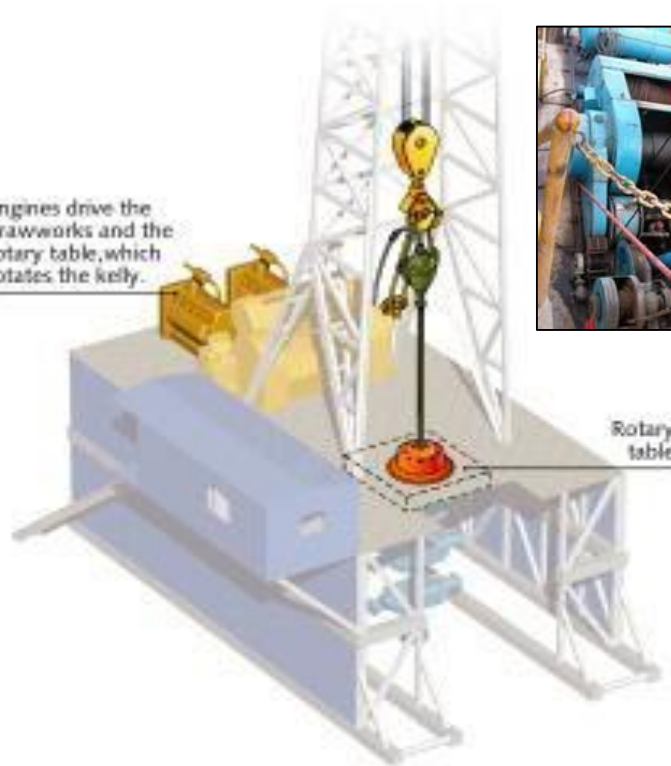


©2001 HowStuffWorks

POWER SYSTEM



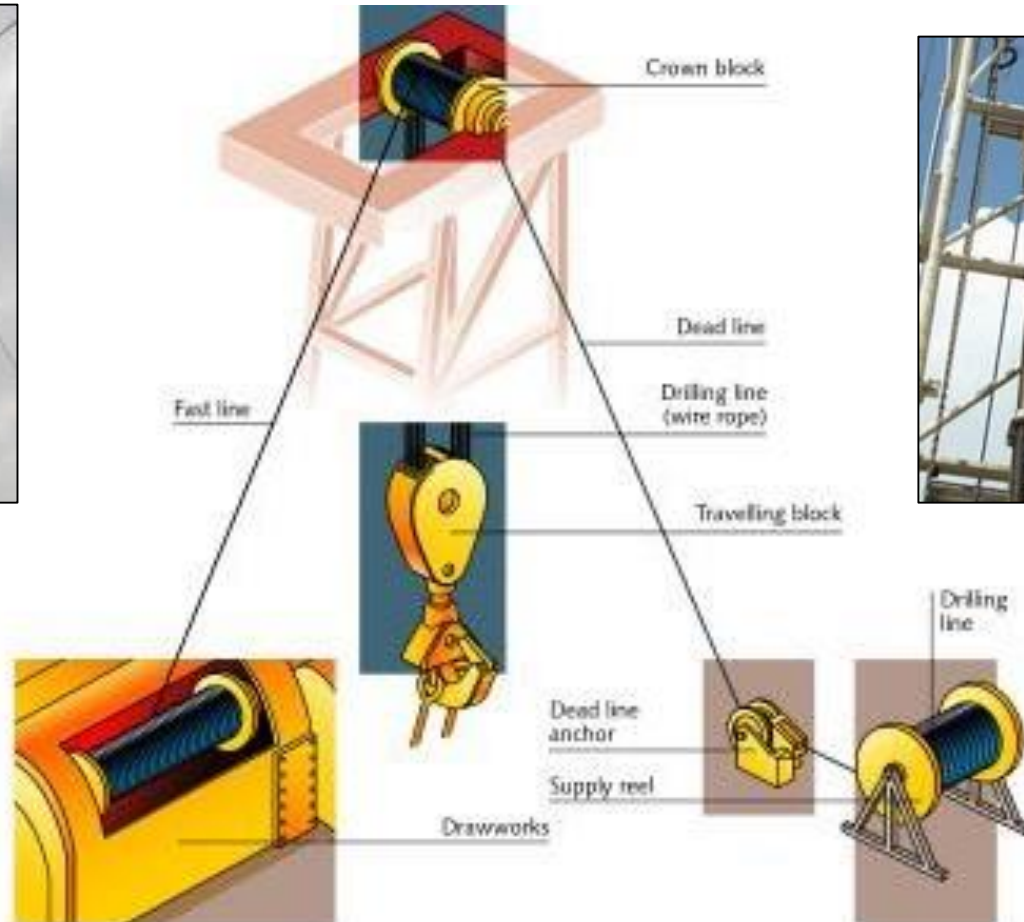
Engines drive the drawworks and the rotary table, which rotates the kelly.



Courtesy of the Petroleum Communication Foundation



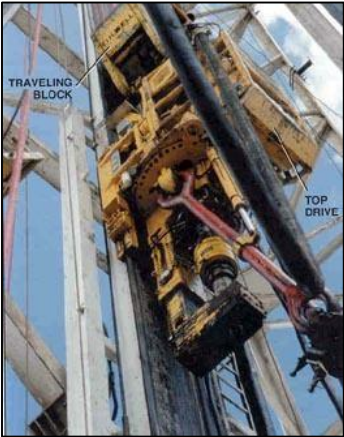
HOISTING SYSTEM



Courtesy of the Petroleum Communication Foundation

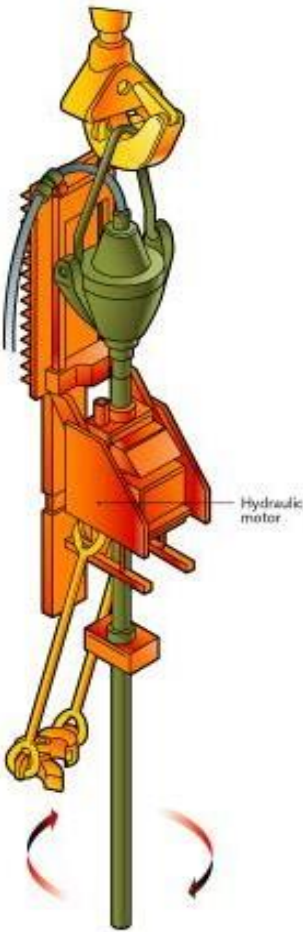
ROTATING SYSTEM

Conventional Kelly system

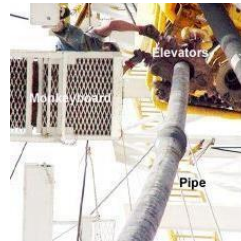
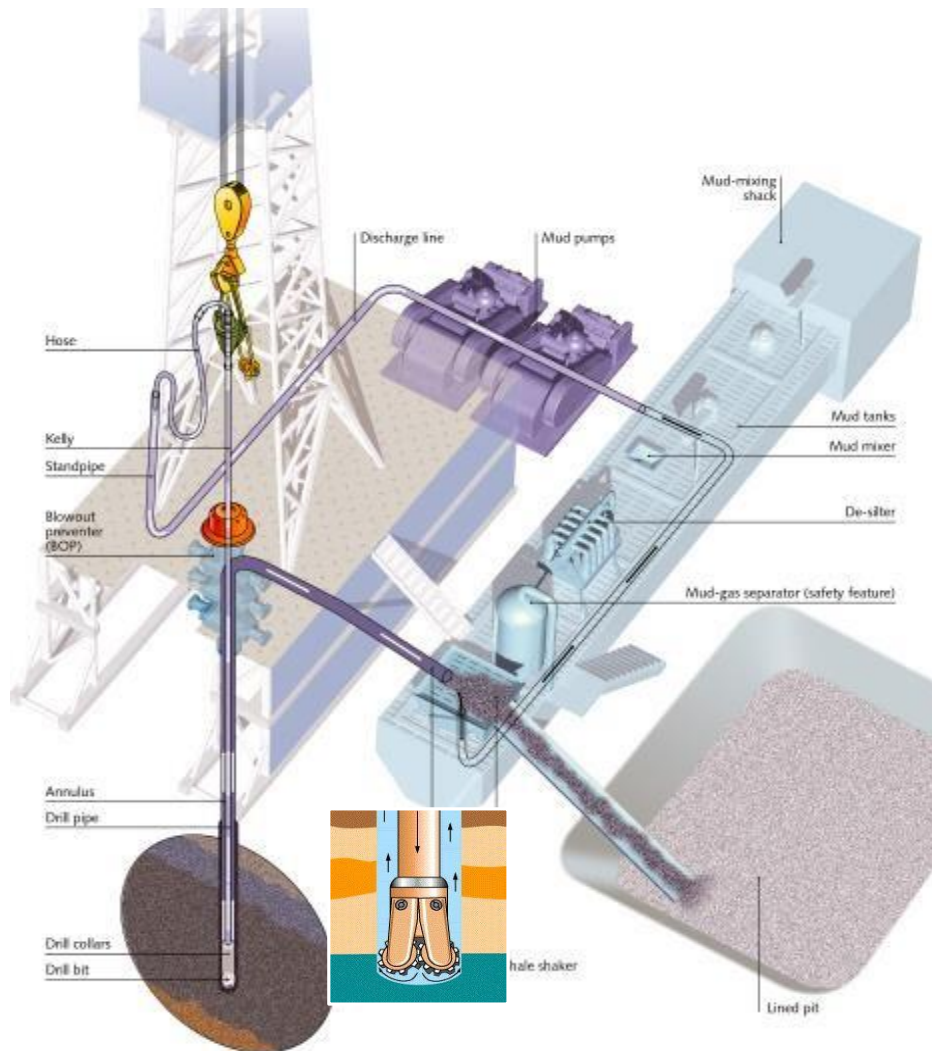


Courtesy of the Petroleum Communication Foundation

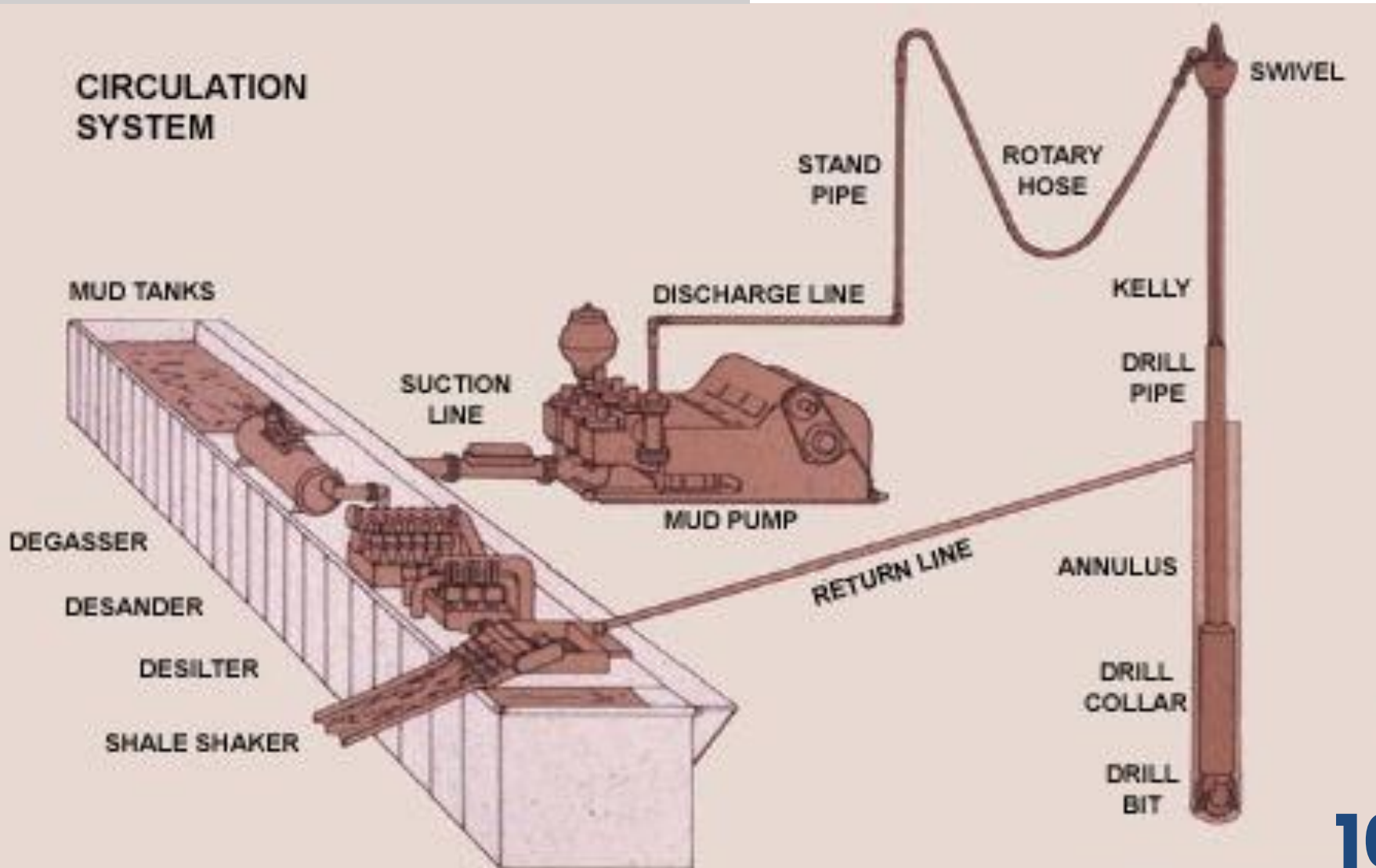
Top Drive System



CIRCULATING SYSTEM



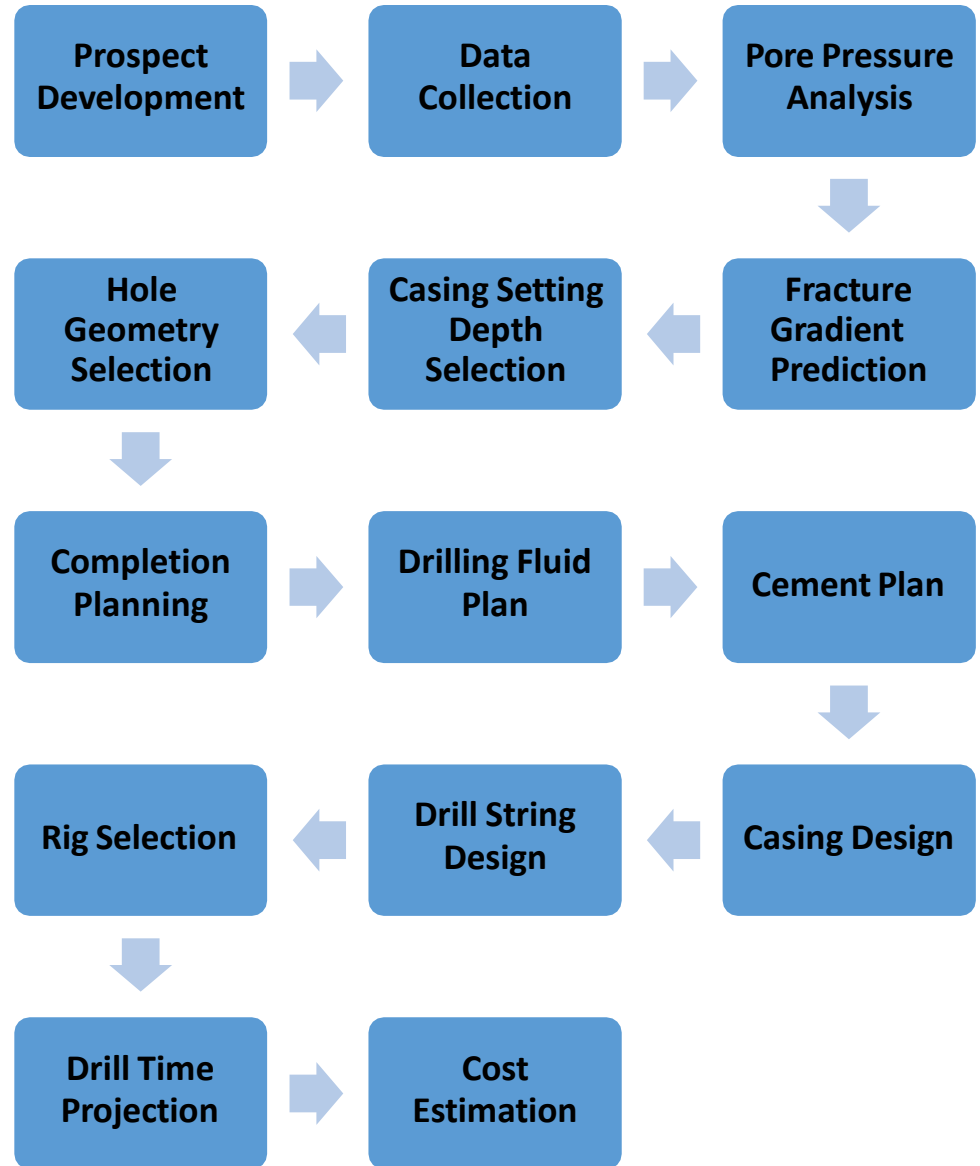
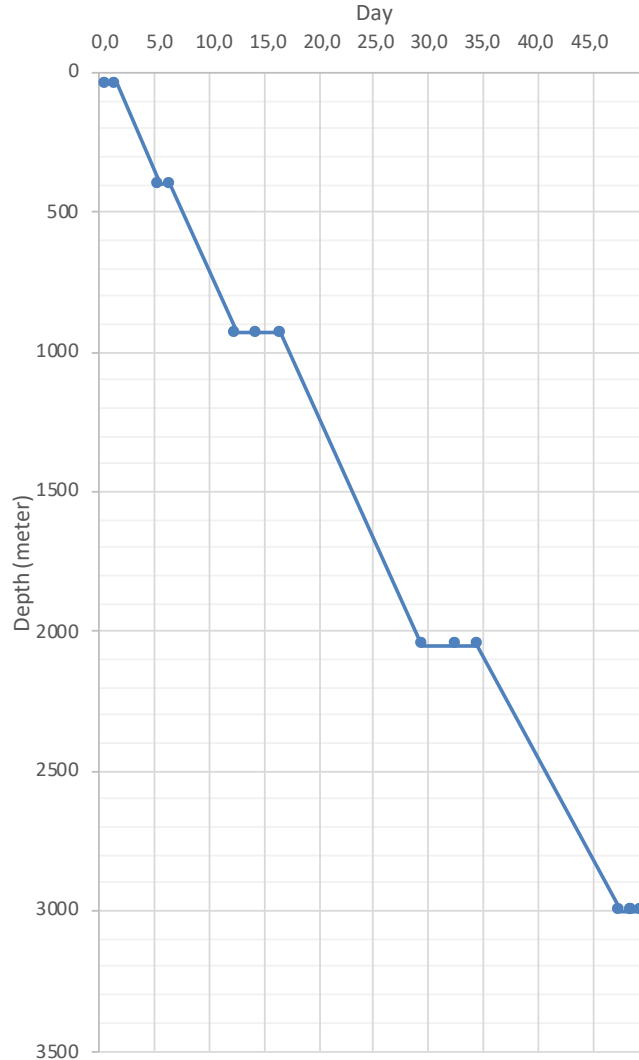
CIRCULATION SYSTEM



WELL CONTROL SYSTEM



Well Planning Workflow



Time Considerations

Drill Rate

Trip Time

Hole Problems

Casing Running

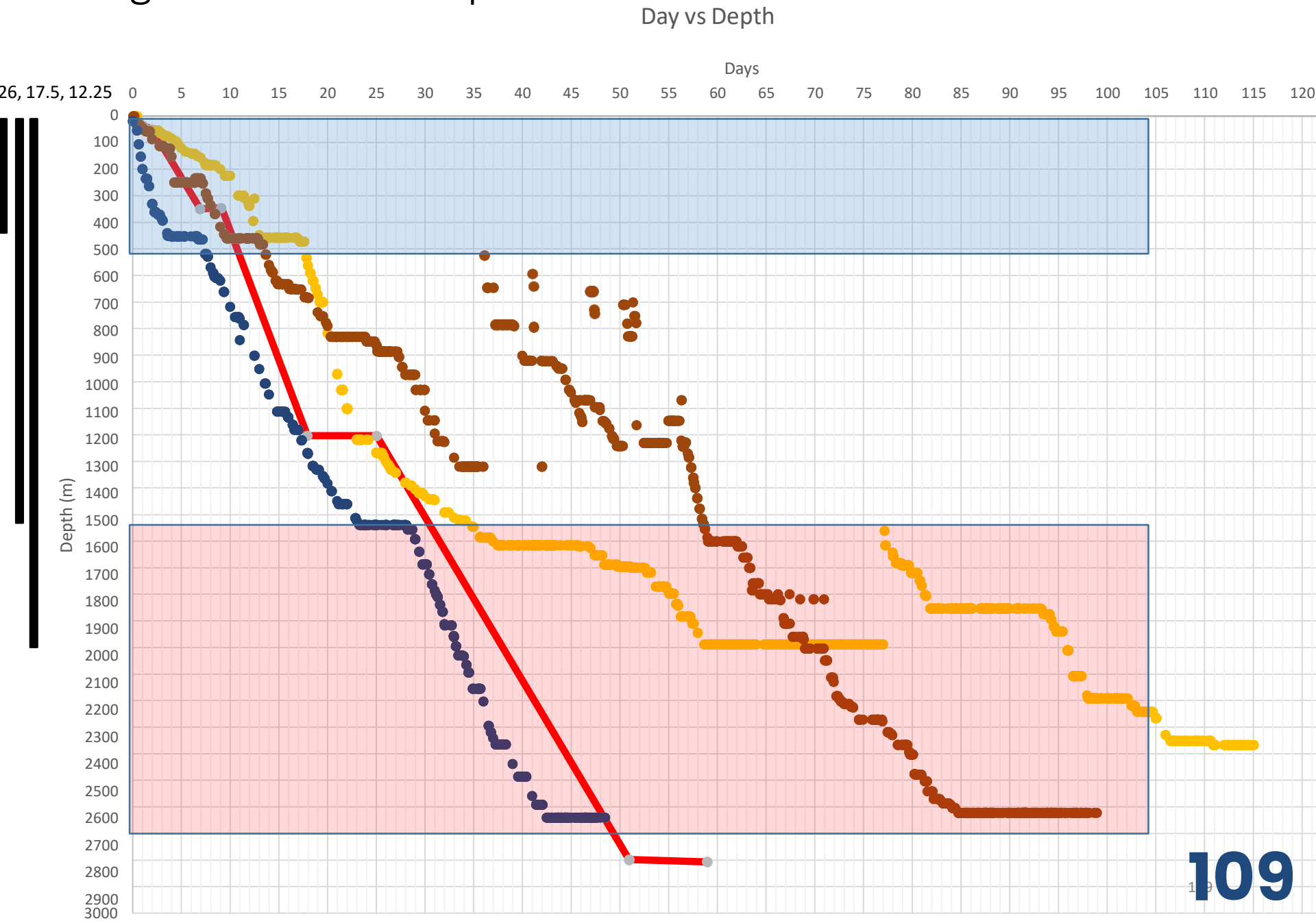
Directional Drilling

Completion Type

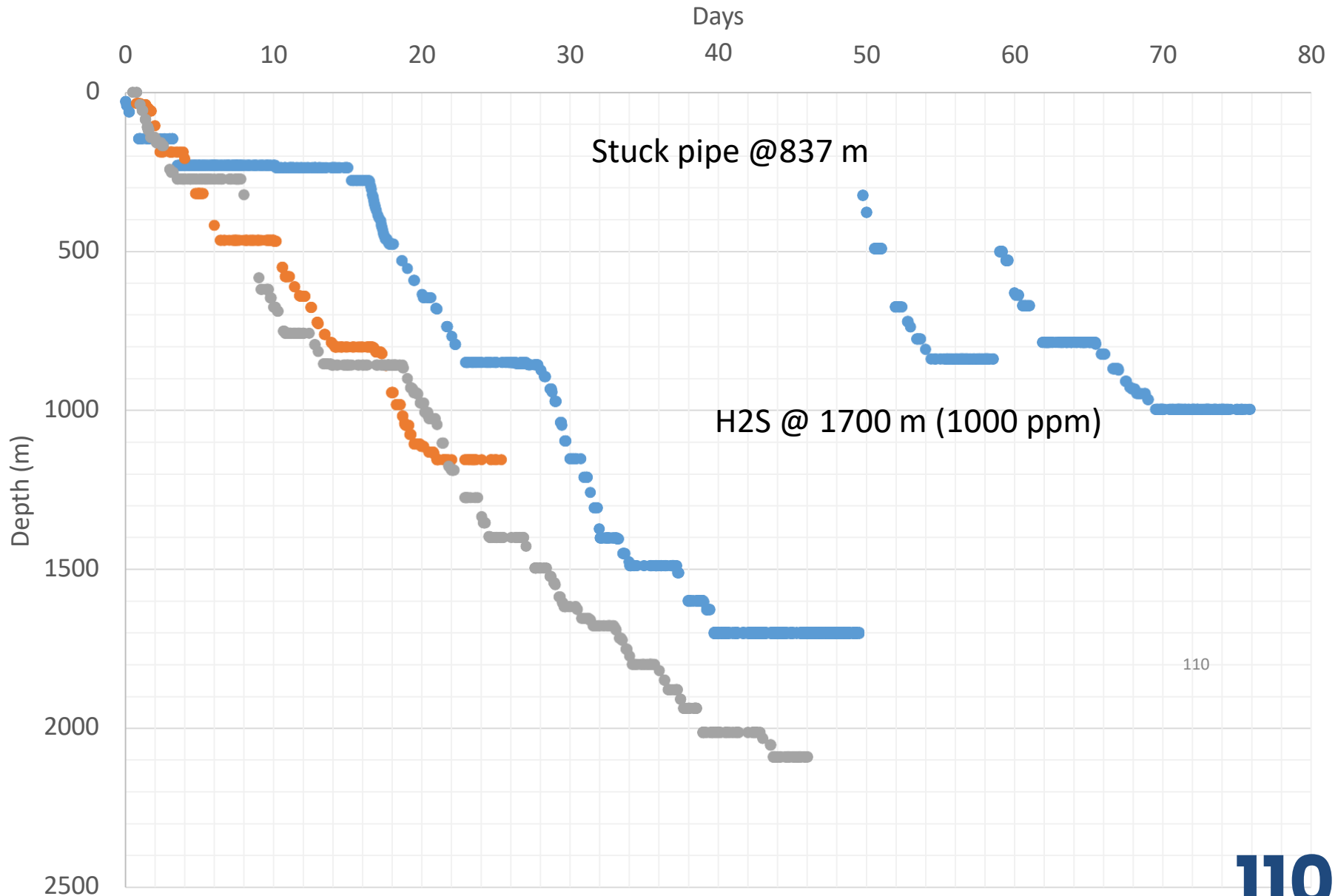
Move-in/out Rig

Weather

Drilling Activities: Example

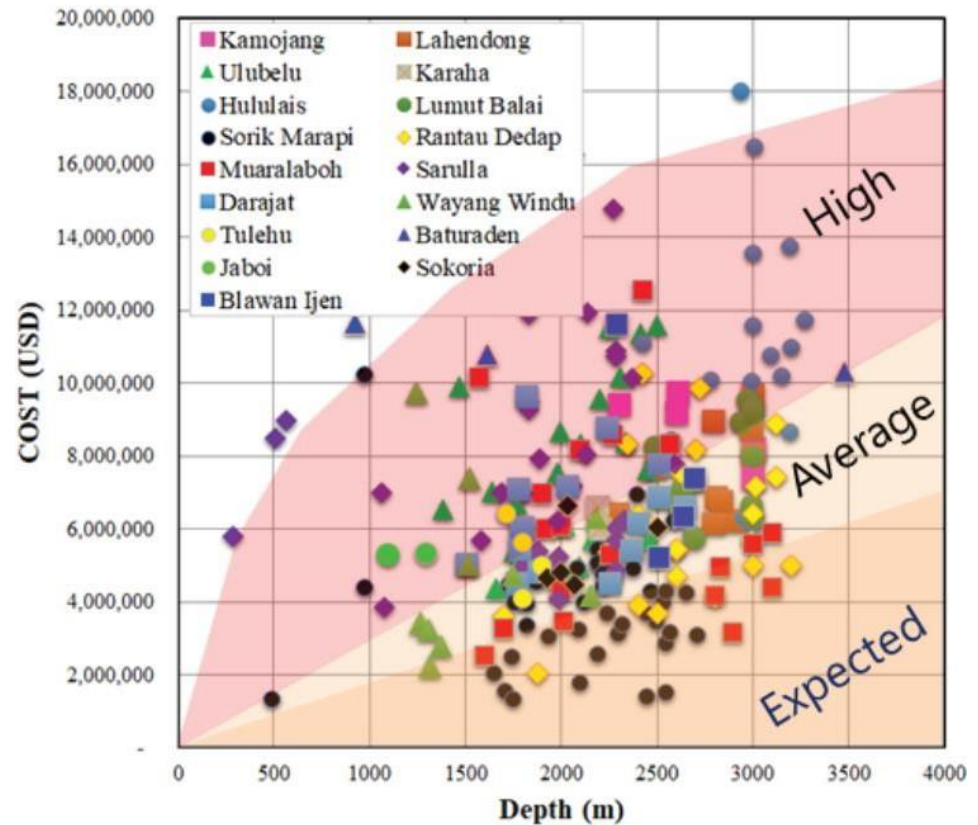


Day vs Depth (Example 2)



Biaya Pengeboran

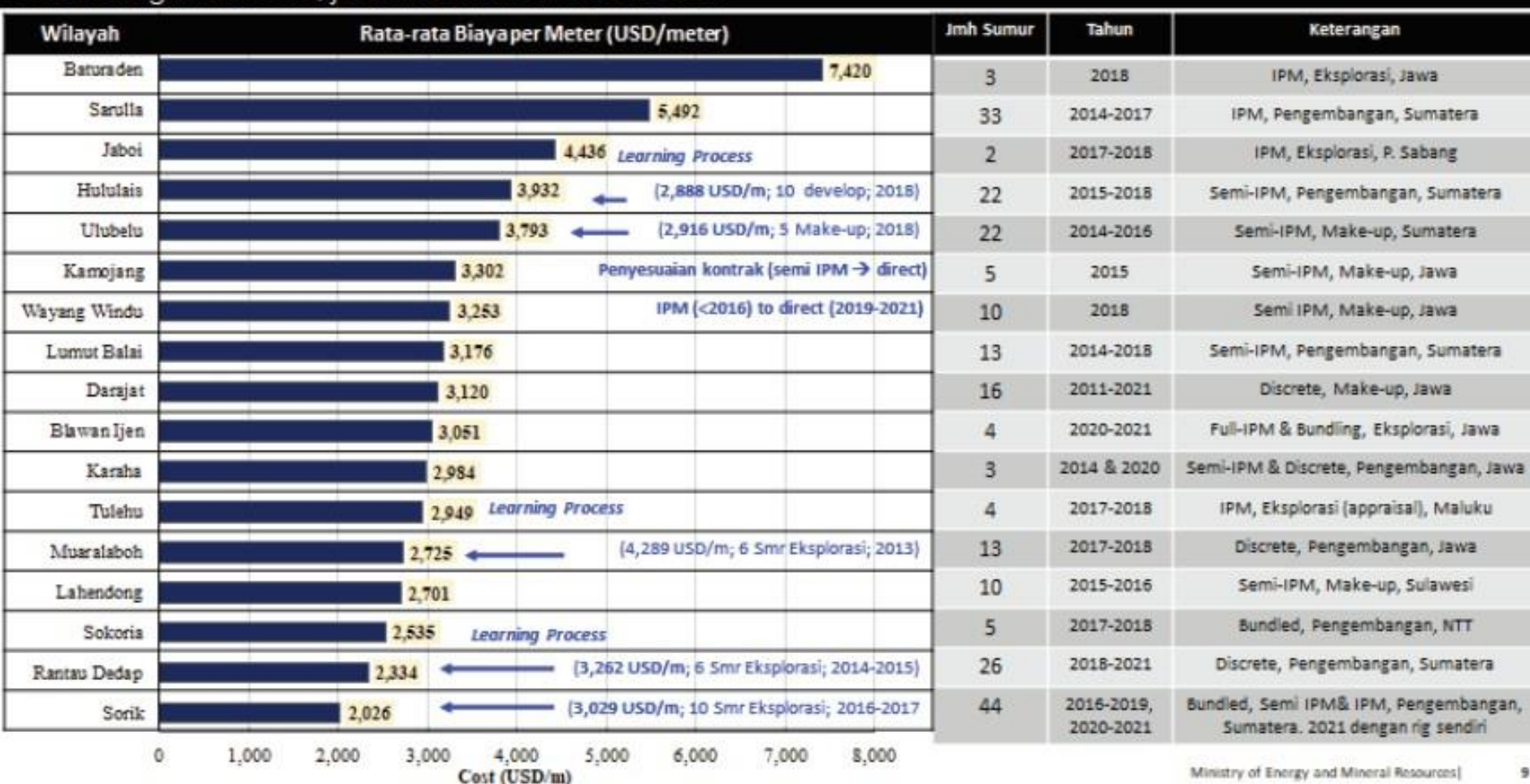
Cost Component	Average %
CONTRACT RIG	32.08%
CASING & INSTALLATION	15.04%
DIRECTIONAL DRILLING	9.40%
CEMENT & CEMENTING	8.65%
BITS, COREHEADS	5.70%
EQUIPMENT RENTAL	5.63%
MUD, ENGINEERING	4.05%
RIG UP / MOB	3.76%
SURFACE EQUIPMENT	2.90%
AIR DRILLING	2.58%
OTHERS	9.77%



Data Pengeboran Sumur Geothermal di Indonesia 2011-2021
Source: KESDM 2023

PERBANDINGAN BIAYA PENGEBORAN PANAS BUMI ANTAR LAPANGAN DI INDONESIA

Perbandingan unit cost, jumlah sumur dan kontraktual



TREND BIAYA PENGEBORAN PANAS BUMI (2011-2021)

Perbandingan fluktuasi biaya pengeboran panas bumi



Tren tahunan biaya pengeboran

Evaluasi unit cost menunjukkan trend :

- Komponen biaya rig pengeboran mengalami penurunan akibat trend penurunan komoditas harga minyak dan berkurangnya frekuensi proyek pengeboran (migas & geothermal). Trend harga minyak turun dari 90-110 USD/bbl (2011-2012) menjadi 50-70 USD/bbl (2017-2019).
- Selain itu, seiring semakin banyaknya tenaga kerja di bidang panas bumi, membuat *developer* dapat merekrut tenaga kerja pada perusahaan masing-masing, dan beralih dari skema kontrak pengeboran Full IPM menjadi Semi-IPM, *bundle* dan *discrete* yang relatif lebih murah serta penggunaan rig sendiri.
- Dengan semakin banyaknya tenaga kerja dan rig pengeboran di bidang panas bumi, proyek pengeboran panas bumi akan semakin terdorong untuk lebih *cost-effective*.
- Kementerian ESDM mendorong biaya pengeboran panas bumi semakin murah (<2000 USD/m)

Trend Harga Minyak Indonesia



Ministry of Energy and Mineral Resources

PERBANDINGAN KONTRAK PENGEBORAN PANAS BUMI (STUDI 2018)

Skema komersial/kontraktual pelaksanaan pengeboran panas bumi



Evaluasi skema kontraktual

- Pengeboran tahun 2012-2016 mayoritas menggunakan skema semi IPM dan full IPM pada kedalaman rata-rata 2180 m.
- Pengeboran tahun 2017-2019 mayoritas menggunakan skema *discrete* dan *bundled* pada kedalaman rata-rata 2290 m.
- Kontrak *discrete* dan *bundled* relatif lebih murah pada kisaran kedalaman yang hampir sama dibandingkan kontrak Semi IPM dan IPM.

Tren tahunan biaya pengeboran

- Baturaden (2018), Sarulla (2014-2016) dan Jaboi dibor hingga kedalaman rata-rata 1700-1800 m dengan menggunakan skema kontrak full IPM.
- Rantau Dedap, Sokoria, Lahendong, Muaralaboh, dan Sorik Marapi dibor hingga kedalaman rata-rata 2100 - 2300 m mayoritas dengan menggunakan *discrete* dan *bundled* kontrak. Sorik Marapi tahun 2021 dibor dengan menggunakan rig sendiri, sehingga menurunkan biaya pengeboran.
- Faktor-faktor operasi seperti gangguan pelaksanaan pengeboran yang spesifik dan berbeda pada masing-masing lokasi (formasi batuan yang dijumpai, *loss circulation*, *shallow gas*, jepitan, dll).

Drilling Contract Comparisons

Description	Discrete	ISM	SIPM	IPM
Operator Risk	Very Low	Low	Medium	High
Project Manager	Company Rep	Company Rep	Contractor Rep	Contractor Rep
Well Cost	5-6 M	4-5 M	7-8 M	7-10 M
Resources	>10 persons	3-8 persons	3-5 persons	2-3 persons
No. of Contract	30-50	<10	2-5	1
Dispute Risk	Very Low	Low	Medium	High
Procurement Time				
Materials	6-9 months	6-9 months	6-12 months	6-12 months
Services	9-12 months	5-8 months	9-18 months	9-15 months
Interface	>15 Reps	3-10 Reps	3-5 Reps	1 Rep
Penalty	Individual	Individual	All Services	All Services
Excess Material	Yes	Yes	Yes	None
Master List Facility	Company to Provide	Company to Provide	Company to Provide	None
Tax over Tax	None	None - Minimum	Sub-Contractors to SIPM	Sub-Contractors to IPM
Double Profit	None	None - Minimum	Sub-Contractors to SIPM	Sub-Contractors to IPM
Mob/Demob	Individual	Individual - Share	Individual - Share	Share
Lesson Learnt	Minimum	Available	Available	Available
Vendor List	Lot	Lot	Adequate	Adequate
Charging Method	Direct to Company	Direct to Company	Direct to Company	To IPM Leader - To Company

Purchase	Services	Support
1. Casing 2. Wellhead 3. Liner Adapter 4. Drilling Bits	1. Drilling Rig 2. Cement and Mud 3. Directional Drilling 4. Air Drilling 5. Mud Coller 6. Wellhead Installation 7. Drilling Personnel 8. Heavy Equipment 9. Drilling Waste Handling 10. VSAT 11. Drilling Report 12. Casing Handling 13. H2S Service 14. Mud Logging 15. Drill String Inspection 16. Hard Bending 17. Solid Control Services 18. Fishing Tool Services 19. Coring Services 20. E-Line & Explosive	1. Diesel Fuel 2. Water Pump Services 3. UKL / UPL 4. Security Services 5. Road Survey 6. Civil work 7. Tubular Inspection 8. Explosive Handling 9. Permits 10. Perforated Liner 11. Logistic Services 12. General Services



Drilling Organization

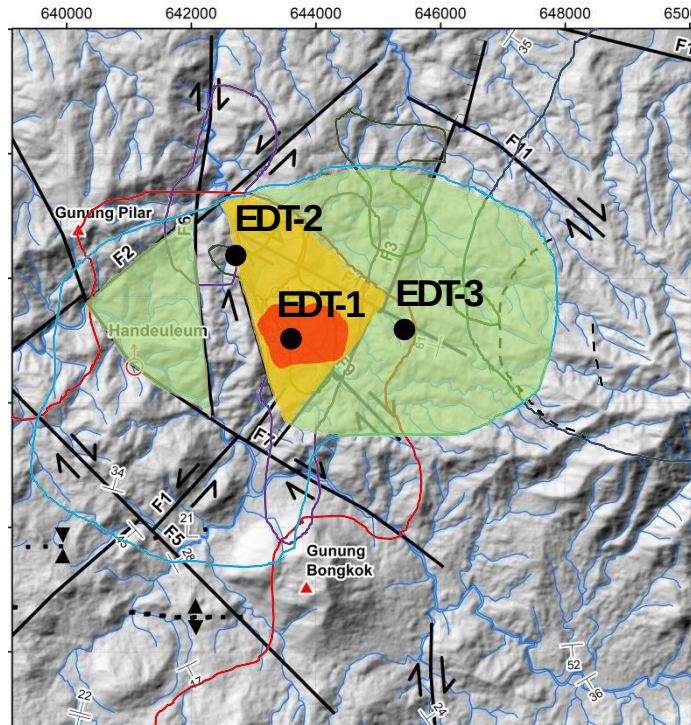


Drilling Cost Example

		Depth	3000
		Days	65
1	Location	\$	400,000
2	Casing	\$	414,972
3	Liner Hanger	\$	109,536
4	Bits	\$	638,500
5	Wellhead	\$	130,000
6	Drilling Rig	\$	1,625,000
7	Cementing Service	\$	400,000
8	Mud Service	\$	400,000
9	Directional Drilling Service	\$	590,000
10	Rig Mob	\$	250,000
11	Fuel	\$	357,000
12	Top Drive	\$	312,000
13	Air Drilling	\$	240,000
14	Solid Removal Equipment	\$	203,000
15	Mud Logging Unit	\$	180,000
16	H2S Services	\$	130,000
17	Engineering	\$	130,000
18	Drillstring and Rig Inspection	\$	100,000
19	Other Equipment and Services	\$	412,000
	Total	\$	7,022,008

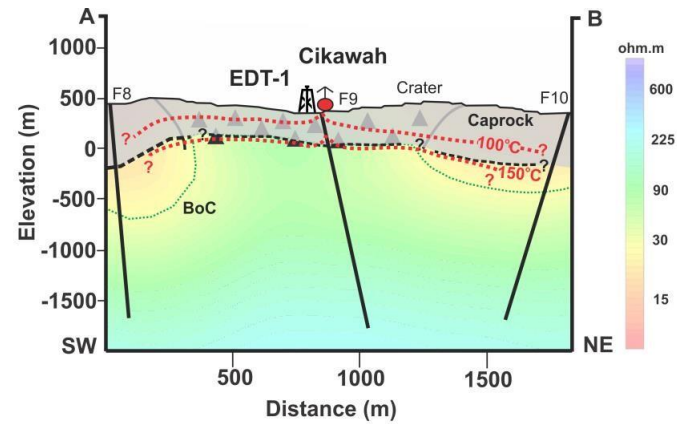
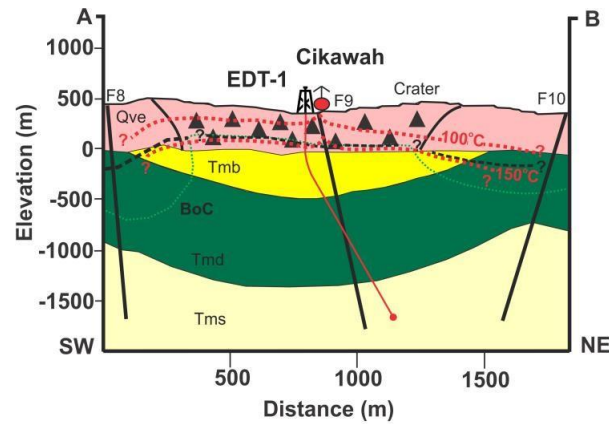
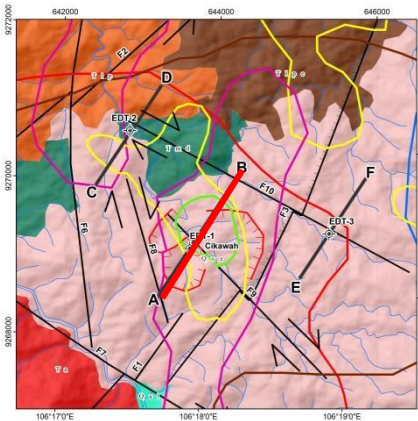
Example of Well Design Project

WELL TARGETING – Well Location

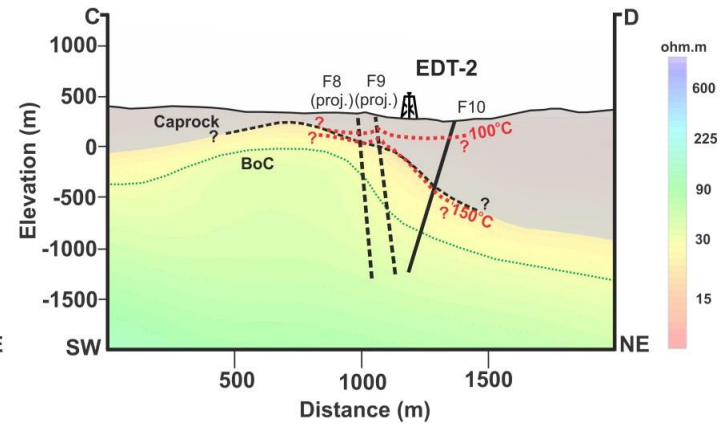
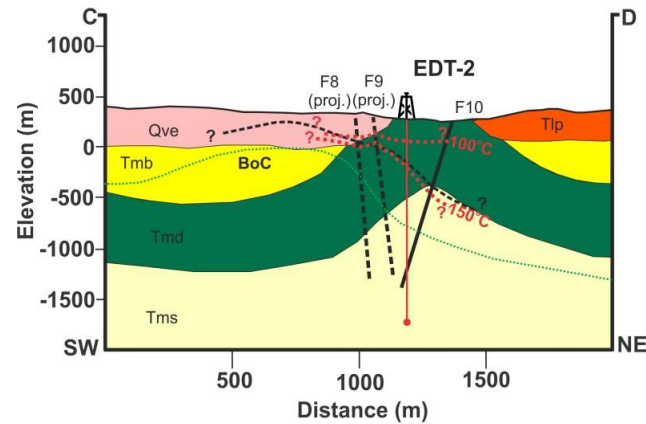
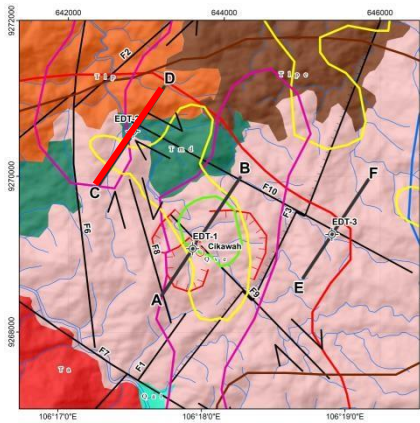


- — — — — Faults (Gravity)
- Hg
- CO₂
- High Gravity
- Low Gravity
- Low Magnetic
- Low Magnetotelluric
- P10 (Probable) = 1.2 km²
- P50 (Possible) = 4.8 km²
- P90 (Plausible) = 23.4 km²
- Exploration Well (proposed)

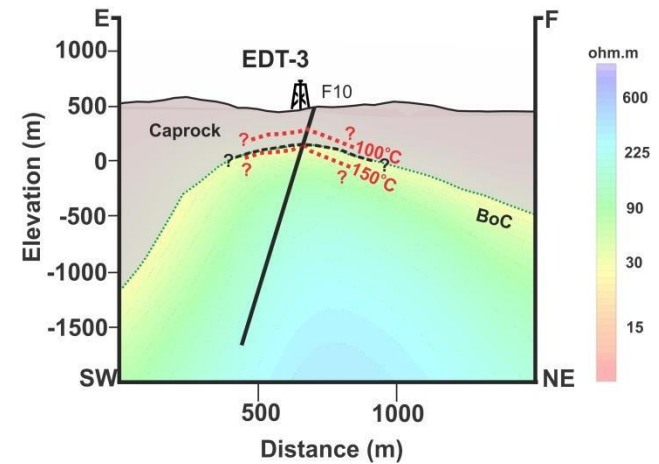
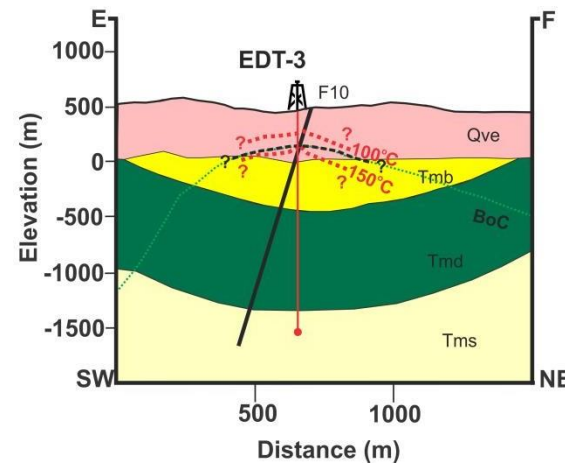
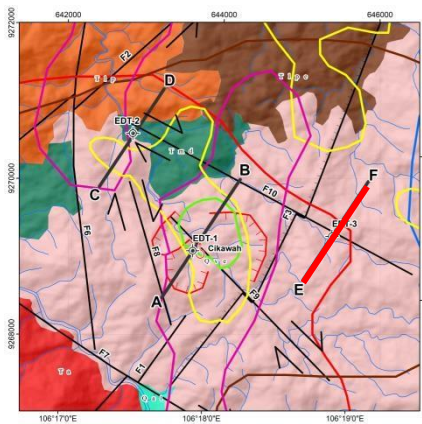
WELL TARGETING – Well Location EDT-1



WELL TARGETING – Well Location EDT-2

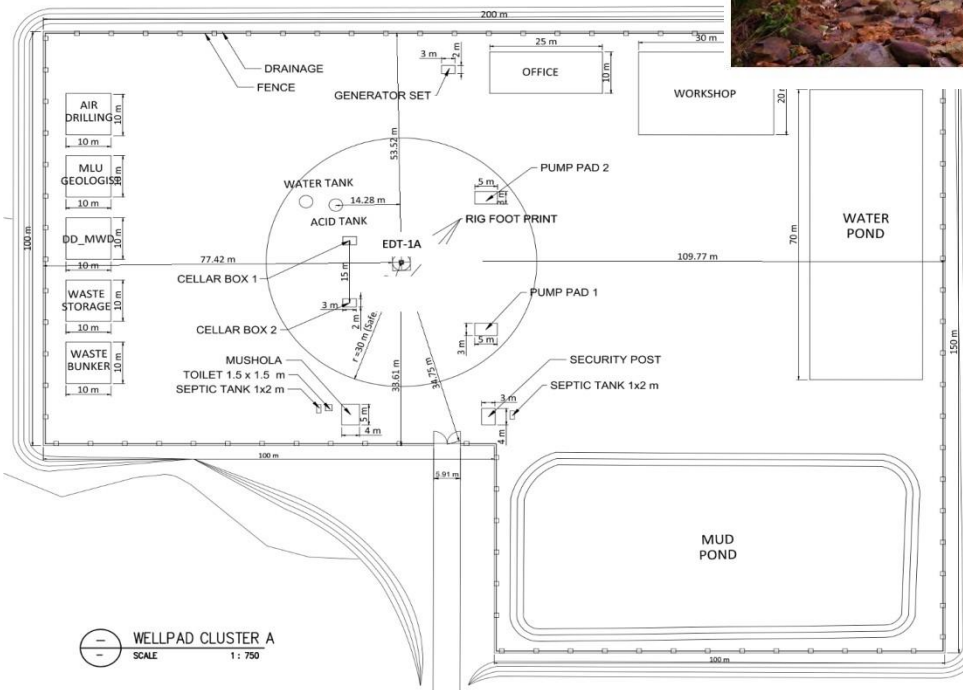


WELL TARGETING – Well Location EDT-3



BASIC WELL DESIGN

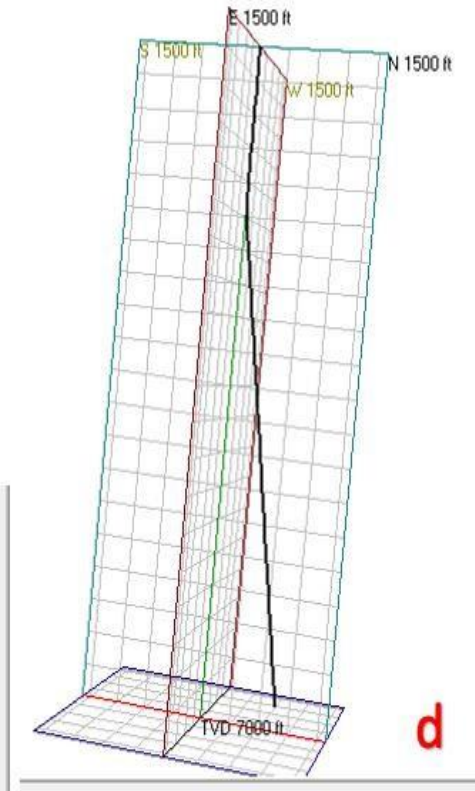
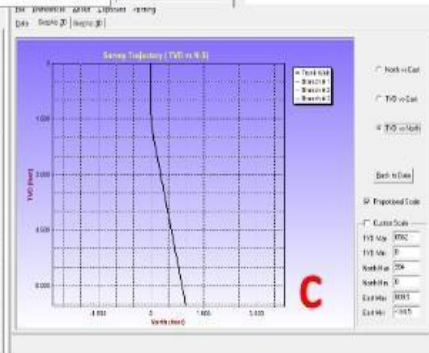
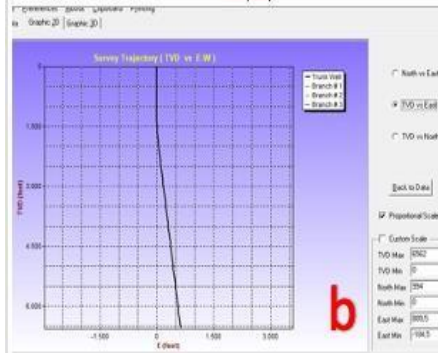
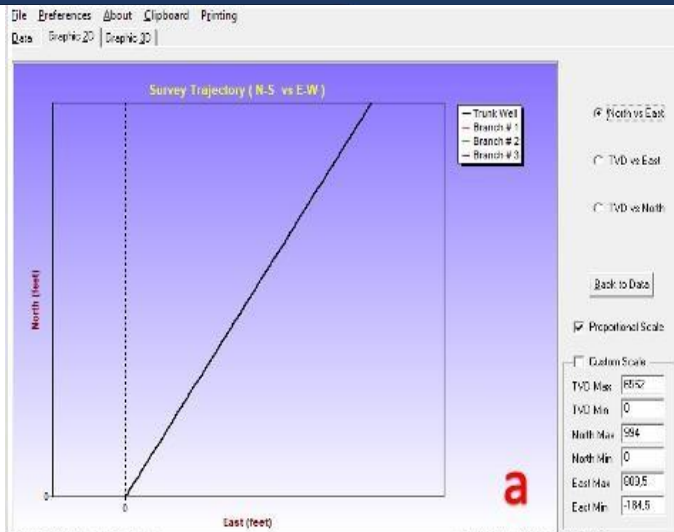
BASIC WELL DESIGN– Wellpad Preparation EDT-1



BASIC WELL DESIGN– Parameters

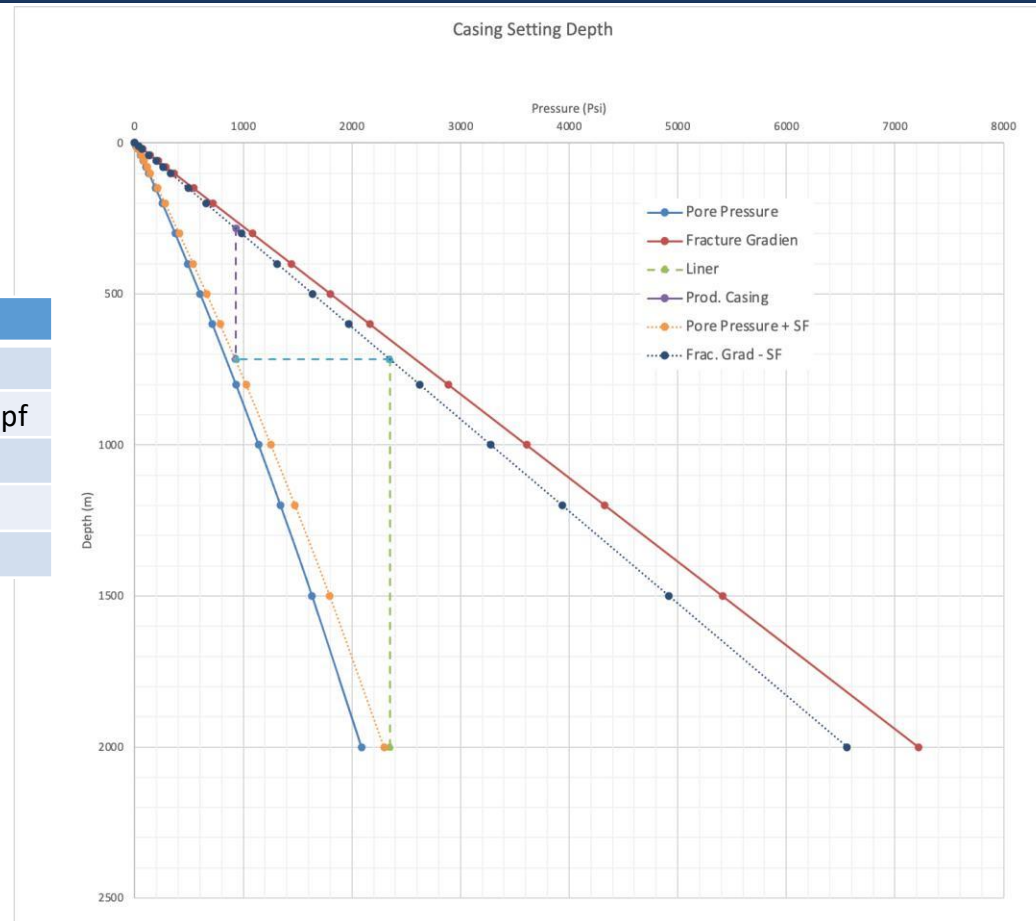
Parameter	EDT-1	EDT-2	EDT-3
Well Coordinate:			
mN			
mE			
Elevation (masl)	365	267	477
Azimuth (°)	32.2	vertical	vertical
Kick-off Point (mMD)	400	0	0
Inclination (°)	13.4	0	0
Buildup Rate (°/100 ft)	2	0	0
Horizontal Offset (m)	358	0	0
Target Total Depth			
Measured (mMD)	2041	2000	2000
Vertical (mTVD)	2000	2000	2000

BASIC WELL DESIGN– Well Trajectory



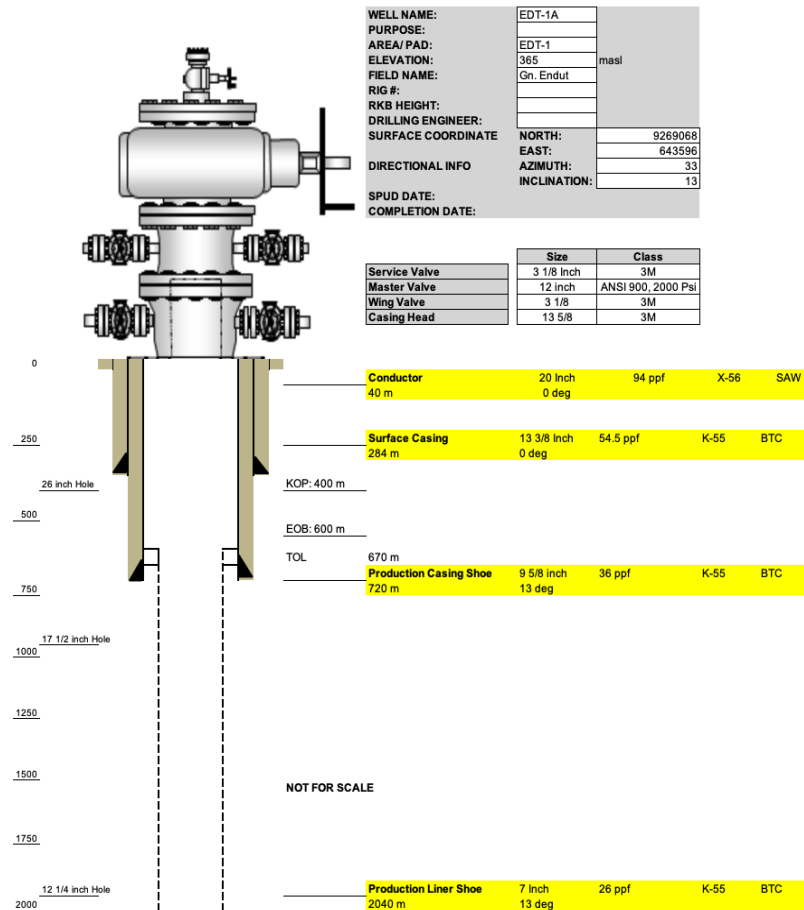
BASIC WELL DESIGN– Casing Setting Depth

Well	Casing Depth (mMD)		
	13 3/8"	9 5/8"	7"
	K-55, 54.5 ppf	K-55, 36 ppf	K-55, 26 ppf
EDT-1	284	720	2040
EDT-2	284	716	2000
EDT-3	284	716	2000



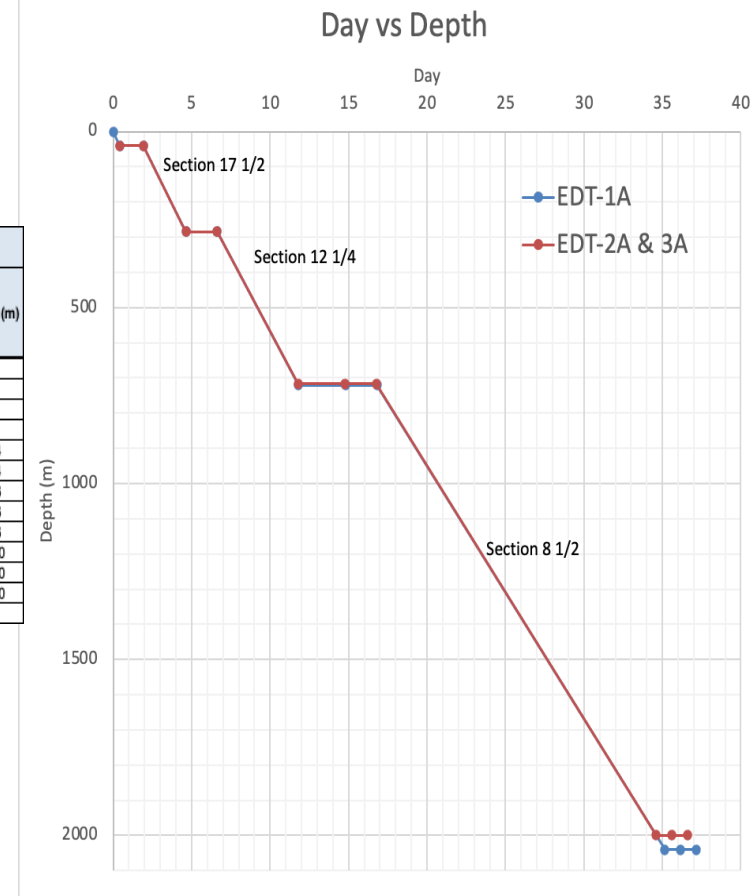
BASIC WELL DESIGN– Casing Configuration

Well EDT-1



BASIC WELL DESIGN– Time Estimation

	Well Name			EDT-1A 2040				EDT-2A 2000				EDT-3A 2000			
	Measured Depth (m)			Hole Section Length (m)	Days	Cum. Day	Depth (m)	Hole Section Length (m)	Days	Cum. Day	Depth (m)	Hole Section Length (m)	Days	Cum. Day	Depth (m)
Operating Activities	Dia.	ROP (m/hr)													
No.	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Drill Surface Hole	26	4	40	0.4	0.4	40	40	0.4	0.4	40	40	0.4	0.4	40
2	Run/Cmnt Surface Casing	20			1.5	1.9	40		1.5	1.9	40		1.5	1.9	40
3	Drilling Anchor Hole	17 1/2	3.75	244	2.7	4.6	284	244	2.7	4.6	284	244	2.7	4.6	284
4	Run/Cmnt Anchor Casing	13 3/8			2.0	6.6	284		2.0	6.6	284		2.0	6.6	284
6	Drill Prod. Hole	12 1/4	3.5	436	5.2	11.8	720	432	5.1	11.8	716	432	5.1	11.8	716
7	Run/Cmnt Prod. Casing	9 5/8			3.0	14.8	720		3.0	14.8	716		3.0	14.8	716
8	Run/Cmnt Prod. Tie-Back Casing	9 5/8			2.0	16.8	720		2.0	16.8	716		2.0	16.8	716
9	Drill Liner Hole	8 1/2	3	1320	18.3	35.2	2040	1284	17.8	34.6	2000	1284	17.8	34.6	2000
10	Run Perforated Liner	7			1.0	36.2	2040		1.0	35.6	2000		1.0	35.6	2000
11	Well Completion				1	37.2	2040		1	36.6	2000		1	36.6	2000
Total Operating Days				37.2					36.6				36.6		



BASIC WELL DESIGN– Well Prognosis EDT-1

Depth (mVD)	Approx. Formations	Brief Rock Description	Notes for Drilling Risks
0 – 401	Gunung Endut Andesitic Volcanic Rock (Qve)	This rock unit consists of andesite lava and pyroclastic rock. The lava is andesite, porphyritic with plagioclase, pyroxene, hornblende, biotite, and opaque mineral. The pyroclastic rock is breccia consists of andesite fragments in the ash-lapilli matrix. The alteration in this zone is quartz, pyrite, and silicification.	The andesite lava is hard due to silicification at shallow depth (possibly continuing to the depth).
401 – 859	Bojongmanik Formation (Tmb)	This rock unit consists of limestone and siltstone. The limestone is wackestone, with foraminifera and algae fragments.	Drilling in carbonate formations with karst and open cracks poses a risk because there may be a sudden and large loss of drilling fluid .
859 – 1708	Badui Formation (Tmd)	This rock unit consists of black to gray carboniferous claystone with parallel lamination and mud crack, and reef limestone with foraminifera fossils.	Some clays, such as montmorillonite, tend to swell when exposed to water , creating a potential drilling hazard when clay-bearing rock formations are exposed to water-based fluids during drilling. In addition, it is necessary to be aware of sudden and large loss of drilling fluid if it encounters karstified reef limestone
1708 - 2000	Sareweh Formation (Tms)	This rock unit consists of claystone, siltstone, and sandstone.	Some clays, such as montmorillonite, tend to swell when exposed to water , creating a potential drilling hazard when clay-bearing rock formations are exposed to water-based fluids during drilling.

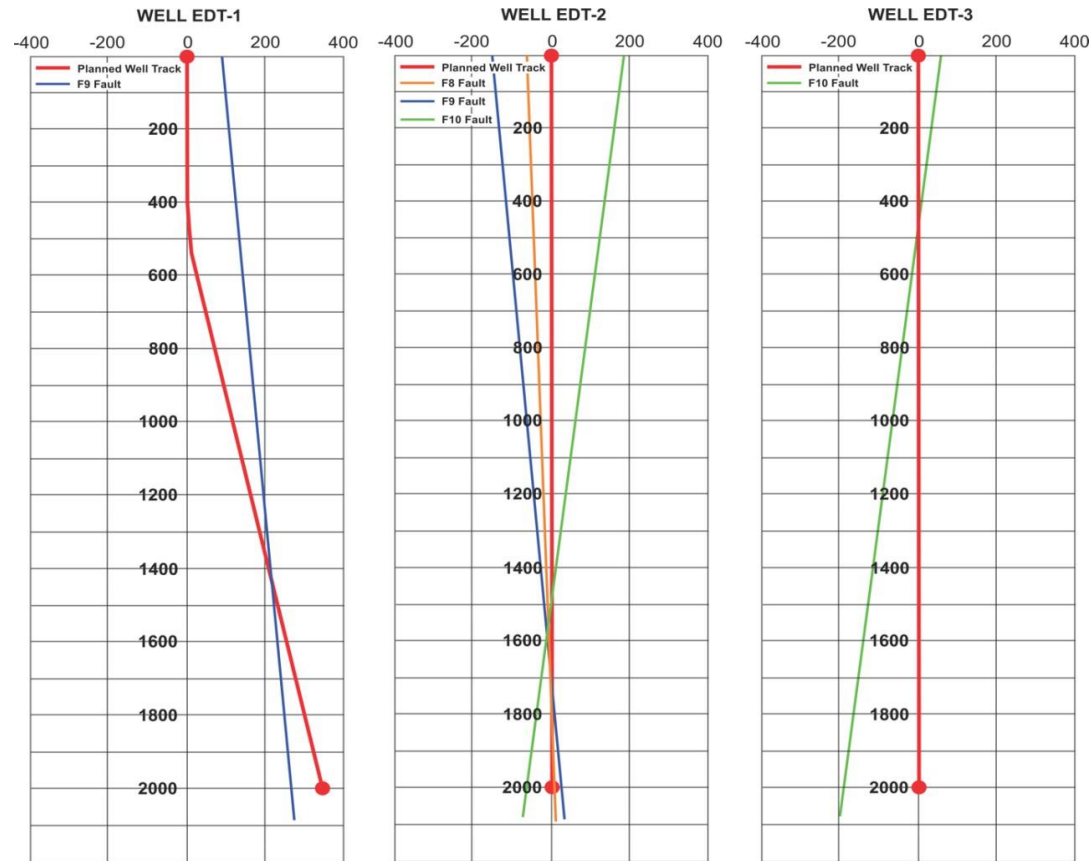
BASIC WELL DESIGN– Well Prognosis EDT-2

Depth (mVD)	Approx. Formations	Brief Rock Description	Notes for Drilling Risks
0 – 796	Badui Formation (Tmd)	This rock unit consists of black to gray carboniferous claystone with parallel lamination and mud crack, and reef limestone with foraminifera fossils.	Some clays, such as montmorillonite, tend to swell when exposed to water , creating a potential drilling hazard when clay-bearing rock formations are exposed to water-based fluids during drilling. In addition, it is necessary to be aware of sudden and large loss of drilling fluid if it encounters karstified reef limestone
796 – 2000	Sareweh Formation (Tms)	This rock unit consists of claystone, siltstone, and sandstone.	Some clays, such as montmorillonite, tend to swell when exposed to water , creating a potential drilling hazard when clay-bearing rock formations are exposed to water-based fluids during drilling.

BASIC WELL DESIGN– Well Prognosis EDT-3

Depth (mVD)	Approx. Formations	Brief Rock Description	Notes for Drilling Risks
0 – 467	Gunung Endut Andesitic Volcanic Rock (Qve)	This rock unit consists of andesite lava and pyroclastic rock. The lava is andesite, porphyritic with plagioclase, pyroxene, hornblende, biotite, and opaque mineral. The pyroclastic rock is breccia consists of andesite fragments in the ash-lapilli matrix. The alteration in this zone is clay minerals: kaolinite, dickite, montmorillonite, quartz, and opal.	The surface lithology in this area is pyroclastic breccia with clay alteration (montmorillonite) in the surface. The present of montmorillonite should be consideration due to swelling potential .
467 – 914	Bojongmanik Formation (Tmb)	This rock unit consists of limestone and siltstone. The limestone is wackestone, with foraminifera and algae fragments.	Drilling in carbonate formations with karst and open cracks poses a risk because there may be a sudden and large loss of drilling fluid .
914 – 1836	Badui Formation (Tmd)	This rock unit consists of black to gray carboniferous claystone with parallel lamination and mud crack, and reef limestone with foraminifera fossils.	Some clays, such as montmorillonite, tend to swell when exposed to water , creating a potential drilling hazard when clay-bearing rock formations are exposed to water-based fluids during drilling. In addition, it is necessary to be aware of sudden and large loss of drilling fluid if it encounters karstified reef limestone
1836 - 2000	Sareweh Formation (Tms)	This rock unit consists of claystone, siltstone, and sandstone.	Some clays, such as montmorillonite, tend to swell when exposed to water , creating a potential drilling hazard when clay-bearing rock formations are exposed to water-based fluids during drilling.

BASIC WELL DESIGN– Fault Intersection



Well	Fault	Dip	Depth of Intersection	Confidence
EDT-1	F9	85° NE	1425 mVD	High confidence to be intersected
EDT-2	F10	83° SW	1480 mVD	High confidence to be intersected
	F9	85° NE	1720 mVD	Low confidence to be intersected
	F8	88° NE	1800 mVD	Medium confidence to be intersected
EDT-3	F10	83° NE	460 mVD	High confidence to be intersected

Geothermal Drilling & Well Problems

Natural Conditions of Geothermal

Hard &
Abrasive

Highly
Fracture zone

Abnormal
Pressure

Source: KS Orka

**LOSS OF
CIRCULATION**

PIPE STICKING

HOLE DEVIATION

**DRILL PIPE
FAILURES**

**BOREHOLE
INSTABILITY**

**MUD
CONTAMINATION**

**FORMATION
DAMAGE**

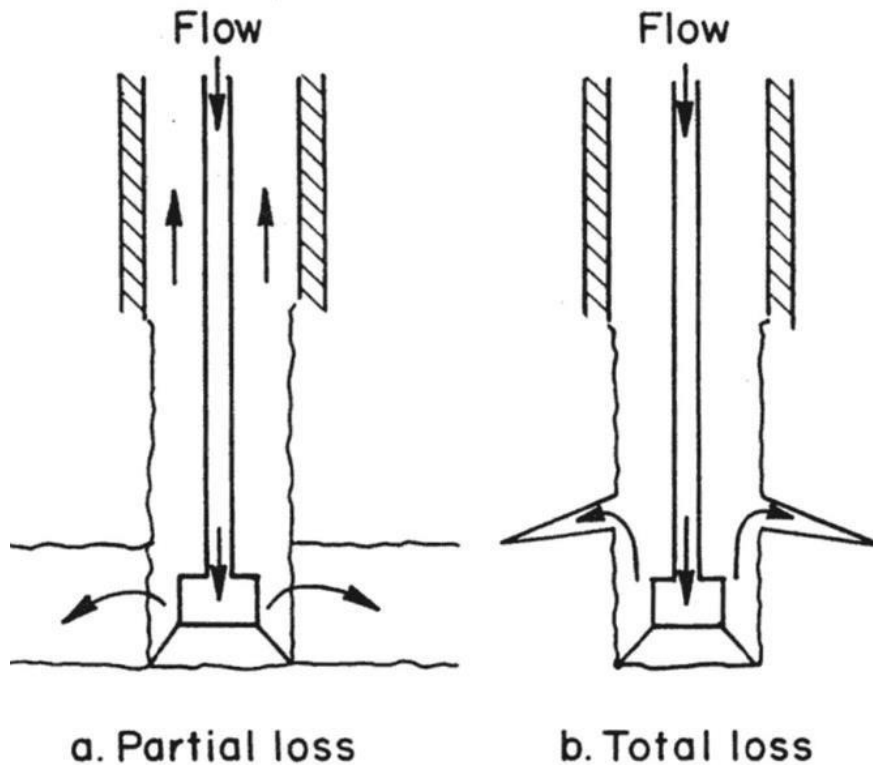
HOLE CLEANING

**H₂S AND
SHALLOW GAS**

**EQUIPMENT &
PERSONNEL
PROBLEMS**

Loss of Circulation

Finger & Blankenship (2010)



- Stuck Pipe
- Well Instability
- Loss of Drilling Fluid
- Well Kick
- Formation Damage
- Difficult Cementing Job

Loss Circulation Handling

1. Blind drilling --> Requires plenty of water, ensure hole cleaning, usually for production zone

2. Lightweight drilling fluid

- Air --> No liquid production
- foam --> To lift heavier cutting
- aerated --> To provide hydrostatic pressure

surfactant added to overcome heat effect

Loss Circulation Handling

3. Loss Circulating material (LCM)

- Fibrous & Flaky --> to catch solid from mud which conducts sealing mechanism. The best option.
- Granular --> seal the tip of fracture to prevent build further
- LCM cannot handle big fracture --> cement plug
- LCM exp: mica flakes, fine rubber curbs, aqueous dispersion rubber

Based on **fracture size**, LoC

- fracture > bit nozzle dia -->

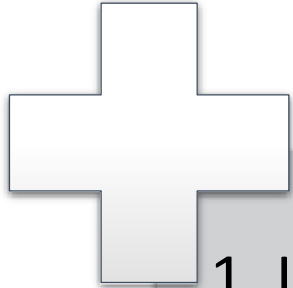
-fracture < bit nozzle dia --> can sealed by LCM

Sandia National Lab developed cementitious mud (bentonite + ammonium polyphosphate, borax, and magnesium oxide) which temperature driven.

Based on **LoC Rate**: (Sidorov, 1985)

- Low (<15 m³/h)
- Medium (< 60 m³/h)
- High (> 60 m³/h)

Drilling with water



- 1. Lower temperature
- 2. High bottom hole pressure → high ROP
- 3. No Mud cake → reduce differential sticking

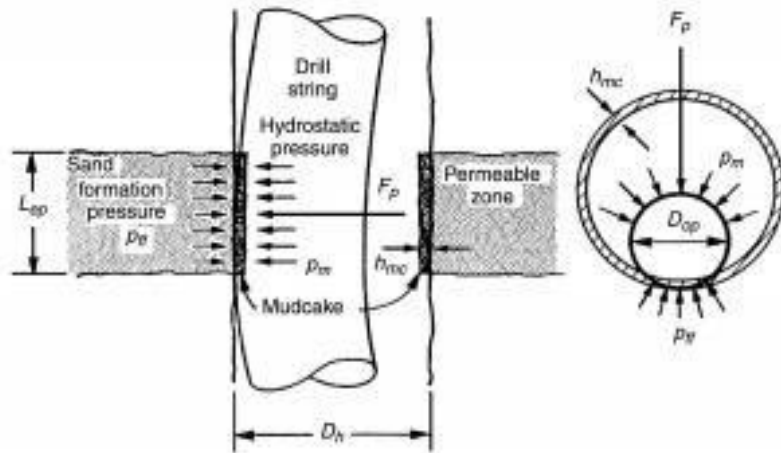
- 1. Large volume water required
- 2. Cannot develop gel strength
- 3. No cutting returned
- 4. Cooling the reservoir

Loss circulation measurement

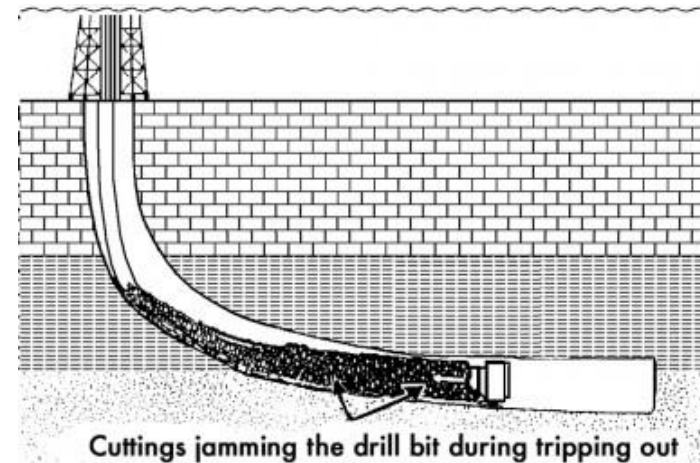
- a) The use of a magnetic, sonic or flapper flow meter on the flow-line: In this method the volumetric flow rate of the fluid being pumped into the well is read from the strokes per minute (SPM) of the mud pumps and the flow out of the well is measured using a flow-meter. The fluid loss is the difference between the total pumping rate and the outflow measured in the flowline.
- b) The tank method: In this method the filling of the tank by make-up water is stopped for a certain interval of time (say 15-30 min.) and then the change in the total mud/water tank volume is measured either by a measuring stick or from the rig instruments. The rate of change in volume (change in volume divided by time) is the fluid loss.
- c) The driller's method: In this method the driller adjusts the pumping rate to keep the well full to the brim after the level has dropped while adding a new drill pipe. The pumping rate (l/s) is then the rate of fluid loss.

PIPE STICKING

Petrowiki.org



Differential-Pressure Sticking

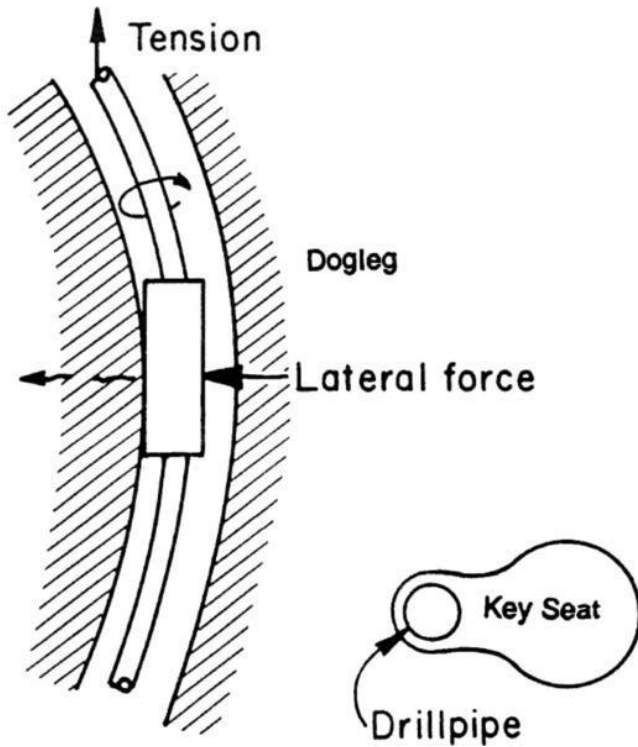


Mechanical: Cuttings Bed

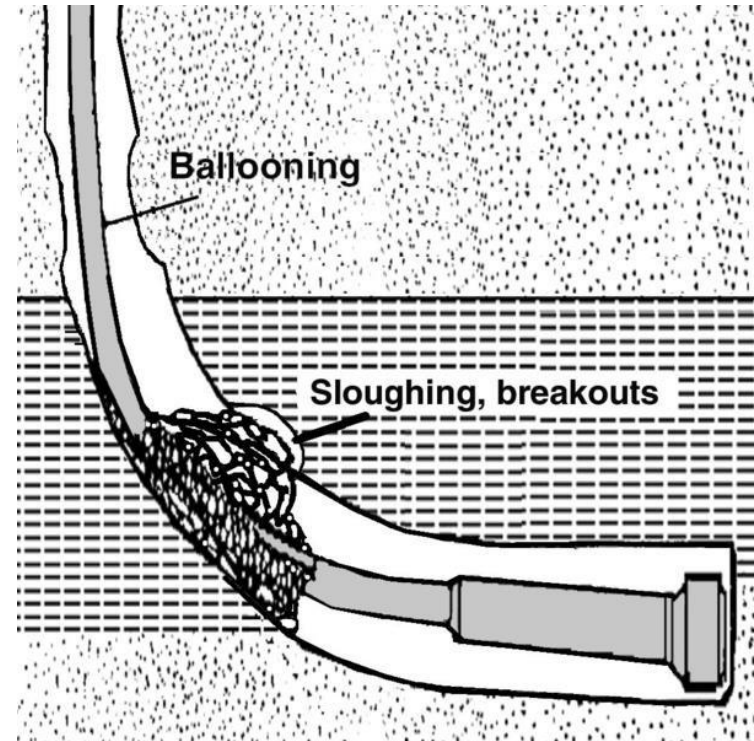
Mechanical Stuck Pipe: Cutting accumulation, formation instability, unconsolidated formation, faulted formation, cement block

Differential pressure sticking: pressure difference formation & wellbore, most of geothermal reservoir are subnormal pressure formation

PIPE STICKING



Mechanical: Key Seating

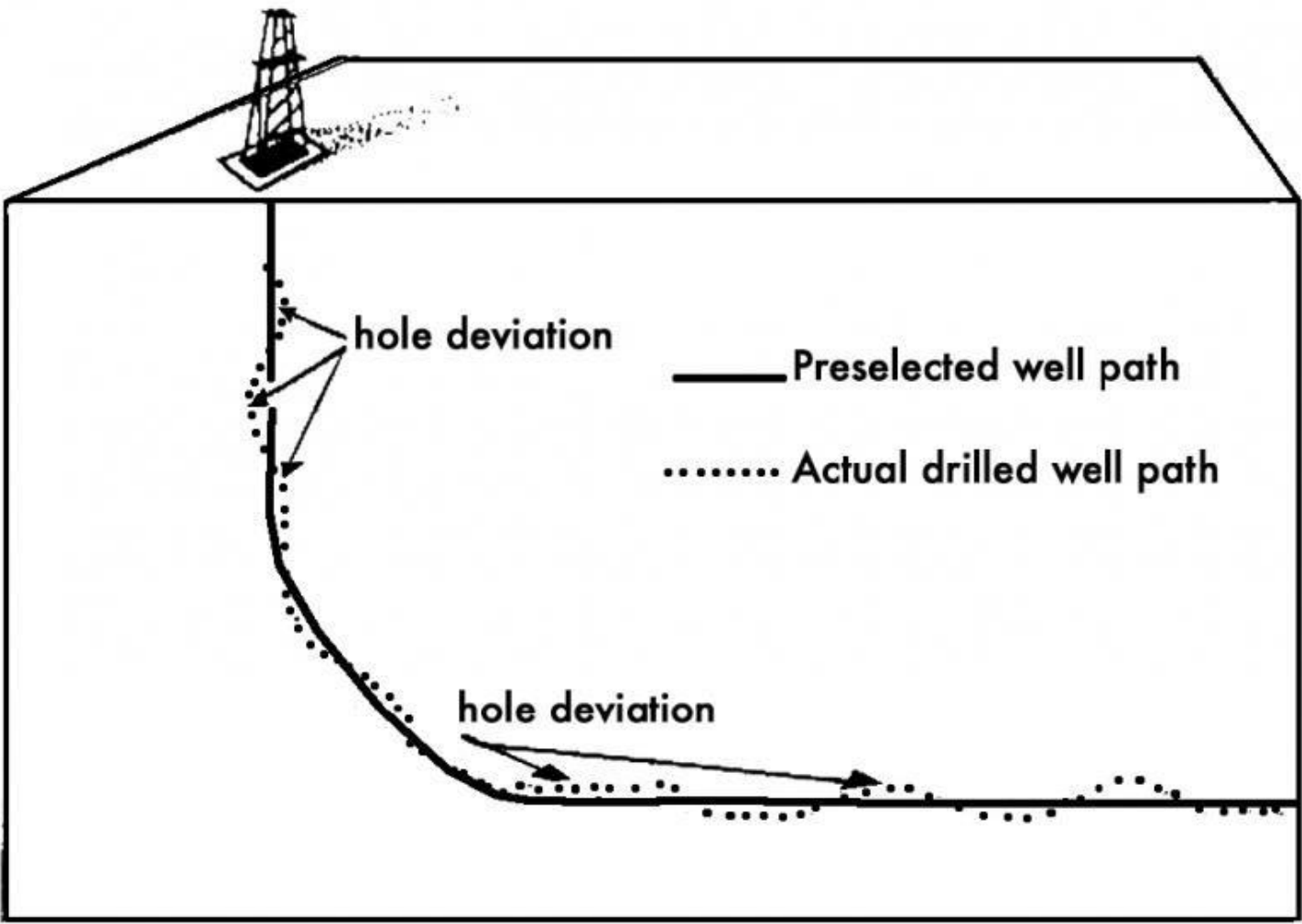


Mechanical: Wellbore Instability

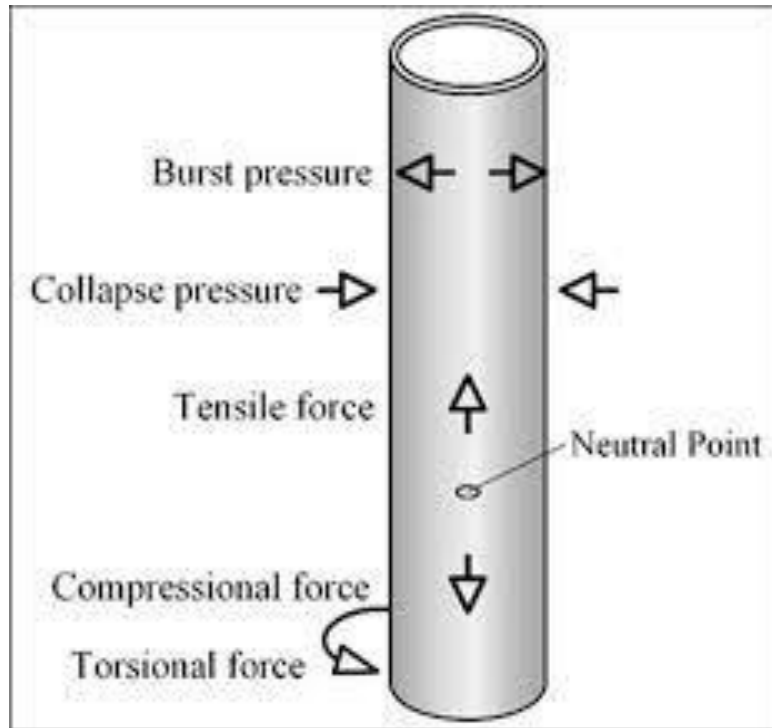
TABLE 1: Stuck pipe indicators during operations (Baker Hughes INTEQ, 1995)

Indicator problem	Torque	Pump pressure	Drill rate
Poor hole cleaning	Increase	Increase	Gradual decrease
High overbalance and permeable sands	Gradual increase	No change	Gradual decrease
Mobile formations	Gradual increase	Increase	Gradual decrease
Fractured and faulted formation	Sudden increase	May be unaffected	Sudden increase
Geo-pressured formation	Increase	Increase	Initial increase with a gradual decrease
Reactive formations	Gradual increase	Increase	Gradual decrease
Unconsolidated formation	Increase	Increase	Decrease
Junk	Sudden increase	No change	Sudden decrease
Cement blocks	Sudden increase	No change	Sudden decrease

Hole
deviation



Drillpipe failures



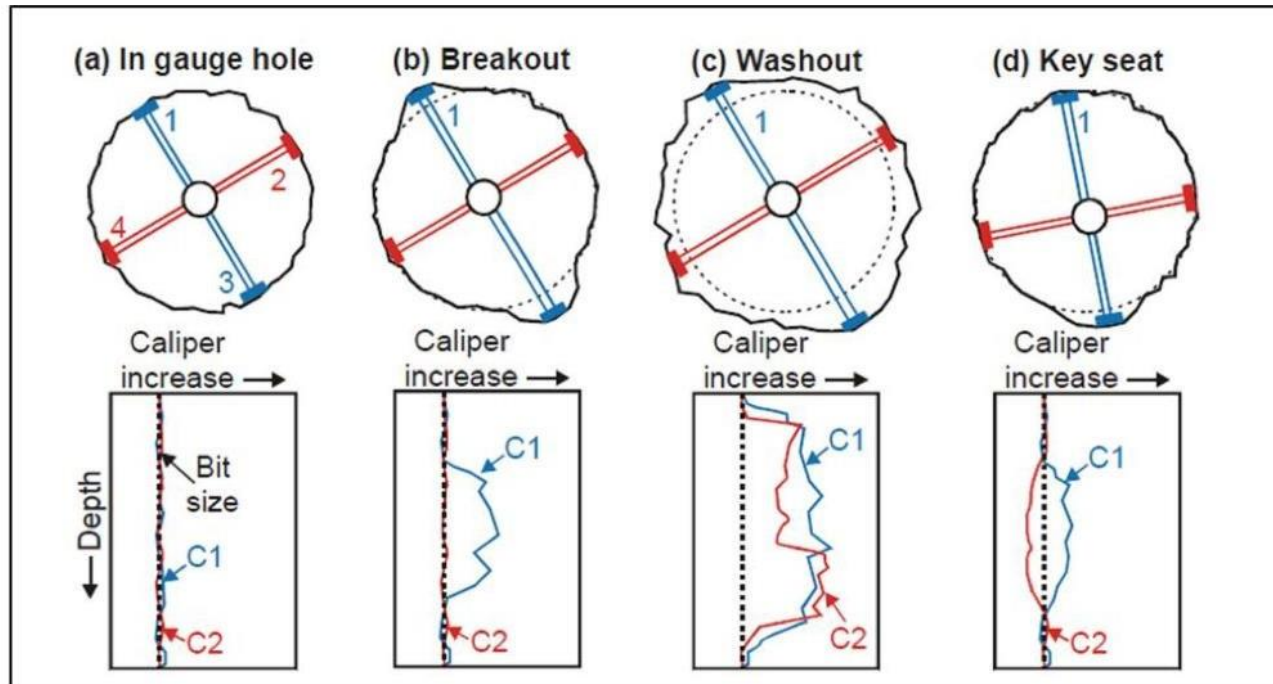
Twist off

Parting

Collapse
& Burst

Fatigue

WELLBORE INSTABILITY

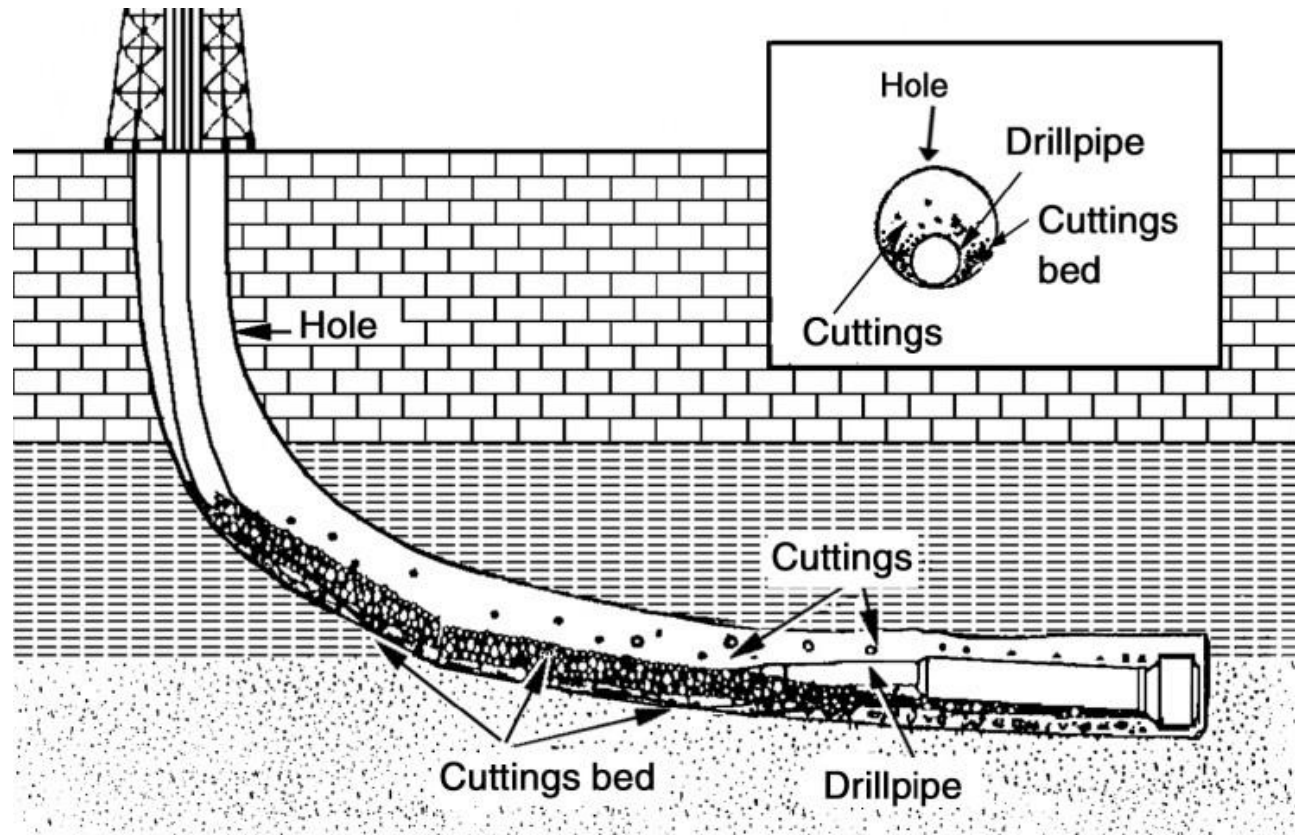


- Often losing wells
- To prevent: needs geomechanics data or model.

Hole cleaning

Cutting accumulated in the area where mud velocity reduced (after drill collar, enlarged hole) and in the wellbore bed (top of drill collar)

- Stuck Pipe monitored from torque and shaker
- Stuck Pipe action: jar and rotate + Circulating maximum rate



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